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Surveying-I.

(Bharat Bahadur Dhalal)

(BBD Sir)

relative
surface
angular

of the

ge, town,

my)

Surveying - I

Surveying - I → Th 100 Pg 50

Surveying - II → 100 50

Survey camp → 100 (viva)

Chapter 1: Introduction: (4 Marks)

→ Surveying is the art of determination of relative position of points on, above or below the surface of the earth by means of linear and angular measurement.

(Distance, elevation, Direction)
(horizontal) ↓ vertical ↓
↓ shortest nt Angle

→ The process of surveying in three stage,
- general view
- observation & Measurement
- presentation of data.

• Objectives of surveying:

- To prepare Map to show relative positions of the objects on the earth surface.
- To draw Map in suitable scale.
- To show Natural features such as village, town, roads, railways, etc.

• History:

Human civilization → Property ownership }

Astronomical positioning (sun observation, astronomy)

Napolean Bonapat →

19th century → Telescope

→ compass, theodolite, total station, GPS, DGPS

→ Robotic total station

□ Uses of Surveying :-

- To prepare topographical map (as such as ground feature)
- To prepare cadastal map (must have boundary map irrespective of topographical map)

Cadastal map: Map showing the field, houses and other properties

- To prepare an engineering map which shows engineering works such as roads, railways, reservoirs, etc.
- To prepare military map
- To prepare contour map
contour: imaginary line joining point of equal elevation.
- To prepare a geological map to determine areas including underground resources.
- To prepare Archaeological map.

□ Principle of Surveying :- (Imp for viva) Theory

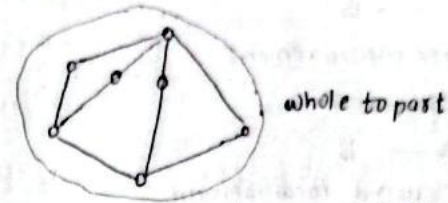
① Working from whole to part.

control point: (x, y, z) known

part to whole → error spread out

- Main idea of working from whole to part is to prevent accumulation of error with less precision method (closeness b/w repeatedly measured value)

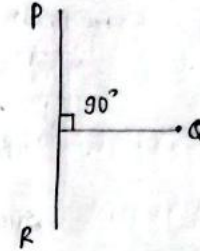
- Sufficient number of points with high degree establish in outer side
- Localize the error within the framework.



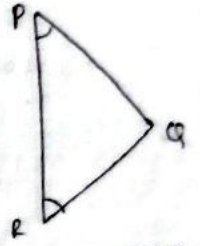
② Location of point by measurement from two control points



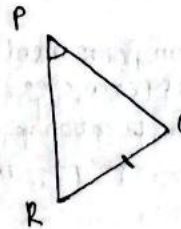
length known
(PQ & RQ known)



Angle known
length



Both Angles known



When angle on one side & length from other is known.

① & ② → Basic principles

③ Independent check:

Length:

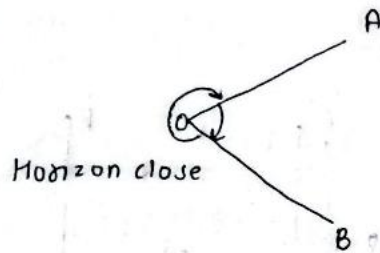
A ————— B

AB = Forward measurement

A ————— B

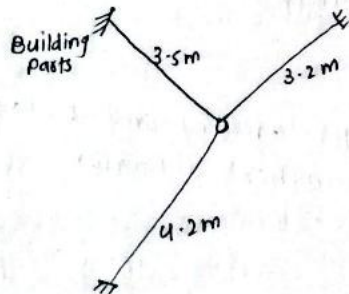
BA = Backward measurement

Angle:



एक Angle measure गर्दा
horizon close गर्ने
horizon close भएको ठाउँबाट
जहाँ start हुन्छ त्यही पुग्ने
Eg: $\angle AOB$ measure गर्ने
Then measure outer
 $\angle AOB$
Such that total angle 360°

Referencing



↑ N

→ From permanent structure
(referencing)

2 pts → location

extra 3rd pt → Independent
check

Permanent str. → fixed

④ Consistency:

x ————— x

Instrument & Surveyor's consistency

⑤ Accuracy & Economy

⑥ Honesty

↳ Proper use of instrument

↳ No data Manipulation

⑦ Safe Guarding

↳ Safe guard instrument / data

⑧ Scientific instrument handling

□ TRUE FACTS:

⇒ Spheroidal surface of earth.

⇒ 12 km long arc is 1 cm greater than chord

⇒ Σ Angles of spherical triangle = Σ Angles of plane triangle

= 1 sec for 195 sq km earth area.

Imp # Classification of Survey

- (A) Primary classification $\rightarrow$ Acc to Earth's curvature
(B) Secondary classification $\rightarrow$ Acc to Methods, Instruments, process, uses, etc.

(A) Primary Classification

- ① Plane Surveying $\rightarrow$ Earth curvature assumed st. line
② Geodetic Surveying
 $\hookrightarrow$ Earth curvature assumed curve
 $\hookrightarrow$ Done by Department of Surveying for Land parcels (न्यायी विभाग)
 $\hookrightarrow$ National level boundary

Scale: भारत $\rightarrow 1:1250$

① Plane Surveying

- $\rightarrow$ Spherical shape of earth neglected. Earth surface assumed plane
- $\rightarrow$ Plane trigonometry used
- $\rightarrow$ Covers area only less than 260km^2 (250km^2)
- $\rightarrow$ Used for highways, railways, canals, bridge, factory, etc [Engineering projects].

② Geodetic Surveying

- $\rightarrow$ Curvature of earth is taken into account
- $\rightarrow$ Line joining two points is considered curved.
- $\rightarrow$ Higher degree of precision
- $\rightarrow$ Covers area greater than 260km^2 (250km^2)
- $\rightarrow$ Spherical Trigonometry is used.
- $\rightarrow$ conducted by ~~the~~ Government Survey Department

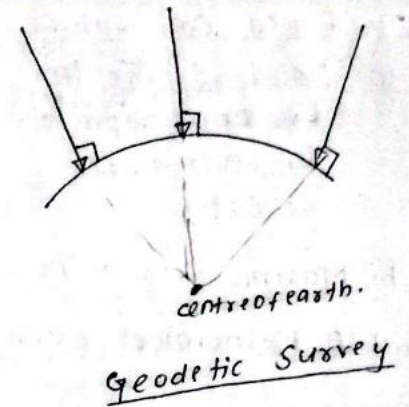
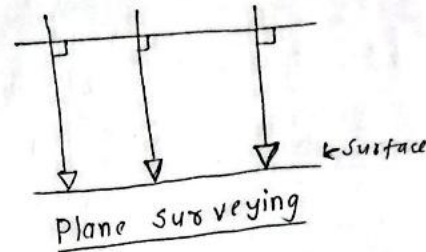


Fig: plumb direction in Surveying

(B) Secondary Classification :-

(I) Based on Instrument

- (a) Chain survey (ताप)
- (b) compass survey
- (c) plane Table survey
- (d) Theodolite survey
- (e) Tacheometric survey
- (f) Photogrammetry survey
- (g) Total Station Survey

(II) Based on method :

- (a) Triangulation surveying
- (b) Traverse surveying

(III) Based on objects

- (a) Geological surveying
- (b) Mine surveying
- (c) Archeological surveying
- (d) Military surveying
- (e) Engineering surveying

(V) Based On Nature of Field

- (a) Land Surveying
 - ↳ Topographical
 - ↳ cadastral
 - ↳ city
- (b) Marine Surveying
- (c) Astronomical Surveying

Chapter 2: Distance Measurement

↳ 2 Numericals + 1 Theory
(16 Marks)

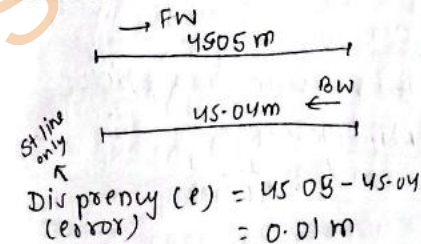
Methods of Linear Measurements

- (1) Pacing
- (2) Odometer
- (3) chaining/Taping
- (4) Tacheometry
- (5) Electronic Distance Measurement (EDM)
- (6) Global Positioning System (GPS)

(1) Pacing :-

Eg:

Precision:

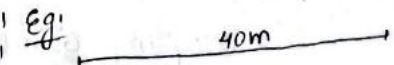


$$\text{Avg} = \frac{45.05 + 45.04}{2} = 45.045 \text{ m}$$

$$\text{Precision} = \frac{1}{\frac{\text{Avg}}{e}} = \frac{1}{\frac{45.045}{0.01}}$$

$$\text{Precision} = \frac{1}{4504.5}$$

4504.5 m Measure dist
1 m error at 4504.5 m



Pace count:

- 1 → 55
- 2 → 54
- 3 → 56

$$\text{Avg} \rightarrow \underline{55}$$

$$\text{Pace factor} = \frac{40 \text{ m}}{55} = 0.73 \text{ m} = 73 \text{ cm}$$

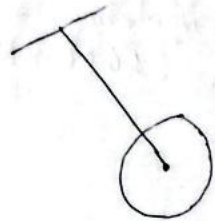
American/Britain standard
Av. pace factor = 80 cm

$$\text{Distance} = \text{Pace} \times \text{Pace factor}$$

Uphill : + some paces (assumption)
Downhill : - some paces from counted pace.

$$\text{Precision} = \frac{1}{200}$$

2. Odometer



distance = no of revolution $\times$
circumference of wheel

(3) Chaining / Taping

① _____ B

Precision: Fairly level ground = $\frac{1}{2000}$

Sloping Ground:

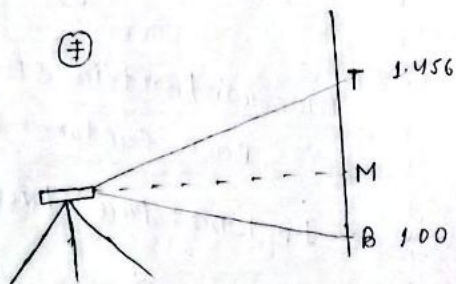
Direct Method: 1. 1000
2. 500

Indirect Method: 1:500

(4) Tacheometry :

4 Measuring hor & ver. distances using optical instrument) principle used,
 $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$



$$T = 1.456$$
$$\theta = 1$$

$$S = T - \beta$$

distance, $D = k \times S$

$$k = 100 \text{ (for given instrument)}$$

$$= 100 \times (1.456 - 1.00)$$

$$= 45.6 \text{ m}$$

(5) EDM

→ Mostly used.

- Mostly used.
- 1948 AD discovered

5) EDM

- Mostly used.
- 1948 AD discovered
- Ray emitted, & the reflected ray is caught up
calculated; velocity of ray known. Thus, distance
of given pt

(G) GPS

(6) GPS → gives (x, y, z) coordinates of given pt
→ gives distance formula to calculate d

5) GPS gives (x, y, z) coordinates of given pt.
→ use distance formula to calculate distance.
→ on Earth

- Google Earth

- Google earth
- runs thr satellite.

□ Significant Figures

→ Significant Figures
→ Depends on Least count of given instrument
→ No. of significant figures in an

- Depends on Least count of given instrument
- The number of significant figures in any observed value indicates the positive (certain) digit plus one (only one) digit that is estimated or round off.

eg: A distance measured with a tape whose L.C. | smallest graduation = 0.01 m graduation recorded 53.27m is said to have four significant figures.

Eg: Two significant figures:
40, 4.2, 0.43, 0.0042, 0.040

eg: Three significant figures:-
836, 83.6, 0.806, 0.0806, 0.00800,
1.02

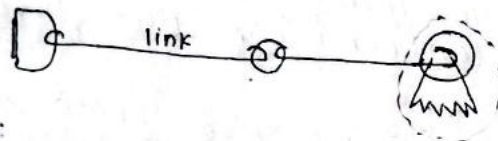
Equipments for chaining/Taping :-

(Accessories for linear measurement).

- chain/Tape
- Arrows
- Pegs
- Ranging Rod
- Plumb bob
- Hand level/Abney level

Chain : made up of Galvanized Iron Rod

↳ Generally 100-150 pieces



link → single unit
(least count)
tally → symbol
kept after certain
no. of links
(marking sp.
points &
after certain
length)

⇒ Types:

- Metric chain : (5, 10, 20m available)
- Gunter's chain : (Surveyor's chain)
[66 ft 100 links]
- Engineer's chain : (100 ft 100 link)
- Revenue chain (33 ft of 16 links)
- Steel band chain (20m or 30m)

Tape :

⇒ Types

- Cloth/Linen tape
- Metallic tape
- Steel tape
- Invar tape (less thermal expansion: more accurate)

Ranging Rod :-

- ↳ Metallic
- ↳ Plastic
- ↳ wooden

→ Use to range intermediate points on survey line.

→ 2m or 3m size.

→ Each Bands of 20cm painted alternate colour
(red & white, black & white)

Ranging Pole :

↳ Usually larger sized than Ranging Rod

→ large size for ranging long distance.

Plumb Bob :

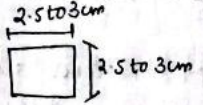
↳ Verticality of Ranging rod

→ Transferring the points to ground.

→ Used for centering purpose

(keeping centre of instrument & peg in same line)
(vertical pb.)

Arrow :-  ← self-made
→ establishing intermediate points.

Peg :-  → Marking/Fixing stations.

Standard size 

Ranging :-

- If Tape length < Measuring Distance
- Process of intermediate establishment of intermediate points
- When tape length is less than measured length, ranging used.

⇒ Types:

- (i) Direct Ranging
- (ii) Indirect Ranging

(I) Direct Ranging :-

- When two end points are visible

Ways:

- Direct eye
- Theodolite
- Total Station
- Level

(II) Indirect Ranging / Reciprocal Ranging

- When two end points are not visible



- M & N are such establish pt by surveyors st both A & B are visible from here



- Repeat the process (usually after 5 to 6 trials) AMNB is made st. line
- A & B not visible
 - Sit on M see A, then again see B st. N is made straight line
 - Again sit on N see B, then again see A, st. M is made st. line

Methods of Chaining / Taping on sloping Ground :-

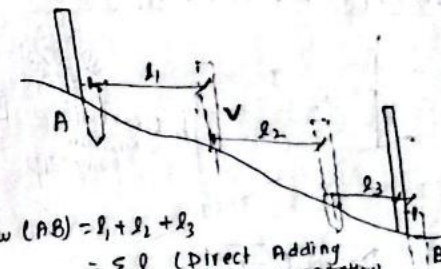
Methods:

- ① Direct Method (Stepping method) { measured lengths are added directly to get total length }
- ② Indirect Method { some computations/calculations are done before adding length of segment }

- (a) With aid of vertical angle measurement
- (b) With aid of difference in level
- (c) Hypotenusal allowances

(I) Direct Method:-

- Applied when ground is very steep.
- Ground is divided into number of horizontal & vertical strips.
- All the horizontal strip added.
- Tape should be horizontal.
- Sufficient pull to avoid sag.
- Precision 1:1000



$$FW(AB) = l_1 + l_2 + l_3 = \sum l \text{ (Direct Adding without computation)}$$

$$\text{Precision} = \frac{1}{\frac{\text{Avg}}{e}} = \frac{1}{1105.36} = \frac{1}{1105}$$

Bw Also

$$\text{'OP'} = 1:1105$$

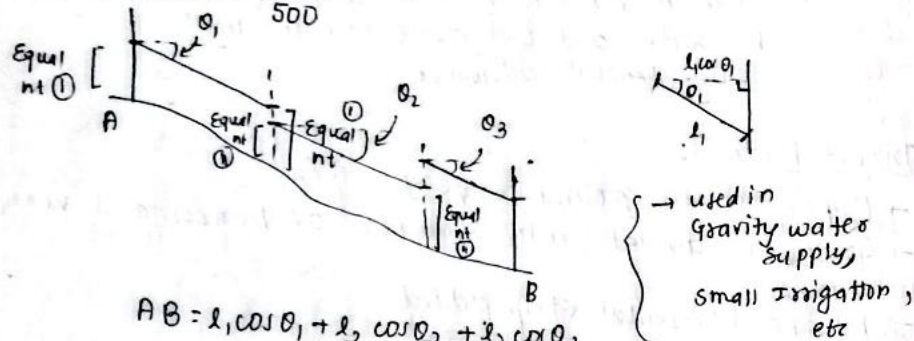
$$\text{'OP'} = 1 \text{ in } 1105$$

(II) Indirect Method:-

(a) Vertical Angle measurement:-

- Abney level to measure vertical angle
- When slope of the ground surface is long & gentle; stepping method is not suitable.

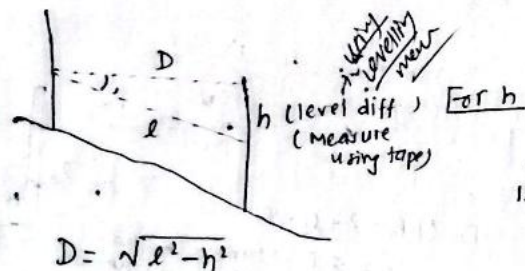
→ Precision $\frac{1}{500}$



$$AB = l_1 \cos \theta_1 + l_2 \cos \theta_2 + l_3 \cos \theta_3$$

$$AB = \sum l_i \cos \theta_i$$

(b) Indirect Method (Difference in level Measured)



$$D = \sqrt{l^2 - h^2}$$

(c) Indirect Method (Hypotenusal Allowances)

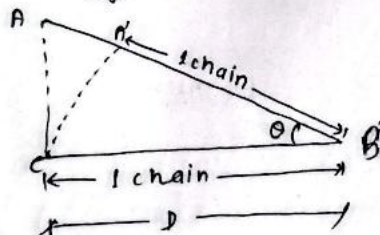
→ used for chain

→ due to slope correction

→ Hypotenusal Allowance AA' (slope correction) is computed for each slope change by

$$BA' = BC = 1 \text{ chain}$$

$$BA = 1 \text{ chain} \times \sec \theta$$



$$AA' = BA - BA' = 1 \text{ chain} (\sec \theta - 1)$$

$\theta \rightarrow 5^\circ \text{ or } 6^\circ$ (usually)
(seeing the slope)

Errors

$e = \text{measured value} - \text{true value}$

→ In engineering, we don't know true value (so, we take discrepancy or error)

→ True value unknown

Discrepancy means deviation in two measured quantity
(eg: $FW - BW$ of same length = Discrepancy)

→ Source of error:

(I) Natural error

(II) Instrumental error

(III) Personal error

(I) Natural error:

→ caused by adverse weather condition

→ Refraction (Bending of line of sight);

Obstacles to measure;

Gravity effects, etc.

→ Eg. Length of steel tape varies with temperature

(II) Instrumental Error:-

→ caused by imperfect construction & Adjustment of the surveying instrument

Eg: Incorrect graduation of steel tape

(III) Personal error:-

→ caused by the inability of the individual to make exact observation due to limitations of human sight, touch & hearing.

Types of error (Classification)

- (1) Mistakes
- (2) Systematic error (Accumulated error or cumulative error)
- (3) Accidental error (Compensating error)

(1) Mistakes

viva

(2) Systematic error

- ↳ same size and sign under some condition.
- ↳ Follow some definite mathematical or physical law.
- ↳ Error can be computed and correction can be applied.
- ↳ Seriously affects the survey work
- ↳ error $\propto$ length
- ↳ Positive error: when measured length is more than actual length.
: chain is too short.
- ↳ Negative error: when measured length less than actual i.e. chain is too long.

(3) Accidental error :-

- Error caused by the regions beyond the capacity of observer to control it.
- Tends to make result larger or smaller (positive or negative)
- Tends to balance each other
- Doesn't follow definite law
- Must be handled by mathematical law of probability
- Do not effect seriously for survey works.

Accuracy :

- ↳ Quality of result
- ↳ Accuracy is the degree of conformity with the standard or accepted value.
- ↳ Accuracy = True value - Most Probable Value
- ↳ Depends on methods, planning and instrument used.

Sum of angles of polygon
 $(2n \pm 4) \times 90^\circ$
+ → exterior angle
- → interior angle
n → number of sides

Precision :

- Also called Apparent accuracy
- ↳ Precision is the degree of perfection used in instrument, methods and observation.
- ↳ Closeness of one measurement to another (repetitive measurement)
- ↳ Good precision $\nrightarrow$ Accurate
- ↳ Good precision doesn't mean that good accuracy
- ↳ Eg: If tape is 29.99m instead of 30m can measure precise but not accurate.

□ Meaning of precision in distance measurement :-

For single measurement } Precision = $\frac{\text{Error in measurement}}{\text{Distance measured}} = \frac{1}{\frac{\text{Avg}}{e}} = \frac{1}{\text{Number}}$

→ $e = Fw - Bw$

→ Avg = $\frac{Fw + Bw}{2}$

→ For tape measurement

(1) Fairly level ground → $\frac{1}{2000}$

(2) Sloping (direct) → $\frac{1}{1000}$

Major → $\frac{1}{2000}$

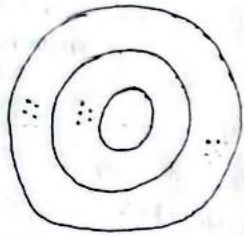
Minor → $\frac{1}{1000}$

(3) Sloping (indirect) $\rightarrow \frac{1}{500}$

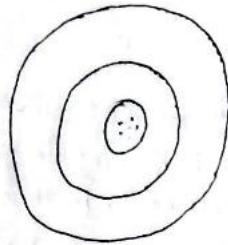
$\rightarrow$ For EDM,

(1) For Major $\rightarrow \frac{1}{5000}$

(2) For minor $\rightarrow \frac{1}{3000}$



Precision



Accuracy
& Precise

□ Relative Precision :

$\rightarrow$ Used to express the precision of various measurement

$\rightarrow$ Relative precision,

$$RP = \frac{\sigma}{M} = \frac{\frac{1}{M}}{\frac{1}{\sigma}} = \frac{1}{\text{Number}}$$

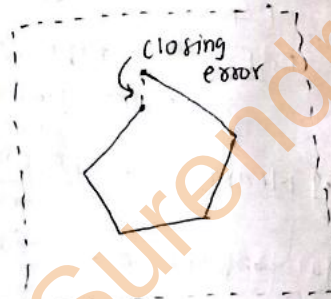
$\sigma \rightarrow$ Standard deviation

$M \rightarrow$ Mean

$\rightarrow$ Relative precision = $\frac{1}{\frac{P}{e}}$

$P =$ Perimeter

$e =$ closing error



□ Map Scales :

$\rightarrow$ For Maps

(a) Large scale : $< 1/200$

(b) Intermediate : $1:2000$ to $1:10,000$

(c) Small scale : $1:10,000$ to $1:100,000,000$

□ GK large scale : $\frac{1}{50}, \frac{1}{100}, \frac{1}{200}$
Small scale : $\frac{1}{500}, \frac{1}{1000}, \frac{1}{10000}$ } comparatively

$\rightarrow$ For Plans

(a) Site Plan : $1:50$ to $1:500$

(b) Detail plan : $1:1, 1:20$

□ Methods of scale :-

(1) Numerical scale

(2) Graphical scale

(1) Numerical Scale :

(a) Engineer's scale : eg. $1\text{cm} = 5\text{m}, 1\text{cm} = 10\text{m}, \text{etc}$

(b) Fraction scale : eg. $1:500, 1:1000, 1:10000$

$\propto$ TO convert Engineer's scale into fraction scale multiply the whole number of meter by 100 (i.e. put in same unit)

E.g. = Engineer's scale is $1\text{cm} = 5\text{m}$

$\therefore$ Fraction scale = 5×100 or $1:500$

Representative Fraction,
 $R.F. = \frac{\text{distance on drawing of object}}{\text{corresponding actual distance of object}}$ (Both in same unit)

$$1 \text{ cm} = 10 \text{ mm}$$

$$\therefore R.F. = \frac{1 \text{ cm}}{10 \times 100 \text{ cm}} = \frac{1}{1000}$$

(2) Graphical scale

↳ Graphical representation

Classification of scales:

- (1) Plain scale
- (2) Diagonal scale
- (3) Scale of chords - Protractor
- (4) Vernier scale

(1) Plain Scale

- Used to represent two dimensions only
 - meter and decimeter
 - kilo meters and hecta meters etc.

Full size → 1:1

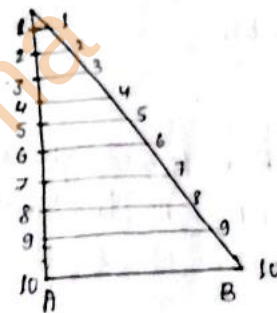
50cm to meter → 1:2

20cm to meter → 1:5

5mm to a meter → 1:200

(2) Diagonal Scale:-

- On diagonal scale it is possible to measure three dimensions such as meters, decimeters and centimeters, etc.
- Short length of plain scale is divided into 10 equal parts.
- Using principle of similar triangle.



$$\text{Distance } 1-1 = \frac{1}{10} AB$$

$$\text{Distance } 2-2 = \frac{2}{10} AB$$

$$\text{Distance } 9-9 = \frac{9}{10} AB$$

Shrink Scale

- Plan drawn shrinks due to variation of atmospheric condition.

→ The shrink scale = $\frac{\text{shrinkage factor} \times \text{original scale}}{\text{shrinkage factor (SF)}}$

→ shrinkage factor (SF) = $\frac{\text{shrink length}}{\text{The actual length}}$

E.g. → Let, shrinkage factor = $\frac{19}{20}$

thus, original scale = $\frac{1}{1900}$

$$\text{Shrink scale} = \frac{19}{20} \times \frac{1}{1900} = \frac{1}{2000}$$

Now, 1cm = 20m

Factors affecting while selecting plotting scale :-

- (1) Sheet size
- (2) Sheet No.
- (3) Precision Required
- (4) Nature of Terrain
→ Flat - large scale

(5) Plotting Accuracy

(6) consistency between scale
(compatible to other map)

(7) time and fund Available

Q Draw a diagonal scale R.F. = 1:2.5 showing centimeters and millimeters and long enough to measure up to 20 centimeters.

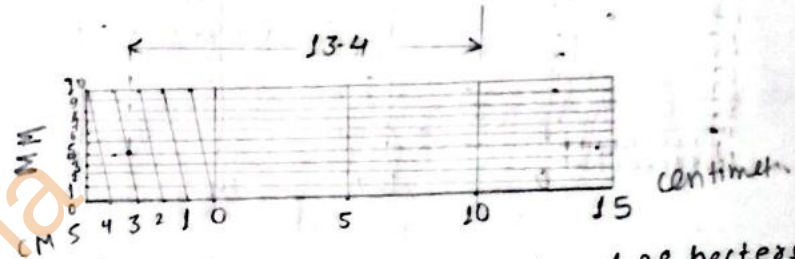
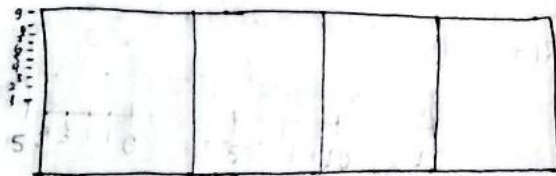
→ Solⁿ:-

$$R.F. = \frac{1}{2.5}$$

$$\text{length of scale} = R.F. \times \text{length to be measured}$$

$$= \frac{1}{2.5} \times 20 = 8\text{cm}$$

→ Divide in 4 equal parts



Q A Rectangular plot of land measuring 1.28 hectares is represented on a map by a similar rectangle of 8 sq. cm. Calculate R.F. Draw diagonal scale to read a single meter. Show a distance of 438m on it.

→ Solⁿ:-

$$1 \text{ hectare} = 10,000 \text{ sq. m}$$

$$1.28 \text{ hectare} = 1.28 \times 10,000 = 12800 \text{ sq. m}$$

$$= 12800 \times 10^4 \text{ sq. cm}$$

$$8 \text{ sq. cm on a map represents} = 12800 \times 10^4 \text{ sq. cm on land}$$

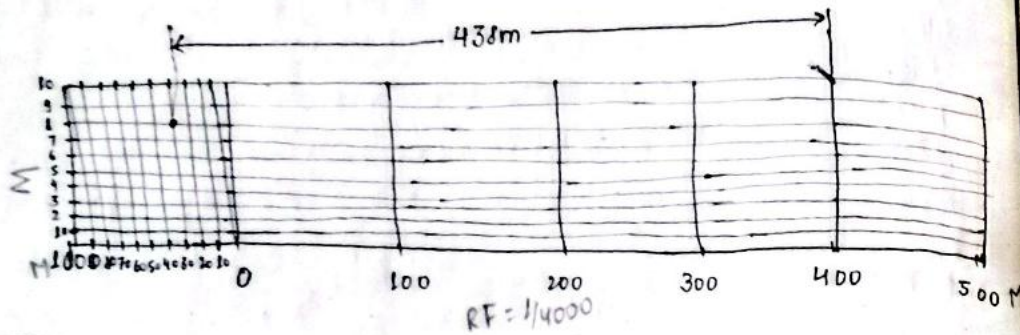
$$1 \text{ sq. cm on map} = 16 \times 10^6 \text{ sq. cm on land}$$

$$1 \text{ cm} = \sqrt{16 \times 10^6 \text{ sq. cm}}$$

$$1 \text{ cm on map} = 4000 \text{ cm on land}$$

$$\therefore R.F. = \frac{1}{4000}$$

$$\text{length of scale} = \frac{1}{4000} \times (600 \times 100) \text{ cm} = 15 \text{ cm}$$



Q. The distance between Kathmandu and Pokhara is 200 km in a highway map. It is represented by a line 5 cm long. Find RF. Draw diagonal scale to show single km, read up to 600 km. Measure distance.

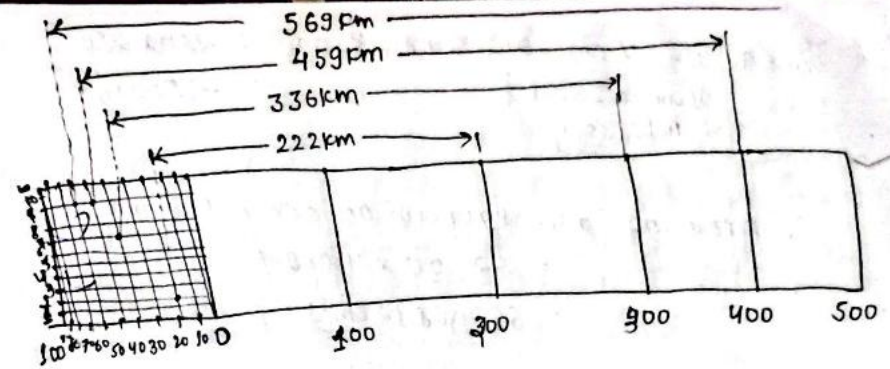
- Soln:-
- (i) 222 km
 - (ii) 336 km
 - (iii) 459 km
 - (iv) 569 km

→ Soln:-

5 cm in map = 200 km on land

$$RF = \frac{5 \text{ cm}}{200 \times 10^3 \times 100 \text{ cm}} = \frac{1}{40,00,000}$$

$$\begin{aligned} \text{length of scale} &= \left(\frac{1}{40,00,000} \times 600 \times 10^3 \times 100 \right) \text{ cm} \\ &= 15 \text{ cm.} \end{aligned}$$



Q. The area of a plot in a map is 10.22 cm². The scale of map was 1:25000 but present it is shrunk such that a line originally 5 cm in map is now 4.8 cm. What is correct field area?

→ Soln:-

$$SF = \frac{\text{Shrunk length}}{\text{Actual length}} = \frac{4.8}{5} = \frac{24}{25} = 0.96$$

$$\begin{aligned} \text{Shrunk scale} &= SF \times \text{original scale} \\ &= \frac{24}{25} \times \frac{1}{25000} \end{aligned}$$

→ The area of plot = 10.22 cm²

$$\begin{aligned} \rightarrow \text{The ratio of shrinkage area} &= \left(\frac{\text{Shrunk length}}{\text{Actual length}} \right)^2 \\ &= \left(\frac{4.8}{5} \right)^2 \\ &= 0.9216 \end{aligned}$$

$$\begin{aligned} \text{Area of plot on shrunk map on } 1:25000 \\ \text{scale} &= \frac{\text{Actual area}}{SF} = \frac{10.22}{0.9216} = 11.0891 \text{ cm}^2 \end{aligned}$$

1. Area of 1 cm^2 on scale 25000 = 250×250
 $\left\{ \begin{array}{l} 1 \text{ cm} = 250 \text{ m} \\ \text{in } 1:25000 \text{ scale} \end{array} \right\}$ = 62500 m^2

$\therefore$ Area of 11.08941 cm^2 on scale 1:25000
 $= 62500 \times 11.08941$
 $= 693088.12 \text{ cm}^2$

□ chain Tape/Correction :-

- (1) Correction for absolute length or standardization.
- (2) Correction for temperature.
- (3) Correction for slope.
- (4) Correction for pull or tension.
- (5) Correction for sag
- (6) Correction for mean sea level
- (7) Correction for misalignment.

(1) Correction for absolute length or standardization :-

$C_a = \text{True length} - \text{nominal length.}$

$C_a = \frac{CL}{L}$

where,

$C_a = \text{Correction for absolute length.}$

हुनुपर्ने value/length

$C = \text{Correction per tape}$

$L = \text{nominal length of line (Designated length)}$

$L = \text{Measured length of line}$

→ If, absolute length > Nominal length
 $\Rightarrow$ correction +ve & vice-versa.

Absolute length > Nominal length $\rightarrow$ Measured length/distance

Nominal length $\rightarrow$ temp/presure/diff. condition $\rightarrow$ Short
 अवांले गर्दा आएको deviated length

with error
 Normal cond. मा measure गर्नुको length.

$\frac{\text{True distance}}{\text{Measured distance}} = \frac{\text{Actual length of tape}}{\text{Nominal length of tape}}$

For Chain,

$\text{True distance} = \frac{L'}{L} \times \text{measured distance.}$

$\text{True area} = \left(\frac{L'}{L} \right)^2 \times \text{measured area}$

Sim. for volume.

where,

$L' = \text{incorrect length of chain}$

$L = \text{correct length of chain}$

For new work : (& for laying out work just vice versa)

If tape too long $\rightarrow$ correction positive
 If tape is too short $\rightarrow$ correction -ve

② Correction for Temperature

$$C_t = \alpha * (t_m - t_0) * L$$

Where,

C_t = correction in temp^r

t_m = mean temp^r

t_0 = standardization temp^r (standard temp)

α = coeff of thermal expansion

L = measured length.

If $t_m > t_0$ is positive correction

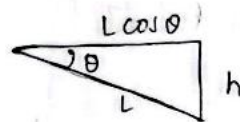
If $t_m < t_0$ is negative correction.

③ Correction for slope

θ = angle of slope

L = measured length

C_s = correction



$$C_s = L - L \cos \theta \rightarrow \text{Actual}$$

$$= L(1 - \cos \theta)$$

$$C_s = L - \sqrt{L^2 - h^2} = L - (L^2 - h^2)^{1/2}$$

$$C_s = L - L \left(1 - \frac{h^2}{L^2} \right)^{1/2}$$

Expanding term $\left(1 - \frac{h^2}{L^2} \right)^{1/2}$ binomially.

$$C_s = L - L \left(1 - \frac{h^2}{2L^2} - \frac{h^4}{8L^4} - \dots \right)$$

$$= L - L + \frac{h^2}{2L} + \frac{h^4}{8L^3} = \frac{h^2}{2L} + \frac{h^4}{8L^3}$$

If $L = 20\text{m}$, $h = 3\text{m}$

$$\frac{h^4}{8L^3} = \frac{3^4}{8 \times 20^3} =$$

Now, neglect the term & approx.

$$C_s = \frac{h^2}{2L} \text{ which is always subtractive}$$

(4) Correction for pull or Tension:-

Hooke's law,

Stress = strain x constant

$$C_p = \frac{P - P_0}{AE} \times L$$

C_p = Pull correction.

P = Pull applied in field

P_0 = Standard pull in kg or N

E = Modulus of Elasticity

L = Measured length

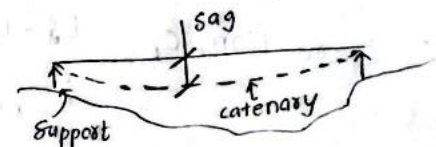
$C_p \rightarrow$ Positive ; for $P > P_0$

$C_p \rightarrow$ Negative ; for $P < P_0$

(A) Tape area = $\frac{\text{weight}}{\text{length} \times \text{sp. wt. of tape}}$

(5) Correction for sag :-

$$C_s = \frac{1}{24} \left(\frac{W}{P} \right)^2 \times L$$



C_s = sag correction

W = weight of chain/tape per span length in N

$$W = w \times L$$

w = weight per unit length

L = Length between two supports.

$$C_s = \pm \frac{n W^2 L^3}{24 p^2}$$

n = no. of length/bays

Note → If tape standardized in flat & used in catenary in field → sag correction negative.

→ If standardized in catenary but used in flat; sag correction positive

If correction for pull neutralize the sag correction,

$$\frac{(P - P_0)L}{AE} = \frac{W^2 L^3}{24 p^2}$$

$$\Rightarrow P = \frac{W \sqrt{AE}}{\sqrt{24(P - P_0)}}$$

⑥ Corrections for Mean Sea Level

$$L = (R + h) \alpha \quad \text{--- (i)}$$

$$L_e = R \alpha \quad \text{--- (ii)}$$

From (i) & (ii)

$$(R + h) \alpha \times e = R \alpha \times e$$

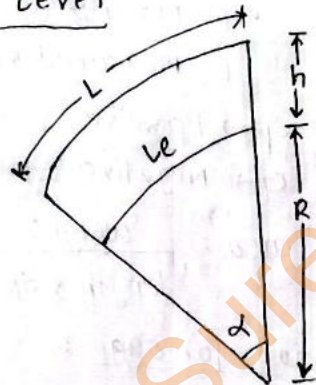
$$e = \frac{R}{R + h}$$

$$\therefore \text{Correction, } C_R = L_e - L$$

$$= \frac{h}{R + h} \times L$$

$$= \frac{h}{R} \times L$$

∴ always, subtractive.



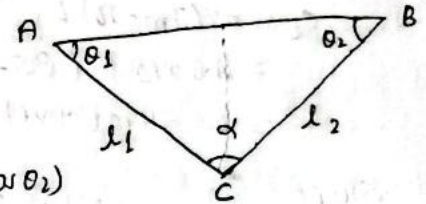
⑦ Correction for incorrect alignment:

$$AB = l_1 \cos \theta_1 + l_2 \cos \theta_2$$

$$\therefore \text{Reqd correction} = (l_1 + l_2) - (l_1 \cos \theta_1 + l_2 \cos \theta_2)$$

$$AB = \sqrt{AC^2 + BC^2 - 2ACBC \cos \alpha}$$

correction always subtractive.



□ Measuring a correct length with a wrong scale:
correct length = $\frac{\text{RF of wrong scale}}{\text{RF of correct scale}} \times \text{Measured length}$.

$$\text{Correct area} = \left(\frac{\text{RF of wrong scale}}{\text{RF of correct scale}} \right)^2 \times \text{measured area}$$

Numericals

(Q) A base line measured with a steel tape gives an approx. length of 1000m. Compute correct length of base line at MSL when the pull at standardization = 15kg. Applied pull = 23kg; cross section of tape 0.0645 cm^2 & $E = 2.11 \times 10^6 \text{ kg/cm}^2$; $t_m = 35^\circ \text{C}$ & $T_0 = 15^\circ \text{C}$.

★ $\alpha = 11.5 \times 10^{-6} / ^\circ \text{C}$. The difference in level of two ends of base line is 2 m, $R = 6400 \text{ km}$, Elevation of base line about MSL = 1000m.

→ Soln:-

$$\text{For pull, } C_p = \frac{(P - P_0) \times L}{AE} = \frac{(23 - 15) 1000}{0.0645 \times 2.11 \times 10^6}$$

$$= 0.05878 \text{ m (+ve)}$$

Temperature,

$$C_t = \alpha (T_m - T_0) L$$
$$= 11.5 \times 10^{-6} (35 - 15) \times 1000$$
$$= 0.24 \text{ m (+ve)}$$

Slope, $C_{sl} = \frac{h^2}{2L} = \frac{2^2}{2 \times 1000} = 2 \times 10^{-3} \text{ (-ve)}$

MSL $C_f = \frac{h}{R} \times L = \frac{1000 \times 1000}{6400 \times 1000} = 0.15625 \text{ (-ve)}$

$$\text{Total correction} = 0.05878 + 0.24 - 2 \times 10^{-3} - 0.15625$$
$$= 0.1405 \text{ m}$$

$\therefore$ Correct base length = $1000 + 0.1405$
Last result = 1000.140 m .
3 digit

Chapter 8-4 :

The compass

14 Marks

→ Traversing is the Framework consists of connected series of lines.

→ Length measured by tape & direction by compass.

□ Direction of Line : Horizontal angle with reference line.

Reference line is called Meridian.

→ Types:

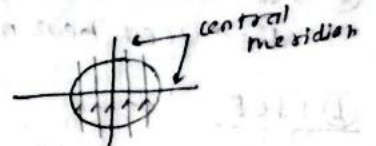
① Astronomic Meridian:

→ with respect to Geographic north and south pole



② Magnetic Meridian:

→ direction of vertical plane shown by a freely suspended magnetic needles.



③ Grid Meridian

→ A line through a point parallel to central meridian or y-axis of rectangular co-ordinate system.

④ Arbitrary Meridian

→ is any convenient direction towards a permanent and prominent mark or signals such as temple top, chimney, etc.

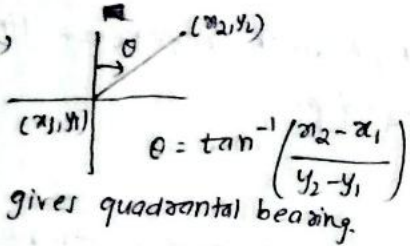
□ Bearing

→ Bearing of line is its direction relative to given meridian.

→ Horizontal angle with respect to given meridian.

Types

- ↳ True bearing
- ↳ Magnetic bearing - (bearing)
- ↳ Grid bearing
- ↳ Arbitrary bearing



Designation Bearing

Two system

- ① Whole circle Bearing (WCB)
- ② Quadrantal Bearing (QB)
Reduced Bearing (RB)

① WCB.

→ Measured with Magnetic North towards line in clockwise direction

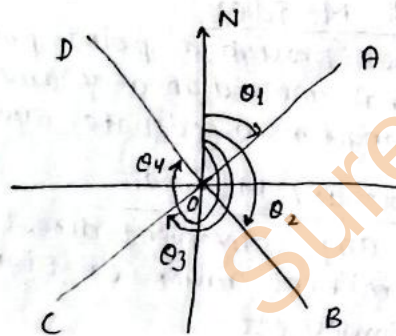
Here,

$$\text{WCB of OA} = \theta_1$$

$$\text{WCB of OB} = \theta_2$$

$$\text{WCB of OC} = \theta_3$$

$$\text{WCB of OD} = \theta_4$$



→ Value of bearing $0^\circ - 360^\circ$

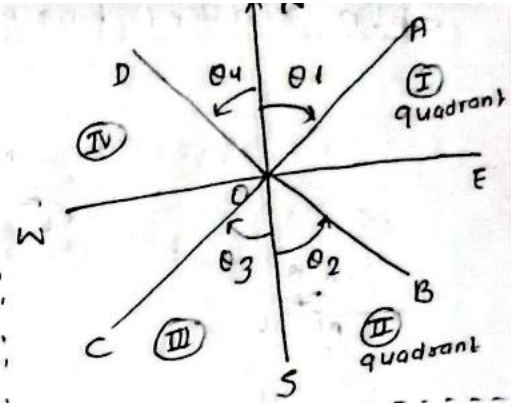
→ Observed by prismatic compass

↳ In Astronomical term, WCB $\Leftrightarrow$ Azimuth

② QB/RB

→ Bearing of line is measured eastward or westward from North or South whichever is nearer.

→ Direction either clockwise or Anticlockwise



→ Here,

$$\text{QB of OA} = \text{N } \theta_1 \text{ E}$$

$$\text{QB of OB} = \text{S } \theta_2 \text{ E}$$

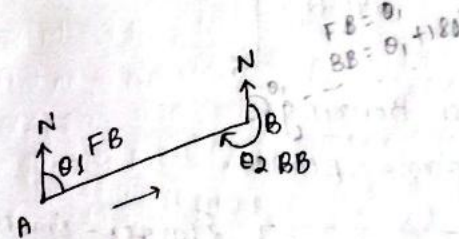
$$\text{QB of OC} = \text{S } \theta_3 \text{ W}$$

$$\text{QB of OD} = \text{N } \theta_4 \text{ W}$$

→ Varies from $0^\circ - 90^\circ$

→ Measured by Surveyor's compass.

Fore Bearing (FB) & Back Bearing (BB)



① In WCB system,

$$\text{BB} = \text{FB} \pm 180^\circ$$

+ve when FB less than 180°

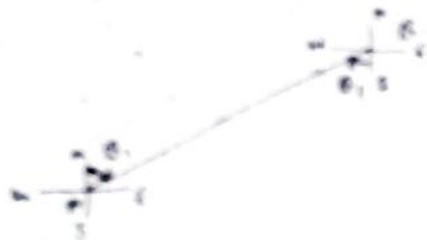
-ve; BB when FB $> 180^\circ$

$$\text{FB} = \theta_1$$

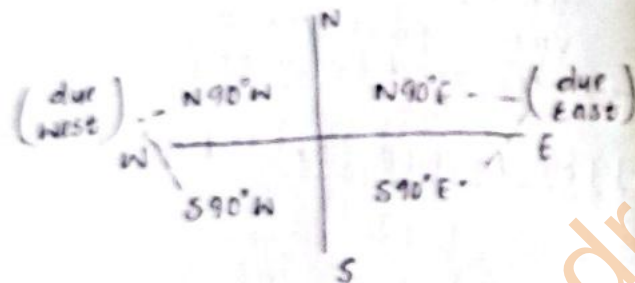
$$\text{BB} = \theta_2 - 180^\circ$$

② In reduced bearing (RB)

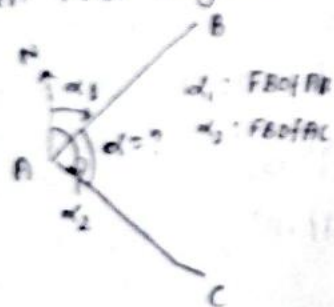
→ FB and BB are numerically equal but quadrant are just opposite



Eg. FB of AB in RB = N 45° E
BB of AB in RB = S 45° W
→ All values are in ideal condition.



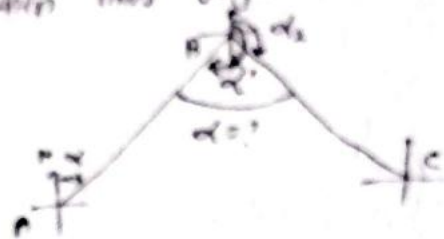
③ Angle b/w two lines from Bearing
④ When lines originated at same points.



$$\alpha = \alpha_2 - \alpha_1 \quad \left\{ \begin{array}{l} \text{larger} - \text{smaller} \end{array} \right\}$$

Common origin.

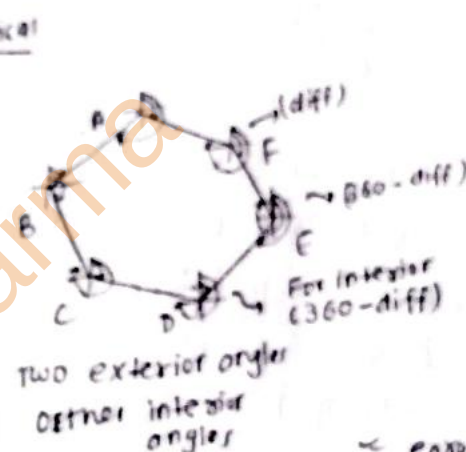
⑤ When lines originated from different points



$$\alpha = \alpha' - \alpha_2$$

$$\alpha = \text{FB of BA} / \text{BB of AB} - \text{FB of BC}$$

→ Practical



$$\angle A, \angle B, \angle C, \angle D, \angle E, \angle F$$

$$\text{Observed sum} = 720' 30' \quad (\text{say})$$

$$\{ (2 \times 6 - 4) \times 90 = 720 \}$$

$$\text{Theoretical sum} = (2n \pm 4) \times 90'$$

+ve → exterior
-ve → interior angle

$$\text{error (e)} = \text{Observed sum} - \text{theoretical sum}$$

$$\text{Permissible error} = L \cdot C \cdot \sqrt{N} \text{ or } C\sqrt{N}$$

L → least count

C → constant

N → Number of station

→ If error within permissible ; distribute error equally
But, not in fraction.

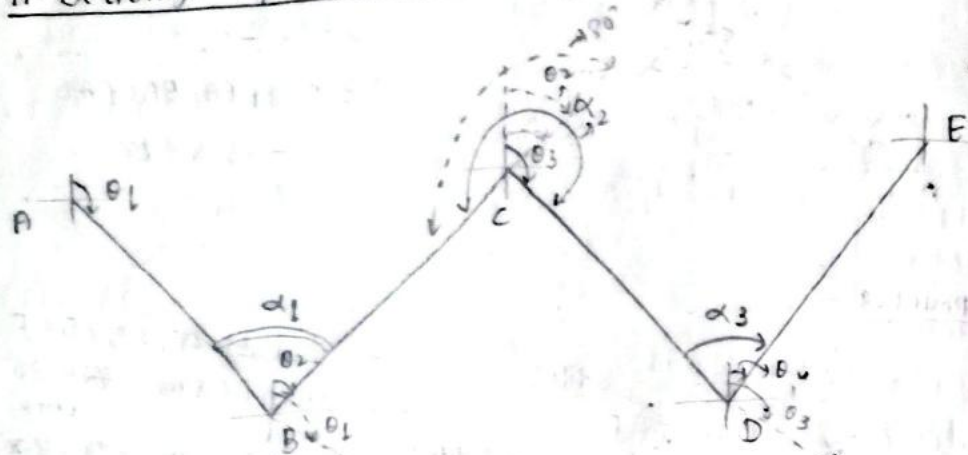
eg. error: 30' : $\frac{30'}{5} = 6'$ { Not fraction no problem distribute error equally }

If error: 27' : $\frac{27'}{5} = \text{Fraction}$ So, distribute equally

① distribute equally

② Then, distribute remaining 3' whole equally

Bearing of line from angles



Angle जलिते clockwise लिजे in Bearing calculation
 ↳ It is also called 'angle to the right'
 ↳ Previous side वर successive side तिर
 right/clockwise तिर को Angle

↳ Data लाई collect मान्ने (Given)

Here,

θ_1 = FB of AB known

Included angles $\alpha_1, \alpha_2, \alpha_3, \dots$ = known

Bearing of lines $\theta_2, \theta_3, \theta_4, \dots$ (unknown) $\rightarrow ?$

$\rightarrow$ Bearing of line BC = $\theta_2 = (\alpha_1 + \theta_1) - 180^\circ$

$\rightarrow$ Bearing of line CD = $\theta_3 = (\theta_2 + \alpha_2) - 180^\circ$

$\rightarrow$ Bearing of line DE = $\theta_4 = (\theta_3 + \alpha_3) - 180^\circ$

* Note:
 FB of line = (FB of previous line + Angle to the right) $\pm 180^\circ$ or -540° clockwise on

$$\pm 180^\circ \text{ or } -540^\circ$$

- If $\text{sum } \theta \geq 180^\circ$; subtract 180°
- If $\text{sum } \theta < 180^\circ$; add 180°
- If $\text{sum } \theta \geq 540^\circ \rightarrow -540^\circ$

$$\begin{aligned} \textcircled{I} : & > 180^\circ \rightarrow -180^\circ \\ & < 180^\circ \rightarrow +180^\circ \\ & > 540^\circ \rightarrow -540^\circ \end{aligned}$$

Q. FB of BC & CD = ?

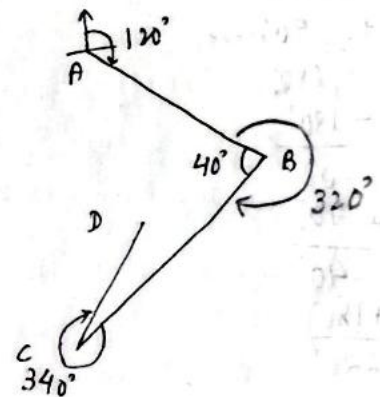
Sol:-

$$\rightarrow \text{FB of AB} = (120^\circ + 320^\circ) - 180^\circ$$

$$\begin{aligned} \text{FB of AB} &= 120^\circ \\ &+ 320^\circ \\ &\hline &440^\circ \\ &- 180^\circ \\ &\hline \end{aligned}$$

$$\begin{aligned} \text{FB of BC} &= 260^\circ \\ &+ 340^\circ \\ &\hline &600^\circ \\ &- 540^\circ \\ &\hline \end{aligned}$$

$$\text{FB of CD} = 60^\circ$$



#

→ Sol:-

$$\text{FB of AB} = 320^\circ$$

$$+ 60^\circ$$

$$\hline 380^\circ$$

$$- 180^\circ$$

$$\text{FB of BC} = 200^\circ$$

$$+ 110^\circ$$

$$\hline 310^\circ$$

$$- 180^\circ$$

$$\text{FB of CD} = 130^\circ$$

$$+ 50^\circ$$

$$\hline 180^\circ$$

$$- 180^\circ$$

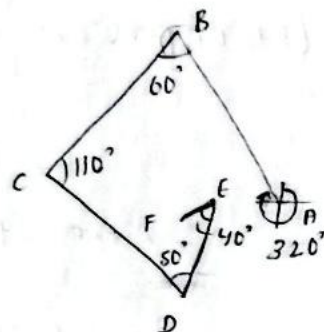
$$\text{FB of DE} = 0^\circ$$

$$+ 40^\circ$$

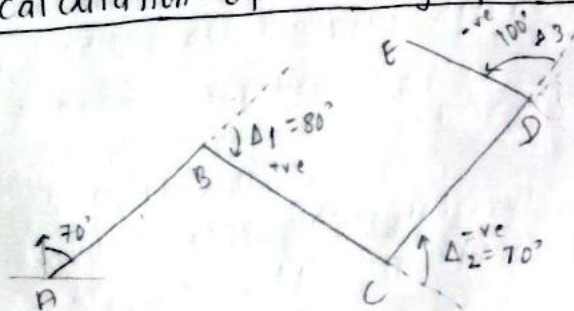
$$\hline 40^\circ$$

$$+ 180^\circ$$

$$\text{FB of EF} = 220^\circ$$



calculation of bearing from deflection Angle:-



$\Delta_1, \Delta_2, \Delta_3 = \text{deflection}$
 Right deflection $\rightarrow +ve$
 Left deflection $\rightarrow -ve$
 Deflection Angle

• FB of line = FB of previous line

If sum $< 360^\circ \rightarrow$ value gives bearing
 If sum $> 360^\circ \rightarrow$ subtract 360°
 If sum $-ve \rightarrow$ Add 360°

For given problem,

$$\text{FB of AB} = 70^\circ$$

$$+ 80^\circ \text{ (Deflection)}$$

$$\text{FB of BC} = 150^\circ (< 360^\circ)$$

$$- 70^\circ \text{ (Deflection)}$$

$$\text{FB of CD} = 80^\circ (< 360^\circ)$$

$$- 100^\circ \text{ (Deflection)}$$

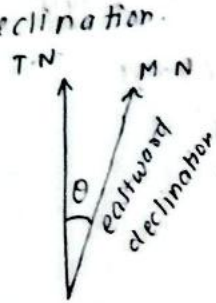
$$- 20^\circ (-ve)$$

$$+ 360^\circ$$

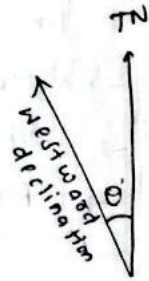
$$\text{FB of DE} = 340^\circ$$

□ Magnetic Declination:

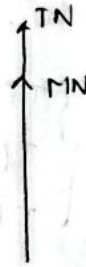
Horizontal angle between the magnetic meridian and true meridian is known as magnetic declination.



Positive declination



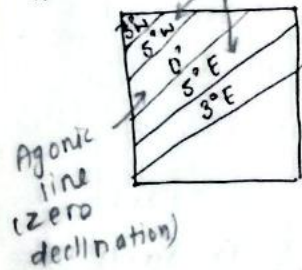
Negative declination



Zero declination

TN = True North
MN = Magnetic North

Isogonic line
(same declination value)



□ Variation of Magnetic declination

① Secular Variation

→ Magnetic Meridian behaves like pendulum w.r.t. true meridian

→ About 100 years interval

→ In Paris 11°E in 1680 to 22°W in 1820.

② Annual variation
→ Varies due to rotation of earth with its axes inclined in an elliptical path around Sun.
→ Amt. is about 1' to 2'.

③ Diurnal variation
→ Varies due to rotation of earth on its axis in 24 hr.
→ Amount of variation 3' to 12'.

④ Irregular variation
→ Suddenly due to natural causes such as earthquakes, volcanic eruption and soon.

□ Determination of True Bearing :-

True bearing = Magnetic bearing $\pm$ declination

+ve → eastward declination
-ve → westward declination.

Local Attraction

Conversion of WCB to FB:

Case	WCB	Rule for FB	Quadrant
NE	0° to 90°	$F.B = WCB$	
SE	90° to 180°	$F.B = 180^\circ - WCB$	
SW	180° to 270°	$F.B = WCB - 180^\circ$	
NW	270° to 360°	$F.B = 360^\circ - WCB$	



Principle of Compass Surveying:

Principle of compass surveying is traversing which includes a series of connected lines.

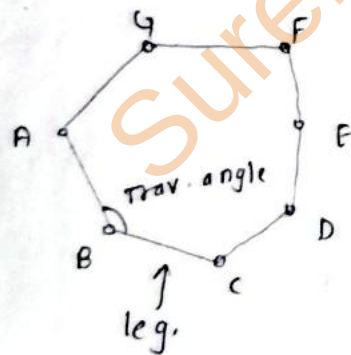
- direction : compass
- length : tape

Traversing:

- ↳ connected series of lines
- ↳ Sides:- traverse legs
- ↳ measurement of length & direction

Types

- ① closed traverse
- ② open traverse



① Closed traverse:

Start from the known and ended at the known.

Further, Types:-

- (i) closed loop closed traverse
- (ii) open loop closed traverse

(i) Closed loop closed traverse

→ starting and ending point same.

→ Mathematically & Geometrically closed

Mathematically,

Observed sum of external angles $> \angle A + \angle B + \angle C + \angle D + \angle E + \angle F$

Theoretical sum = $(2n \pm 4) \times 90^\circ$

+ → exterior

- → interior

error (e) = Observed sum - Theoretical sum

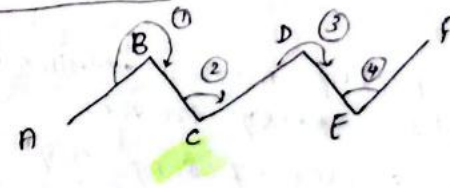
Permissible = $\frac{L \cdot C \cdot \sqrt{N}}{\text{least count}}$ or $\frac{C \cdot \sqrt{N}}{\text{constant}}$

N = Number of station.

If error within permissible, distribute error equally but not in fraction.

(ii) Open loop closed traverse

→ starting and ending point not same but starting and ending point bearing known.



→ Eg: coordinate of A & F known.

currently, open loop closed traverse } Bearing of AB and EF are known (First line & last line bearing known)

Mathematically closed but geometrically open

Angular error (e) = $\left(\begin{matrix} \text{calculated} \\ \text{bearing} \end{matrix} \right) - \left(\begin{matrix} \text{Standard or} \\ \text{Given bearing} \end{matrix} \right)$

→ permissible = $C\sqrt{N}$

N = No of affected angle to compute bearing.

For given figure: $N=4$

If, error lies within permissible then, distribute error by

• For first bearing = $\frac{\text{coefficient}}{N} \times \text{error}(e)$

coefficient = 1, 2, 3, 4, ... N

Thus,
for first bearing = $\frac{1}{N} \times e$

• For second bearing = $\frac{2}{N} \times e$

• For third bearing = $\frac{3}{N} \times e$

• For 4th bearing = $\frac{4}{N} \times e$

For given figure,

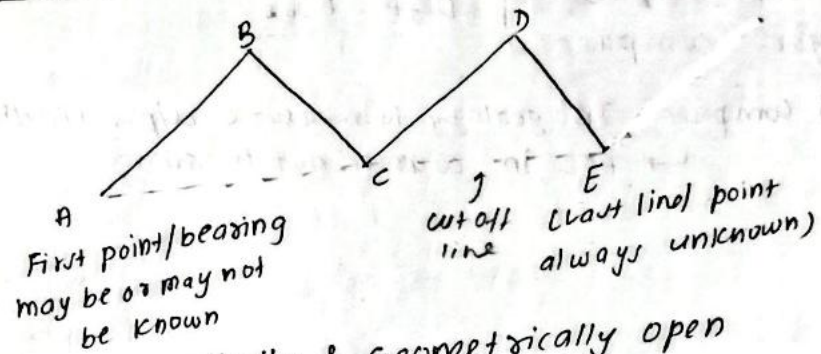
For 4th bearing = $\frac{4}{4} \times e$

Uses:

(1) To pickup internal details in side closed loop closed traverse

(3) For survey of long strip layout (eg: Road survey)

② Open traverse :-



↳ Mathematically & Geometrically open

↳ No direct check can apply.

↳ If visible, cut-off lines are used to check.

Viva ★ According to Nature of work

- ① Major traverse (CLCT)
- ② Minor traverse/Link traverse (OLCT)

angular error (e) = $\left(\begin{matrix} \text{calculated} \\ \text{bearing} \end{matrix} \right) - \text{standard bearing}$

↳ Permissible (P) = $C\sqrt{N}$

↳ N = Number of affected angle to compute bearing

For given figure,

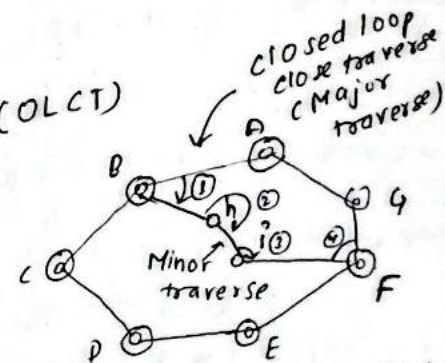
$N=4$

$N = n + 2 = 4$

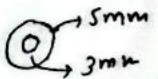
↳ n = number of minor station.

$N = n' + 1 = 4$

↳ n' = number of traverse leg
length of minor traverse



Major → Double circle



Inside → 0

* types of compass

- ① Prismatic compass
 - ② Surveyor's compass
- } Diff Book

Baunton compass: In geology to measure dip & dip dir.
↳ not in course (not in survey)

Temporary adjustment of prismatic compass

- ① Fixing the compass with tripod stand
 - ② Centering
 - ③ Levelling
 - ④ Adjustment of prism
- } Temporary Adjustment

If said field procedure +

Method of elimination of local attraction:

① By included angles.

- compute included angles from bearing
- start from unaffected line to calculate correct bearing

• If non line have difference 180°

$$FB \pm BB \neq 180^\circ$$

In this case every line affected by local attraction.

Steps:

- calculate interior (exterior) angle & check their sum.

$$\text{error (e)} = \text{Observed sum} - \frac{\text{Theoretical sum}}{(2n \pm 4) \times 90^\circ}$$

Permissible = check.

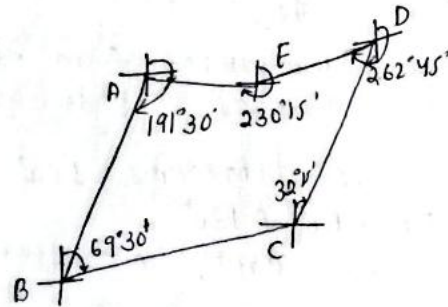
- If error within permissible distribute error equally but not in fraction.
- locate least difference to 180° in WCB & 0° in QB.
- calculate bearing distributing half error to each bearing on that line.
- Other process same as previous.

Q. Find which station are affected by attraction & correct bearings.

Line	F.B.	B.B.
AB	191° 30'	130°
BC	69° 30'	246° 30'
CD	32° 15'	210° 30'
DE	262° 45'	80° 45'
EP	230° 15'	53° 00'

→ Soln:

Rough sketch:



Calculation of interior angle

$$\begin{aligned}\angle A &= \text{FB of AB} - \text{BB of AB} \\ &= 191^\circ 30' - 53^\circ 00' \\ &= 138^\circ 30'\end{aligned}$$

Similarly,

$$\begin{aligned}\angle B &= \text{FB of BC} - \text{BB of BA} \\ &= 69^\circ 30' - 13^\circ 0' \\ &= 56^\circ 30'\end{aligned}$$

$$\begin{aligned}\angle C &= \text{FB of CD} - \text{BB of BC} \\ &= 32^\circ 15' - 246^\circ 30' \\ &= (360^\circ - 214^\circ 15') \\ &= 145^\circ 45'\end{aligned}$$

$$\begin{aligned}\angle D &= 262^\circ 45' - 210^\circ 30' \\ &= 52^\circ 15'\end{aligned}$$

$$\begin{aligned}\angle E &= 230^\circ 15' - 80^\circ 45' \\ &= 149^\circ 30'\end{aligned}$$

$$\begin{aligned}\text{Observed sum} &= 138^\circ 30' + 56^\circ 30' + 145^\circ 45' + 52^\circ 15' + 149^\circ 30' \\ &= 542^\circ 30'\end{aligned}$$

$$\begin{aligned}\text{Theoretical sum} &= (2n - 4) \times 90^\circ \\ &= (2 \times 5 - 4) \times 90^\circ \\ &= 540^\circ\end{aligned}$$

Angular errors

$$\begin{aligned}e &= \text{Obs. sum} - \text{theoretical sum} \\ &= 542^\circ 30' - 540^\circ \\ &= 2^\circ 30'\end{aligned}$$

$$\begin{aligned}\text{Correction for each angle} &= \frac{-2^\circ 30'}{5} \\ &= -30'\end{aligned}$$

} permissible = ~~15' 45'~~
= 15' 45'

corrected angle.

$$\angle A = 138^\circ 30' - 30' = 138^\circ$$

$$\angle B = 56^\circ 30' - 30' = 56^\circ$$

$$\angle C = 145^\circ 45' - 30' = 145^\circ 15'$$

$$\angle D = 52^\circ 15' - 30' = 51^\circ 45'$$

$$\angle E = 149^\circ 30' - 30' = 149^\circ$$

$$\text{sum} = 540^\circ$$

Comparison : diff

$178^{\circ} 30' (-1^{\circ} 30') \rightarrow \text{least}$
 $177^{\circ} 00'$
 $178^{\circ} 15'$
 $182^{\circ} 00'$
 $177^{\circ} 15'$

AB line differs from 180° by least

Thus,

$$\text{Correct bearing of AB} = 191^{\circ} 30' + \frac{1^{\circ} 30'}{2} = 192^{\circ} 15'$$

$$\text{Correct bearing of AB} = 192^{\circ} 15'$$

$$\begin{aligned} \text{(corrected) } \angle B &= \frac{+56^{\circ}}{248^{\circ} 15'} \\ &\quad - 180^{\circ} \end{aligned}$$

$$\begin{aligned} \text{Correct } \angle B &= 68^{\circ} 15' \\ + \angle C &= 145^{\circ} 15' \\ \hline &213^{\circ} 30' \\ &\quad - 180^{\circ} \end{aligned}$$

$$\begin{aligned} \text{FB of CD} &= 33^{\circ} 30' \\ + \angle D &= 51^{\circ} 45' \\ \hline &85^{\circ} 15' \\ &\quad + 180^{\circ} \end{aligned}$$

$$\begin{aligned} \text{FB of DE} &= 265^{\circ} 15' \\ + \angle E &= 149^{\circ} 00' \\ \hline &414^{\circ} 15' \\ &\quad - 180^{\circ} \end{aligned}$$

$$\begin{aligned} \text{FB of EA} &= 234^{\circ} 15' \\ + \angle A &= 136^{\circ} \\ \hline &370^{\circ} 15' \end{aligned}$$

$$\text{FB of AB} = \frac{-180^{\circ}}{192^{\circ} 15'} \text{ (checked)}$$

Q.7 The following with compass.

FB & BB were observed in traversing

Line	FB	BB
PQ	S $37^{\circ} 30'$ E	N $37^{\circ} 30'$ W
QR	S $43^{\circ} 15'$ W	N $44^{\circ} 15'$ E
RS	N $73^{\circ} 00'$ W	S $72^{\circ} 15'$ E
ST	N $12^{\circ} 45'$ E	S $13^{\circ} 15'$ W
TP	N $60^{\circ} 00'$ E	S $59^{\circ} 00'$ W

Calculate internal angles and correct them.

Internal angles

$$\begin{aligned} \angle P &= 59^{\circ} 00' + 37^{\circ} 30' \\ &= 96^{\circ} 30' \end{aligned}$$

$$\begin{aligned} \angle Q &= 180^{\circ} - 37^{\circ} 30' - 43^{\circ} 15' \\ &= 99^{\circ} 15' \end{aligned}$$

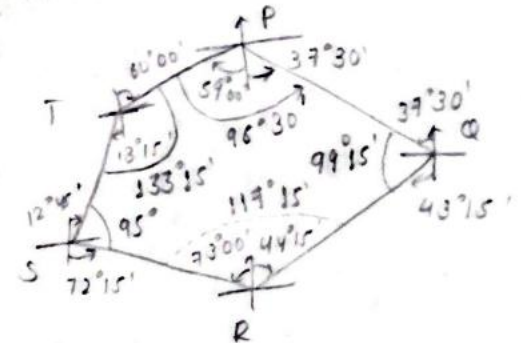
$$\begin{aligned} \angle R &= 73^{\circ} + 44^{\circ} 15' \\ &= 117^{\circ} 15' \end{aligned}$$

$$\begin{aligned} \angle S &= 180^{\circ} - 12^{\circ} 45' - 72^{\circ} 15' \\ &= 95^{\circ} \end{aligned}$$

$$\angle T = 180^{\circ} - 60^{\circ} + 13^{\circ} 15' = 133^{\circ} 15'$$

$$\begin{aligned} \text{Observed sum} &= \angle P + \angle Q + \angle R + \angle S + \angle T \\ &= 541^{\circ} 15' \end{aligned}$$

$$\begin{aligned} \text{Theoretical sum} &= (5n - 4) \times 90^{\circ} \\ &= 540^{\circ} \end{aligned}$$



Angular error = Obs. sum - Theoretical sum
 $= 541^{\circ}15' - 540^{\circ}$
 $= 1^{\circ}15' (+ve)$ [correction] $\rightarrow -ve$

Permissible = $30' \sqrt{n} = 30' \sqrt{5} = 1^{\circ}4'$

correction = $\frac{-1^{\circ}15'}{5} = -15'$

Thus,

corrected internal angles

$\angle P = 96^{\circ}30' - 15' = 96^{\circ}15'$

$\angle Q = 99^{\circ}15' - 15' = 99^{\circ}$

$\angle R = 117^{\circ}15' - 15' = 117^{\circ}$

$\angle S = 95^{\circ} - 15' = 94^{\circ}45'$

$\angle T = 133^{\circ}15' - 15' = 133^{\circ}$

sum = 540°

Q. What is the WCB & QB of a line joining points A (1023.25m E, 2134.3m N) to point C (1141.71m E, 2012.89m N)?

$\rightarrow$ Solⁿ:

$\theta_{AC} = \tan^{-1} \left(\frac{E_C - E_A}{N_C - N_A} \right)$

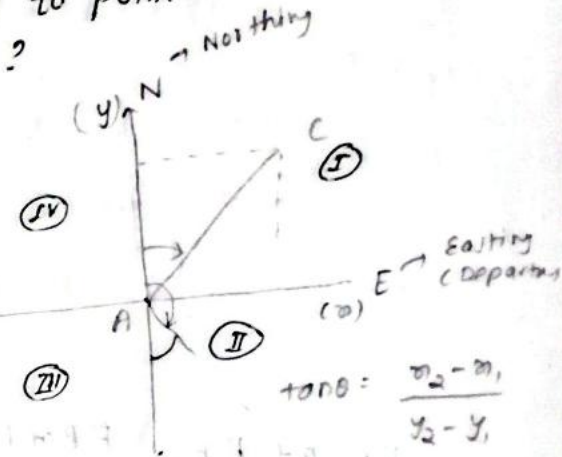
$= \tan^{-1} \left(\frac{1141.71 - 1023.25}{2012.89 - 2134.3} \right)$

$= -44^{\circ}17'43''$ (true)
-ve

So,

QB = $544^{\circ}17'43''$ E

WCB = $180^{\circ} - 44^{\circ}17'43''$
 $= 135^{\circ}43'19''$



Nothing $\rightarrow$ Latitude
 Easting $\rightarrow$ Departure

(Q) Compute the angles misclosure and adjusted bearing in the following link traverse PAB CQ.

Bearing of line PA = $30^{\circ}15'00''$

Bearing of line QC = $226^{\circ}0'0''$

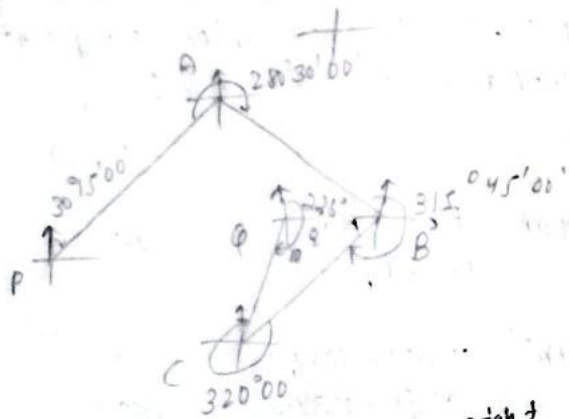
Angle to right,

$\angle A = 280^{\circ}30'00''$

$\angle B = 315^{\circ}45'00''$

$\angle C = 320^{\circ}00'00''$

Left count of instrument = 30° .



$$\begin{aligned} \text{FB of AB} &= \text{FB of PA} + \text{Angle to the right} \\ &\quad \pm 180^\circ \text{ or } -540^\circ \\ &= (30^\circ 15' 00') + (280^\circ 30' 00'') \\ &\quad - 180 \end{aligned}$$

$$\begin{aligned} &= \frac{130^\circ 45' 00'}{446^\circ 30'} \\ &\quad - 180^\circ \end{aligned}$$

$$\text{FB of BC} = \frac{266^\circ 30' 00''}{586^\circ 30' 0''}$$

$$\begin{aligned} \angle C &= +320^\circ \\ &\quad - 540^\circ \end{aligned}$$

$$\text{FB of CQ} = 46^\circ 30' 0''$$

$$\begin{aligned} \text{Angular error} &= \text{Calculated bearing of AC} \\ &\quad - \text{Standard bearing of AC} \\ &= (46^\circ 30' + 180^\circ) - (226^\circ 0' 0'') \\ &= 80' \quad (\text{correction} \rightarrow -ve) \\ \text{Permissible} &= 30' \sqrt{N} = 30' \sqrt{3} = 51' 57.09'' \\ &\quad (> 80'). \end{aligned}$$

✓ Method I: Direct bearing correction
Here, $N = 3$

$$\text{correction in bearing} = \frac{\text{coeff.}}{N} \times \text{Total error (e)}$$

$$\text{correction for bearing AB} = \frac{1}{3} \times 30' = 10'$$

$$\text{correction for bearing BC} = \frac{2}{3} \times 30' = 20'$$

$$\text{correction for bearing CA} = \frac{3}{3} \times 30' = 30'$$

$$\begin{aligned} \text{Corrected bearing of AB} &= 130^\circ 45' - 10' \\ &= 130^\circ 35' 0'' \end{aligned}$$

$$\begin{aligned} \text{Corrected bearing of BC} &= 266^\circ 30' - 20' \\ &= 266^\circ 10' \end{aligned}$$

$$\begin{aligned} \text{corrected bearing of CA} &= 46^\circ 30' - 30' \\ &= 46^\circ 0' \end{aligned}$$

Method II: By angle correction:

$N = 3$

Correction for first angle = $\frac{1}{3} \times 90' = 30'$

" " second angle = $\frac{2}{3} \times 30' = 20'$

" " 3rd angle = $\frac{2}{3} \times 30' = 30'$

Correction for first angle = $280^\circ 30' - 10' = 280^\circ 20'$

" " 2nd angle = $315^\circ 45' - 20' = 315^\circ 25'$

" " 3rd angle = $320^\circ 0' - 30' = 319^\circ 30'$

correct FB of AB = $280^\circ 20'$

Correct FB of AB = $(30^\circ 15' + 280^\circ 20') - 180^\circ = 130^\circ 35'$

Correct $\angle B = +315^\circ 25'$
 $446^\circ 0' 0''$
 $- 180^\circ$

Correct FB of BC = $226^\circ 10' 0''$

$\angle C = 319^\circ 30'$
 $585^\circ 30'$
 $- 540^\circ$
 $45^\circ 30'$

Correct FB of CD = $45^\circ 30'$

Corrected Bearing of

FB of AB = $30^\circ 15' + 280^\circ 20' - 180^\circ = 130^\circ 35'$
 ↑
 incorrect correct angle

FB of BC = $130^\circ 45' + 315^\circ 25' - 180^\circ = 226^\circ 10'$
 ↑
 incorrect

FB of CD = $226^\circ 30' + 319^\circ 30' - 540^\circ = 46^\circ 00'$

#Q. The FB & BB of the lines of a closed compass traverse are as follows:

Line	FB	BB
AB	$32^\circ 36'$	$214^\circ 30'$
BC	$124^\circ 30'$	$303^\circ 15'$
CD	$181^\circ 00'$	$1^\circ 00'$
DA	$289^\circ 30'$	$108^\circ 45'$

Correct the bearings for local attraction and determine the true bearings of lines if magnetic declination at the place is $3^\circ 30' W$.

→ Solⁿ

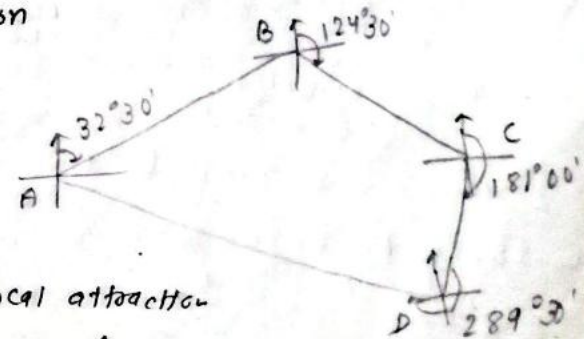
True Bearing = Magnetic bearing $\pm$ Declination.

From the above, observation line CD differs 180° .

So, C and D are free from local attraction.

(D & C pt. etc)

सिर्फे reading local attraction free है।



Observed bearing of CD = $181^{\circ}00'$
Declination = $-3^{\circ}30'$

[or] correct Actual Bearing of CD = $177^{\circ}30'$
 $+ 180^{\circ}$

correct bearing of DC = $357^{\circ}30'$

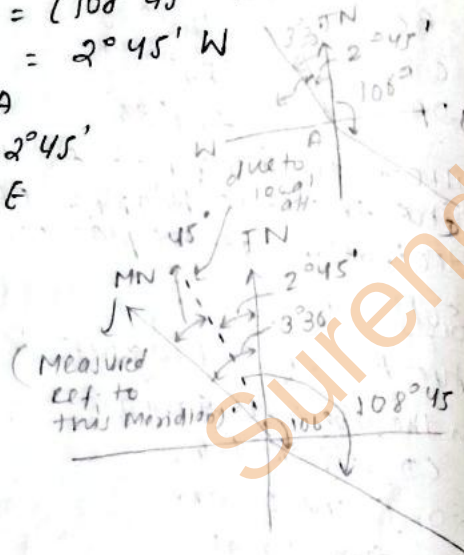
Observed bearing of DA = $289^{\circ}30'$
Declination = $-3^{\circ}30'$

correct bearing of DA = $286^{\circ}00'$
 $- 180^{\circ}$

correct bearing of AD = $106^{\circ}00'$

But, It difference from $108^{\circ}45'$ by
 $= (108^{\circ}45' - 106^{\circ}00')$
 $= 2^{\circ}45' W$

$\therefore$ Local attraction of A
 $= 3^{\circ}30' - 2^{\circ}45'$
 $= 0^{\circ}45' E$



Correct Bearing of AB = $32^{\circ}30' - 2^{\circ}45'$
 $= 29^{\circ}45'$

$+ 180^{\circ}$ (For finding Bt of AB)

correct bearing of BA = $209^{\circ}45'$

Observed bearing of BA = $214^{\circ}30'$ { $\frac{y}{x}$ towards west attracted reading }

Correction at B = $214^{\circ}30' - 209^{\circ}45'$
 $= 4^{\circ}45' W$

Local attraction of at B = $4^{\circ}45' - 3^{\circ}30'$
 $= 1^{\circ}15' W$

Correct bearing of BC = $124^{\circ}30' - 4^{\circ}45'$
 $= 119^{\circ}45'$

$+ 180^{\circ}$

correct bearing of CB = $299^{\circ}45'$

Observed bearing of CB = $303^{\circ}15'$

Correction at C = $303^{\circ}15' - 299^{\circ}45'$
 $= 3^{\circ}30'$

Local attraction at C = $3^{\circ}30' - 3^{\circ}30'$
 $= 0$ (No local attraction)

Since, point C is local attraction free.

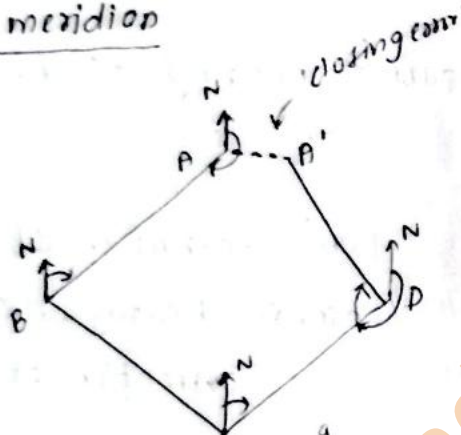
Error due to only declination

$\therefore$ correct bearing of CB = $303^{\circ}15' - 3^{\circ}30'$
 $= 299^{\circ}45'$ ✓
(check)

Methods of plotting of compass traverse:

- (1) Plotting by Parallel Meridian
- (2) Included Angle
- (3) Rectangular co-ordinates
- (4) Chords
- (5) Paper protector
- (6) Tangents

(1) Plotting by Parallel meridian



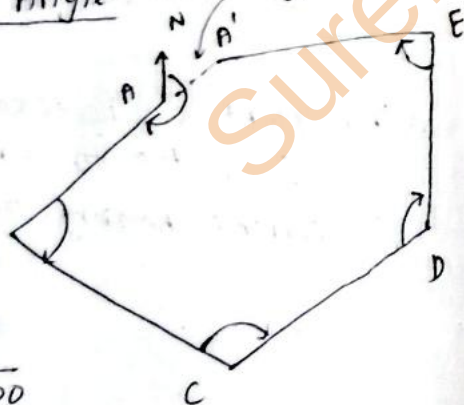
(2) Plotting by Included Angle

$$\text{Relative precision} = \frac{1}{\frac{P}{e}}$$

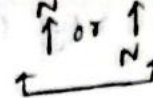
P → Perimeter
(A B C D E A')

e → closing error

↳ Should be better than
 $\frac{1}{600}$ or $\frac{1}{800}$ or $\frac{1}{1000}$



plotting map Plot → A1
1:200



Legends:

Sr	Description	Symbol
1.	Traverse	

1. Traverse station

2. Traverse leg

3. Buildings

4. Road

5. Footpath

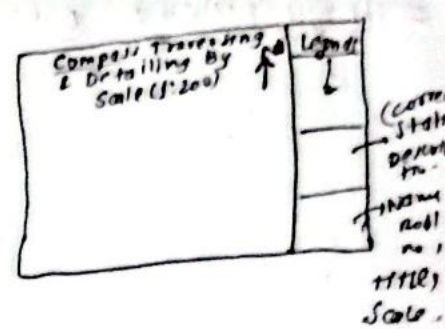
6. Chautari

7. Electric pole

8. Container

→ Trees
→ Grass, shrubs

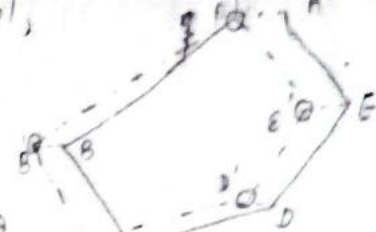
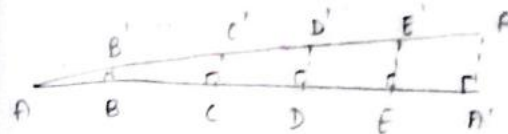
Ad



Adjustment of closing error:



If closing error horizontal,



Errors in Compass surveying :-

① Instrumental errors:

- ↳ compass out of level.
- ↳ Needle not straight
- ↳ Movement of needle sluggish.
- ↳ Magnetism of needle weak.
- ↳ plane of sight not vertical.
- ↳ line of sight not passing through centre of graduated ring.

② Personal errors

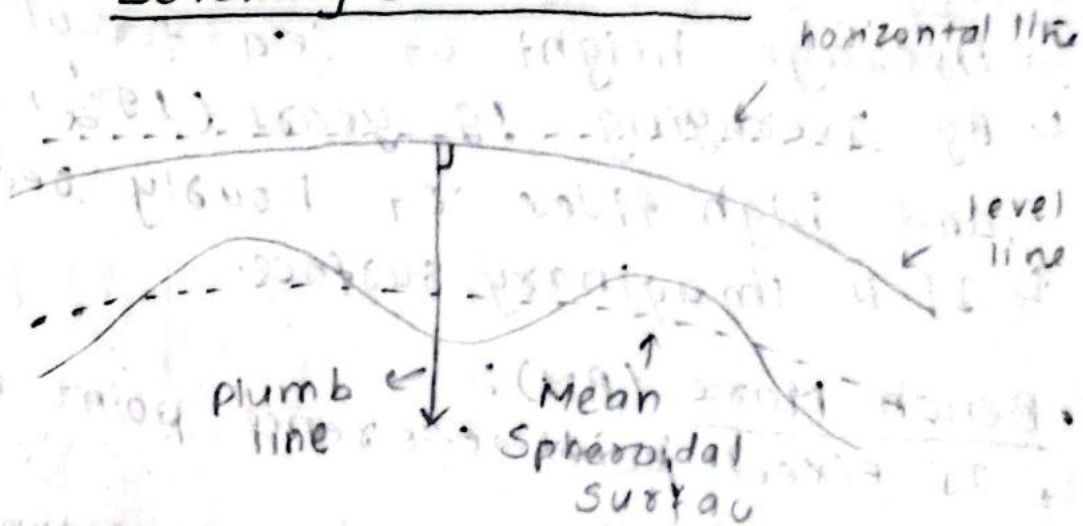
- ↳ Compass not properly levelled.
- ↳ compass not properly centred over the station
- ↳ Ranging Rod or signal not accurately bisected.
- ↳ Incorrect reading and recording of the graduated ring.

③ Natural errors :-

- ↳ Variation in declination.
- ↳ local attraction.
- ↳ Magnetic changes in atmosphere due to clouds and storms.
- ↳ Irregular variation.

Chapter-5:

Levelling (18-24 marks)



→ Art of determining the relative height or elevation of a point or object on the earth surface.

→ Level deals in vertical plane.

Terms (Definitions):

• Level surface:

↳ Any surface parallel to mean spheroidal surface of the earth.

↳ Such surface is curved.

↳ Mean Sea Level (MSL) is assumed as spheroidal.

↳ Eg: Still water in lake considered level surface.

• Level line:

→ Any line on the level surface

↳ This is assumed normal to plumb line.

↳ This is assumed normal to plumb line.

(direction of gravity) at all points.

• Datum:

↳ Is any imaginary surface to which elevation are referred.

eg: Mean sea level.

• Mean Sea level

- Average height of sea for all stages of tides.
- ↳ By averaging 19 years ($19\frac{1}{2}$, 20) low tides and high tides of hourly recorded data.
- ↳ It is imaginary surface.

• Bench Mark (BM):-

- ↳ Is fixed point reference point of known elevation.

Types: (PBM)

- ① Permanent BM → reference to datum; Survey Department (higher order of survey)
 - ② Temporary BM (TBM) → reference to PBM; self requirement transferred from PBM
 - ③ Site BM
 - ④ Arbitrary BM → SBM: on the vicinity of workplace for working purpose from TBM
- ↓
(ABM)
↳ Assumed

• Vertical line:- → plumb line (dir. of gravity)

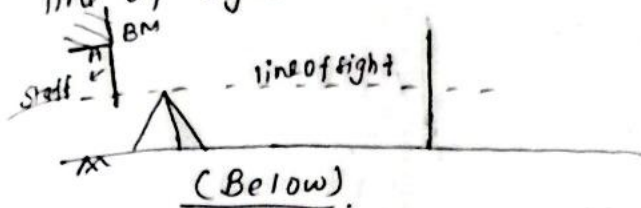
• Horizontal line:- → Perpendicular to plumb line

• Reduced level (RL):-

- ↳ Height or depth of a point above or below the assumed datum.

Principle of levelling:-

- Furnishing a horizontal line of sight and finding the vertical distances of the points above or below the line of sight.



- Line of sight produced by level machine.
- Vertical distance measured by staff.

Uses of levelling:-

- Preparation of topographic map of an area.
- Design of highways, railways, canal, sewer, etc. (All Engineering Projects)
- Locating the gradient lines for drainage characteristics of area.
- Lay out of construction projects.
- To calculate volume of earthwork, reservoir, etc.
- To determine attitudes of different important points.

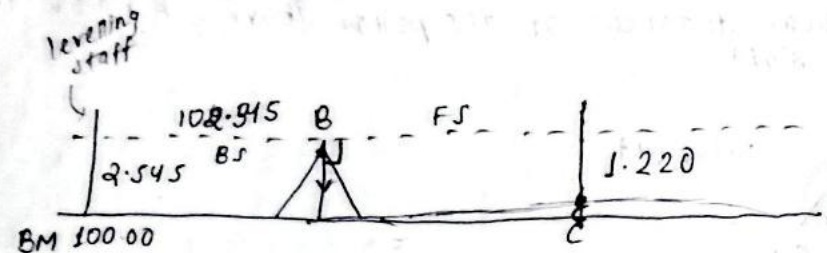
Methods of Levelling:-

1. Spirit levelling (Direct levelling)
2. Trigonometric levelling (Indirect levelling)
3. Barometric levelling

1. Spirit Levelling (Direct levelling):-

- ↳ Vertical distances with respect to horizontal line.
- ↳ Level Machine & staff used.
- ↳ Line of sight fixed.
- ↳ Relative different in elevation between two adjacent points.

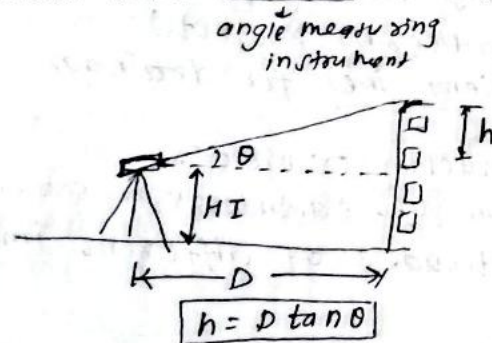
→ More precise method used by surveyors.



(2) Trigonometric levelling

- ↳ Inaccessible BM (where direct levelling can't be used)
- ↳ Elevation of points are computed from vertical & horizontal distances.

↳ Instrument used: Theodolite & staff



- ↳ Less accurate than direct levelling.
- ↳ Line of sight movable.

(3) Barometric levelling

- ↳ Used by travellers
- ↳ Gives less accurate

□ Terms used in levelling:-

1. Instrument Station:

- ↳ The point where instrument is set.

2. Station:-

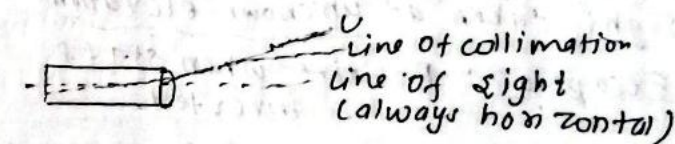
- ↳ Point where the level staff is put taking observation.

3. Height of instrument (H.I.)

- ↳ Elevation of plane of sight w.r.t. assumed datum.

4. Line of collimation

- ↳ Is an imaginary line joining the optical centre of objective with the intersection of cross hair and its continuation.
- ↳ Line through optical centre of objective traversing eye piece & entering eye is known as line of sight.



5. Back sight (BS)

- ↳ BM में निम्ने reading
- ↳ known point निम्ने reading
- ↳ staff upright $\rightarrow$ +ve sight
- ↳ staff inverted $\rightarrow$ -ve sight

- $\rightarrow$ First reading taken at a station at a station of known elevation e.g. BM, turning point or change point.
- $\rightarrow$ It is plus sight
- $\rightarrow$ Gives ht. of instrument
- $\rightarrow$ $H.I. = R.L. \text{ of known elevation} + BS$
- Exception:- It is minus sight when staff inverted.

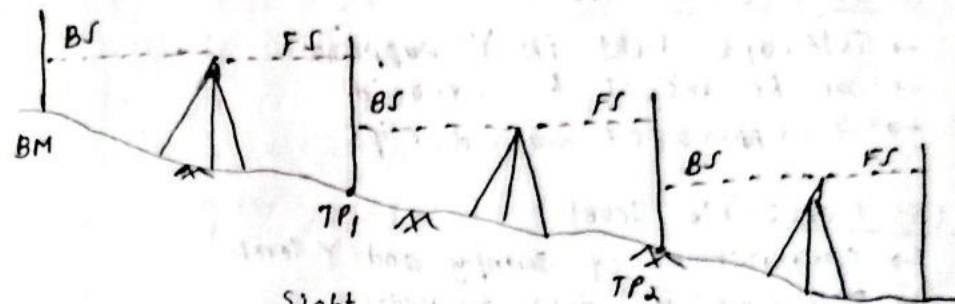
6. Fore sight (FS)

- $\rightarrow$ Last reading taken before shifting on instrument to a new instrument station.
- $\rightarrow$ It is also called minus sight
- $\rightarrow$ R.L. of ground point = $H.I. - FS$
- $\rightarrow$ Sight taken at unknown elevation

Exception: Positive when staff inverted.

7. Turning Point or Change point (TP or CP)

- $\rightarrow$ Change point connects one setup of instrument reading to next setup.
- $\rightarrow$ Both BS & FS are taken on this point.
- $\rightarrow$ Temporary station created by turning plate



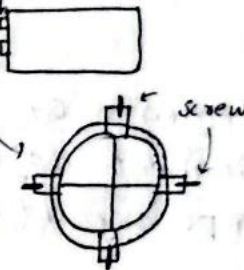
8. Intermediate point (IS)

- $\rightarrow$ Reading taken between first reading (BS) and last reading before shifting the instrument to new station
- $\rightarrow$ Sight taken at unknown elevation.

Types of levels:

- (1) Dumpy Level (Telescope fixed/hinged)
- (2) Wye (Y) level (Y-type support, can be reversed)
- (3) Reversible level (can work as both Y & Dumpy level combined)
- (4) Tilting level (can move line of sight vertically as well, quick reading)
- (5) Automatic level (Auto compensator, automates the level to 90° with plumb line)

Diaphragm



(1) Dumpy Level

- $\rightarrow$ Designed by Gravatt
- $\rightarrow$ Telescope rigidly fixed in supports.
- $\rightarrow$ Bubble tube attached in telescope

(2) Wye (Y) level:

- Telescope held in Y supports
- Can be moved & reversed.
- 4 support of curved clips

(3) Reversible level

- combination of dumpy and Y level.
- Telescope in two sockets

(4) Tilting levelling

- Line of sight can be tilted without tilting the vertical axis
- Quickly levelling instrument, rough level can be used.

(5) Automatic level

- When instrument levelled axis are auto adjusted.

Adjustments of level instrument:-

- ① Permanent adjustment
- ② Temporary adjustment

① Permanent Adjustment:

- Establishment of desired relationship between fundamental lines of levelling instrument.
- Rectification of errors in instrument.

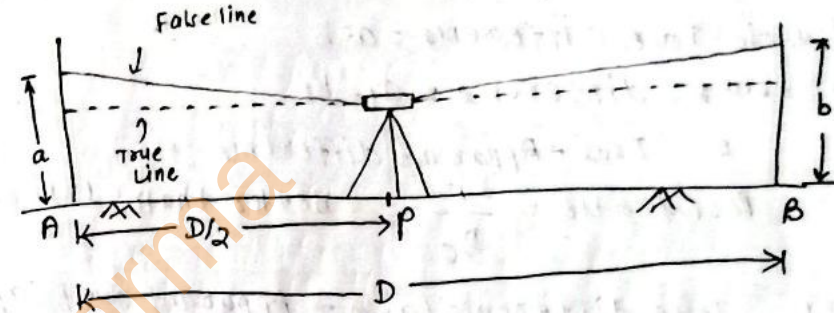
Fundamental lines & its relationship

- (i) Line of collimation should be parallel to axis of bubble.
- (ii) Line of collimation should coincide with axis of telescope (line of sight)

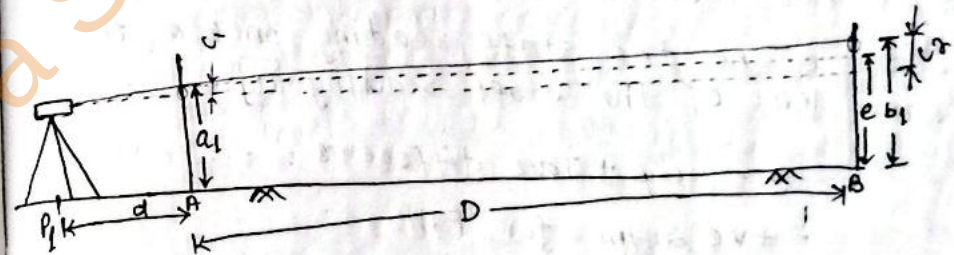
(iii) Axis of bubble should be perpendicular to vertical axis.

Numerical Two peg test:

- To make the line of collimation parallel to the axis of bubble.
- Fairly level ground (40-60m)



• only mid reading



$$d = \frac{D}{10}$$

• If considered no error at nearest staff (mid --- line)

Procedure:-

- ① Pegs A & B are placed at known distance D in a fairly level ground.
- ② Instrument at P (midway) $\rightarrow$ readings are a & b
- ③ Shift instrument at P_1 (nearer to one peg) & again readings are a_1 & b_1 .

④ Calculate True difference $= a - b$

⑤ Apparent difference $= a_1 - b_1$

⑥ $e = \text{True} - \text{Apparent difference}$

Permissible $= \frac{1}{D/e} \pm$ Better than 1:10,000

- ⑦ If True difference $(a - b) = \text{Apparent diff } (a_1 - b_1)$
 $\rightarrow$ NO collimation error.

If not there is ~~the~~ collimation error.

- ⑦ Let 'e' be the staff reading on B at the same level of the staff reading a_1 .

$e = a_1 \pm \text{True difference}$

• +ve sign for Fall

• -ve sign for Rise

- ⑧ If $b_1 > e \rightarrow$ line of collimation is inclined upward.
 If $b_1 < e \rightarrow$ line of collimation is inclined downward.

- ⑨ By applying principle of similar triangles,
 correction on peg A $(C_1) = \frac{d}{D} (b_1 - e)$

correction on peg B $(C_2) = \frac{D-d}{D} (b_1 - e)$

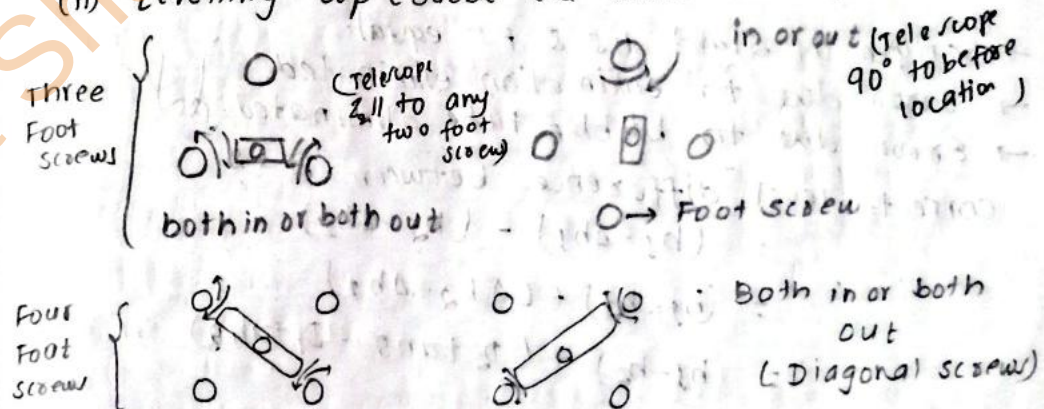
- ⑩ Correct staff reading on A $= a_1 \pm C_1$
 correct staff reading on B $= b_1 \pm C_2$
 +ve for inclined downward.
 -ve for inclined upward.
- ⑪ Then, the cross hair is brought to the calculated correct staff reading by raising or lowering the diaphragm by means of the diaphragm screw.

{ Level मा Centering सुद्धेन }

② Temporary Adjustment

- (i) Setting up (Tripod मा instrument set गर्ने)

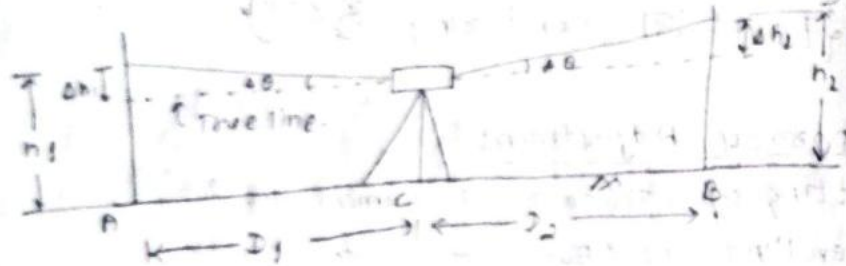
- (ii) Levelling up (Bubble लाई centre मा ल्याउने)



⑤ Elimination of Parallax

- (a) Focusing the eye piece
- (b) Focusing the objectives

Balancing Back sight & Fore sight: *mediate point*
(Balancing sight)



- Length of sight BS & FS equal.
- Error due to collimation eliminated.
- Error due to bubble tube eliminated.

Correct level difference between A & B

$$= (h_1 - \Delta h_1) - (h_2 - \Delta h_2)$$

$$= (h_1 - h_2) + (\Delta h_2 - \Delta h_1)$$

$$= (h_1 - h_2) + (D_2 \tan \theta - D_1 \tan \theta)$$

If $D_1 = D_2$

$$\text{Correct level diff b/w A \& B} = h_1 - h_2$$

$$= \text{BS at A} - \text{FS at B}$$

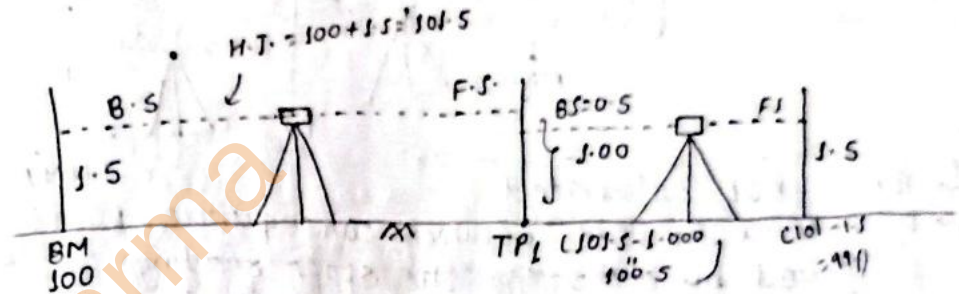
- If sight balanced, no correction required, direct staff reading can be used.

⑥ Method of calculation of Reduced level:

- ① The collimation system or HI system.
- ② The rise and fall method.

① Height of instrument method (HI method)

$$H.I. = \text{Elevation of BM} + B.S.$$



$$\text{Elevation} = H.I. - FS \text{ or } IS$$

→ For next instrument setup,

→ Calculate new HI for next setup.

$$H.I. = RL \text{ at } TP_1 + B.S. \text{ at } TP_1$$

$$(H.I. = 100.50 + 0.50 = 101.00)$$

$$\text{Elevation} = H.I. - FS \text{ at } TP_2$$

$$= 101.00 - 1.50$$

$$= 99.5$$

→ Process continue till last RL

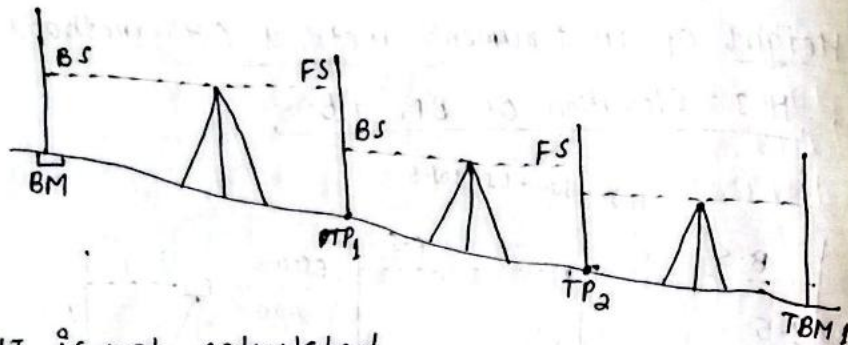
→ For IS, RL is obtained by subtracting its respective staff reading from H.I. for same instrument setup.

→ Arithmetic check,

$$\Sigma BS - \Sigma FS = \text{Last RL} - \text{First RL}$$

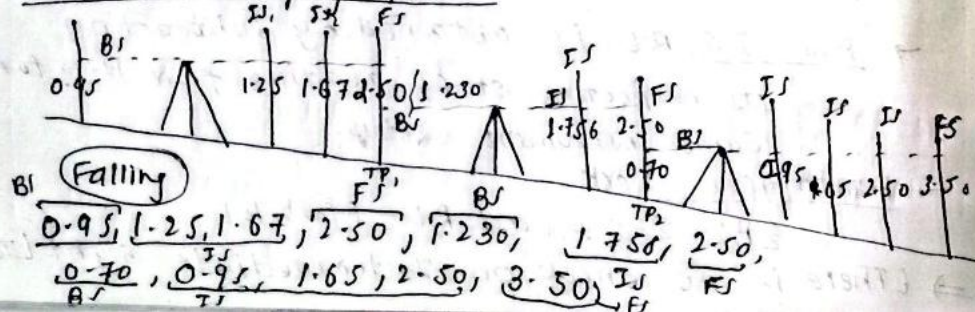
⇒ (There is no check on Intermediate sight (IS).)

2. Rise and Fall Method :-



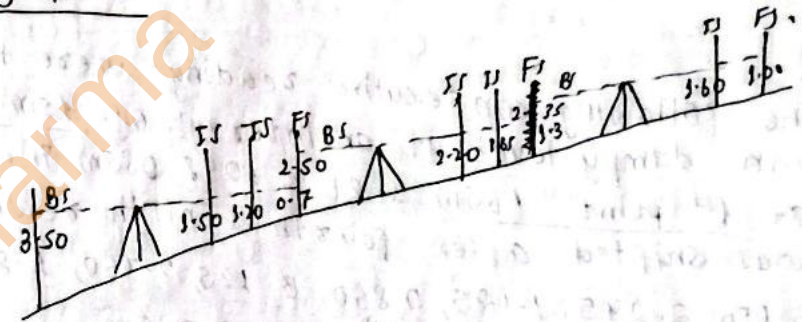
- HI is not calculated
- Difference of level between consecutive point is found by comparing the staff readings on two points for the same setting of instrument.
- Differences gives rise and fall.
- All the points gives vertical distances of each point above or below the preceding one.
- If above rise (+ve),
If below fall (-ve)
- Arithmetic check: $\sum BS - \sum FS = \sum Rise - \sum Fall$
 $= Last RL - First RL$
- Complete check on IS.

Continuous sloping Ground (either rise or fall)



- ✓ Falling Ground If staff reading must increase continuously for same instrument setup.
If any decrease/break in data, it must be BS for next instrument setup.
- ✓ Rising Ground If staff reading must decrease continuously for same instrument setup.
Break/increase indicates next instrument setup's BS.

Rising Ground



$$\underbrace{3.50}_{BS}, \underbrace{1.50}_{IS}, \underbrace{1.20}_{IS}, \underbrace{0.7}_{FS}, \underbrace{2.50}_{BS}, \underbrace{2.20}_{FS}, \underbrace{1.65}_{FS}, \underbrace{1.3}_{FS}, \underbrace{2.235}_{BS}, \underbrace{1.60}_{IS}$$

$$\therefore \text{No. of } BS = FS$$

[illegible]

calculate $\frac{q_m}{RL}$ of all point
method (M.I. = RL of BM + BS)

Calculate RL of HI Method (HI = RL of BM + BS)							
Station point	Chainage	BS	IS	FS	HI	RL	Remarks
					1101.235	1098.085	Starting pt
1	165	3.150			1099.94	1098.990	
2	180		2.245			1100.110	
3	195		1.125				
4	210	3.125		0.860	1103.580	1100.375	CP1
5	225		2.760			1100.448	
6	240		1.835			1101.965	
7	255		1.470			1102.830	
8	270	1.225		1.965	1102.760	1101.625	CP2
9	285		2.390			1100.800	
10	300			3.035		1099.725	End pt

$$\text{Elevation} = HI - (IS \text{ or } FS)$$

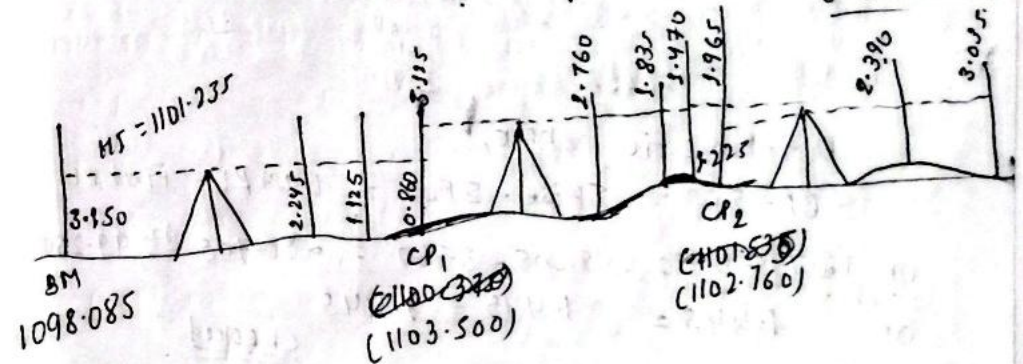
single instrument (अपनी ओर जोड़कर)

Station Point	Chainage	BS	IS	FS	Rise (+)	Fall (-)	RL	Remark
							1098.085	Fixed Pt.
1	165	3.150					1098.990	
2	180		2.245		0.905		1100.110	
3	195		3.125		1.120		1100.110	
4	210	3.125		0.860	0.265		1100.375	CP1
5	225		2.760		0.365		1100.740	
6	240		1.835		0.425		1100.665	
7	255		1.470		0.365		1102.030	
8	270	1.225		1.965		0.495	1101.535	CP2
9	285		2.390			1.165	1100.370	
10	300			3.035		0.645	1099.725	Last Point
		7.500		5.860	3.945	2.305		

90,

$$1.640 = 1.640 \text{ checked}$$

(Ari Next)
 3-4-5-6-7-8-9-10-11-12-13-14-15-16-17-18-19-20-21-22-23-24-25-26-27-28-29-30-31-32-33-34-35-36-37-38-39-40-41-42-43-44-45-46-47-48-49-50-51-52-53-54-55-56-57-58-59-60-61-62-63-64-65-66-67-68-69-70-71-72-73-74-75-76-77-78-79-80-81-82-83-84-85-86-87-88-89-90-91-92-93-94-95-96-97-98-99-100-101-102-103-104-105-106-107-108-109-110-111-112-113-114-115-116-117-118-119-120-121-122-123-124-125-126-127-128-129-130-131-132-133-134-135-136-137-138-139-140-141-142-143-144-145-146-147-148-149-150-151-152-153-154-155-156-157-158-159-160-161-162-163-164-165-166-167-168-169-170-171-172-173-174-175-176-177-178-179-180-181-182-183-184-185-186-187-188-189-190-191-192-193-194-195-196-197-198-199-200-201-202-203-204-205-206-207-208-209-210-211-212-213-214-215-216-217-218-219-220-221-222-223-224-225-226-227-228-229-230-231-232-233-234-235-236-237-238-239-240-241-242-243-244-245-246-247-248-249-250-251-252-253-254-255-256-257-258-259-260-261-262-263-264-265-266-267-268-269-270-271-272-273-274-275-276-277-278-279-280-281-282-283-284-285-286-287-288-289-290-291-292-293-294-295-296-297-298-299-300-301-302-303-304-305-306-307-308-309-310-311-312-313-314-315-316-317-318-319-320-321-322-323-324-325-326-327-328-329-330-331-332-333-334-335-336-337-338-339-340-341-342-343-344-345-346-347-348-349-350-351-352-353-354-355-356-357-358-359-360-361-362-363-364-365-366-367-368-369-370-371-372-373-374-375-376-377-378-379-380-381-382-383-384-385-386-387-388-389-390-391-392-393-394-395-396-397-398-399-400-401-402-403-404-405-406-407-408-409-410-411-412-413-414-415-416-417-418-419-420-421-422-423-424-425-426-427-428-429-430-431-432-433-434-435-436-437-438-439-440-441-442-443-444-445-446-447-448-449-450-451-452-453-454-455-456-457-458-459-460-461-462-463-464-465-466-467-468-469-470-471-472-473-474-475-476-477-478-479-480-481-482-483-484-485-486-487-488-489-490-491-492-493-494-495-496-497-498-499-500-501-502-503-504-505-506-507-508-509-510-511-512-513-514-515-516-517-518-519-520-521-522-523-524-525-526-527-528-529-530-531-532-533-534-535-536-537-538-539-540-541-542-543-544-545-546-547-548-549-550-551-552-553-554-555-556-557-558-559-560-561-562-563-564-565-566-567-568-569-570-571-572-573-574-575-576-577-578-579-580-581-582-583-584-585-586-587-588-589-590-591-592-593-594-595-596-597-598-599-600-601-602-603-604-605-606-607-608-609-610-611-612-613-614-615-616-617-618-619-620-621-622-623-624-625-626-627-628-629-630-631-632-633-634-635-636-637-638-639-640-641-642-643-644-645-646-647-648-649-650-651-652-653-654-655-656-657-658-659-660-661-662-663-664-665-666-667-668-669-670-671-672-673-674-675-676-677-678-679-680-681-682-683-684-685-686-687-688-689-690-691-692-693-694-695-696-697-698-699-700-701-702-703-704-705-706-707-708-709-710-711-712-713-714-715-716-717-718-719-720-721-722-723-724-725-726-727-728-729-730-731-732-733-734-735-736-737-738-739-740-741-742-743-744-745-746-747-748-749-750-751-752-753-754-755-756-757-758-759-760-761-762-763-764-765-766-767-768-769-770-771-772-773-774-775-776-777-778-779-780-781-782-783-784-785-786-787-788-789-790-791-792-793-794-795-796-797-798-799-800-801-802-803-804-805-806-807-808-809-810-811-812-813-814-815-816-817-818-819-820-821-822-823-824-825-826-827-828-829-830-831-832-833-834-835-836-837-838-839-840-841-842-843-844-845-846-847-848-849-850-851-852-853-854-855-856-857-858-859-860-861-862-863-864-865-866-867-868-869-870-871-872-873-874-875-876-877-878-879-880-881-882-883-884-885-886-887-888-889-890-891-892-893-894-895-896-897-898-899-900-901-902-903-904-905-906-907-908-909-910-911-912-913-914-915-916-917-918-919-920-921-922-923-924-925-926-927-928-929-930-931-932-933-934-935-936-937-938-939-940-941-942-943-944-945-946-947-948-949-950-951-952-953-954-955-956-957-958-959-960-961-962-963-964-965-966-967-968-969-970-971-972-973-974-975-976-977-978-979-980-981-982-983-984-985-986-987-988-989-990-991-992-993-994-995-996-997-998-999-1000-1001-1002-1003-1004-1005-1006-1007-1008-1009-1010-1011-1012-1013-1014-1015-1016-1017-1018-1019-1020-1021-1022-1023-1024-1025-1026-1027-1028-1029-1030-1031-1032-1033-1034-1035-1036-1037-1038-103



#

Arithmetic check (RL)

$$\sum BS - \sum FS = \sum Rise - \sum Fall = \text{Last RL} - \text{First RL}$$

$$7500 - 5860 = 3945 - 2305 = 1099.725 - 1098.085$$

$$1.640 = 1.640 = 1.640 \text{ checked}$$

5. The following is the page of level book where some readings are missing fill missing data & calculate RL.

S.NO.	BS	IS	FS	Rise(+)	Fall(-)	RL	Remarks
1	3.250					249.260	BM
2	1.755		4.200	0.750	0.750	247.510	CP
3		1.950			0.195	247.315	
4	3.940		1.920	0.030		248.345	CP
5		2.340		1.500		249.845	
6		1.240		1.000		250.845	
7	1.750		2.125		0.845	250.000	CP
8		1.575		0.275		250.275	
9		3.945			1.970	248.305	
10	2.800		1.895	1.650		249.955	CP
11			1.350	0.750		250.705	

$$\begin{array}{cccc} \sum BS & \sum FS & \sum Rise & \sum Fall \\ 12.795 & 11.350 & 5.205 & 3.760 \end{array}$$

Arithmetic check,

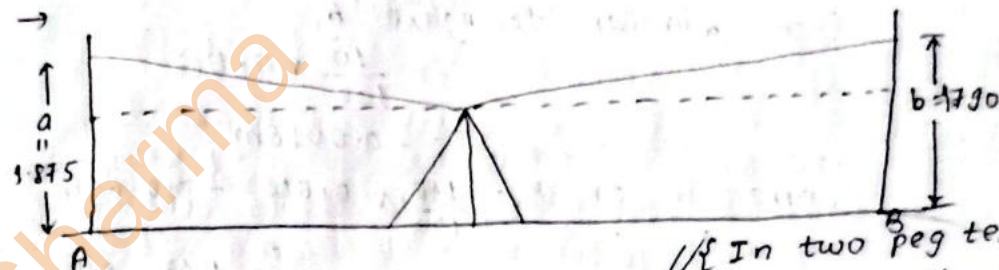
$$\sum BS - \sum FS = \sum Rise - \sum Fall = \text{Last RL} - \text{First RL}$$

$$\text{or, } 12.795 - 11.350 = 5.205 - 3.760 = 250.705 - 249.260$$

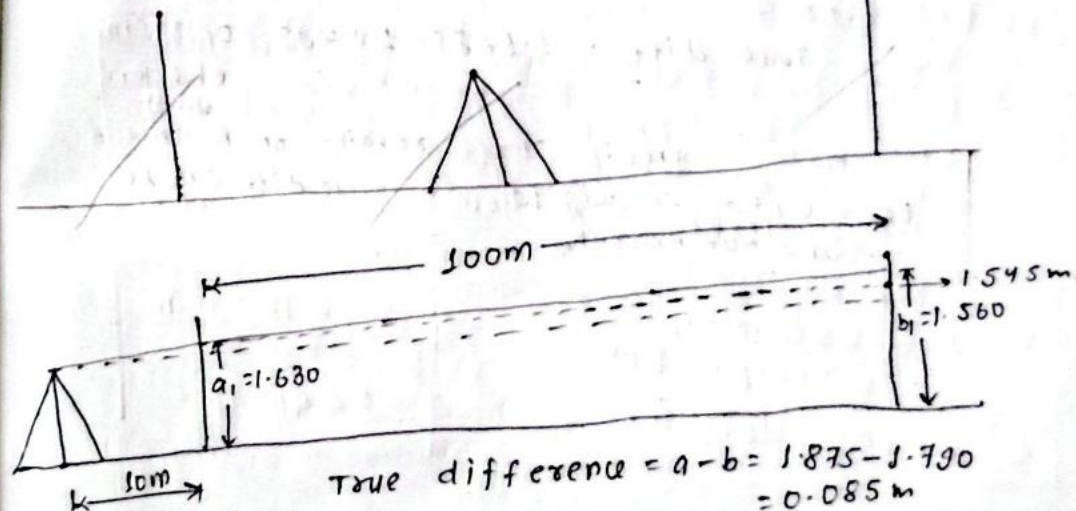
$$\text{or, } 1.445 = 1.445 = 1.445$$

checked

Levelling instrument was set up between two pegs A and B 100m apart. The staff reading on A and B were 1.875 and 1.790 respectively. The instrument was then set up at a distance 10m from A on the line AB. The respective staff readings were 1.630 and 1.560. Calculate, the correct staff readings on A and B, when the line of collimation is exactly horizontal.



In two peg test, if not mentioned where the instrument is set, then it is always outside.



$$\text{True difference} = a - b = 1.875 - 1.790 = 0.085 \text{ m}$$

$$\text{Apparent diff.} = a_1 - b_1 = 1.630 - 1.560 = 0.070 \text{ m}$$

Now, Reading on B at the same level of staff
reading A = $1.630 - 0.085$

$$= 1.545$$

Since, observed reading = 1.560 on B > 1.545

So, line of collimation upward.

Collimation error in $100m$

$$= 1.560 - 1.545 = +0.015m$$

Error positive: correction -ve

From similar triangles on A,

$$= \frac{10}{100} \times 0.015$$

$$= -0.0015m$$

$$\text{Correction on B} = \frac{110}{100} \times 0.015 = -0.0165m$$

$$\text{Correct staff reading on A} = 1.630 - 0.0015 = 1.6285m$$

$$\text{Correct staff reading on B} = 1.560 - 0.0165 = 1.5435m$$

$$\text{True diff} = 1.6285 - 1.5435 = 0.085m$$

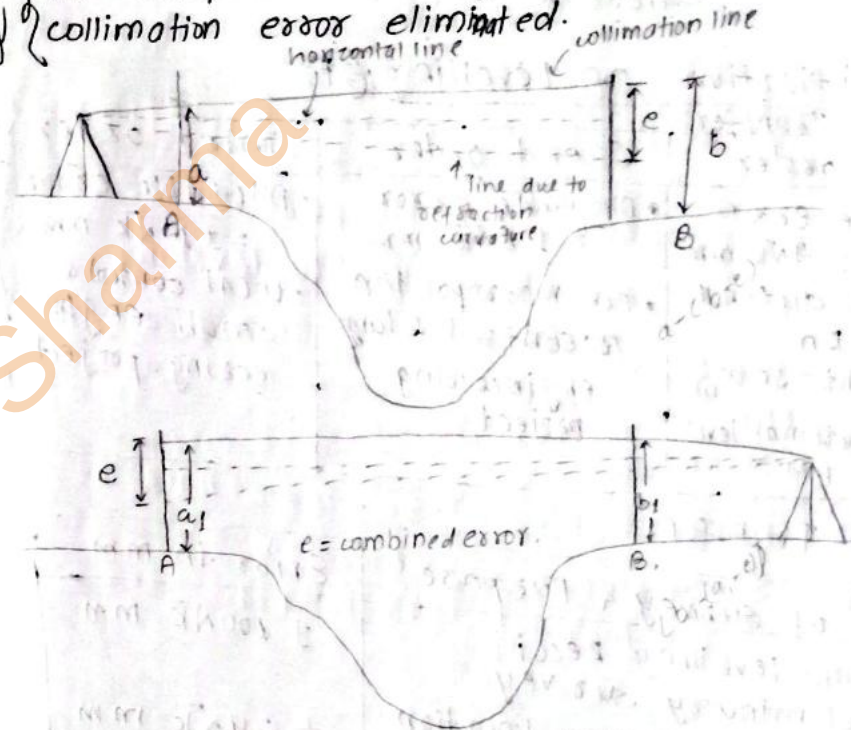
Note: correct staff reading on A and B
kept upto 4 digits so, true diff. can be
found out exactly.

Reciprocal levelling

→ Elevation determined by two set of reciprocal operations.

→ If there is not possible instrument station in between by some obstacles such as river, deep gorge, wide reservoir, etc.

→ In reciprocal levelling curvature, refraction and collimation error eliminated.



correct staff reading on A = a
correct staff reading on B = $b - e$
Height difference b/w A and B

$$H_1 = [a - (b - e)]$$

Height difference between A & B

For next setup,

$$H_2 = (a_1 - e) - b_1$$

#

$$\text{Average height (H)} = \frac{H_1 + H_2}{2}$$

$$= \frac{1}{2} [a - (b - e) + \{(a_1 - e) - b_1\}]$$

$$= \frac{1}{2} [(a - b) + (a_1 - b_1)]$$

It seems direct reading can be utilized no correction required.

Classification of Levelling:

For BM Transfer:-

First order	Second order	Third order
- Allowable error = $\pm 4\sqrt{k}$ mm k = loop distance in km - Max. sight = 50 to 60 m - For national level BM	- Allowable error = $\pm 8\sqrt{k}$ mm - For metropolitan control & large engineering project	- Allowable error = $\pm 12\sqrt{k}$ mm - Local control & small engineering project

For BM Transfer:-

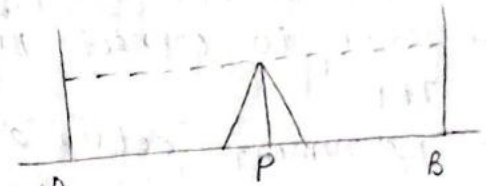
Type of survey & Purpose	Error in mm
1. Rough levelling, Reconnaissance survey	$\pm 100\sqrt{k}$ mm
2. Ordinary levelling, Location and construction survey	$\pm 24\sqrt{k}$ mm
3. Accurate levelling, principle BM	$\pm 12\sqrt{k}$ mm
4. Precise levelling	$\pm 4\sqrt{k}$ mm

Types of levelling operations :-

- Simple levelling
- Differential levelling
- Fly levelling
- Profile levelling
- Cross-section levelling
- Check levelling

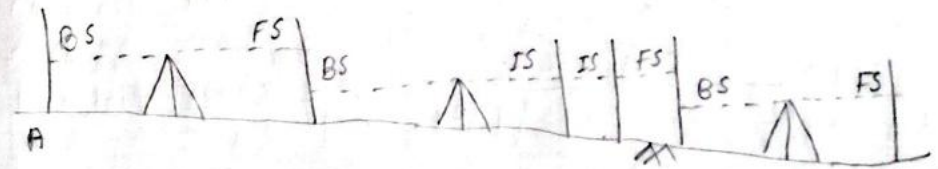
① Simple Levelling :-

- Instrument at mid point & one set up.
- Staff reading gives level difference.



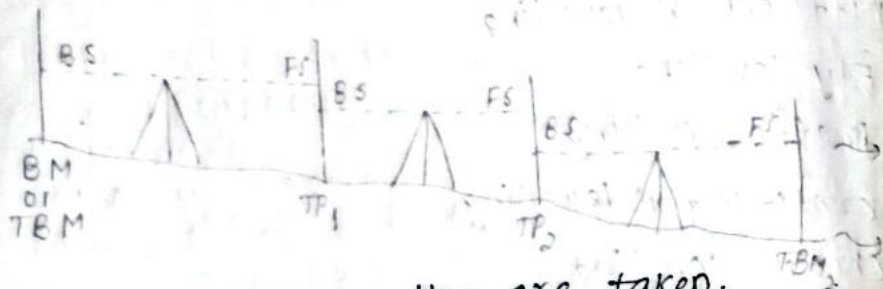
② Differential level :-

- compound levelling; IS
- Compound levelling or continuous levelling
- Level set up number of points & staff reading taken at all of these.
- If two points are far distance
- The difference between two points large.



- In differential levelling there may be or may not be IS.

③ Fly Levelling.



- Only BS & FS reading are taken.
- Used to connect BM or TBM to another BM or TBM.
- Instrument setup at mid point of BS & FS.

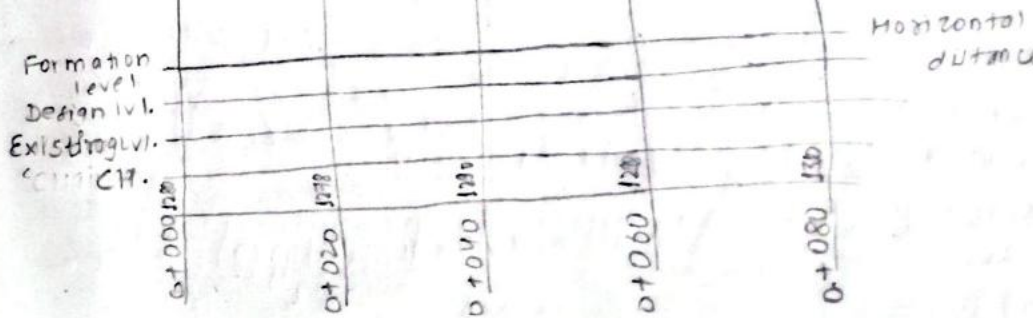
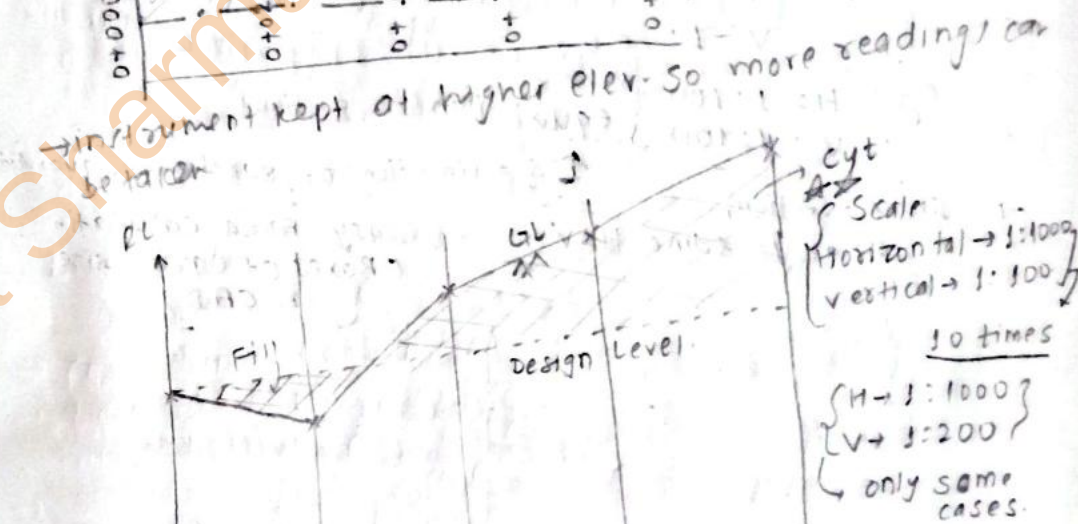
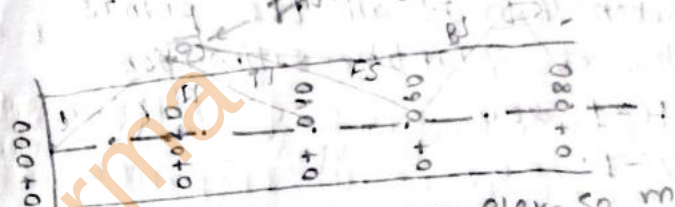
□ Points to be considered in fly levelling

- ① Staff reading in between 0.600m to 2.00m.
- ② Mean - Mid = $\pm 3\text{mm}$
→ Mean = $\frac{T+M+B}{3}$
- ③ Sight balance should be maintained at $\pm 1\text{m}$.
- ④ permissible error = $\pm 2\sqrt{K}$ mm
K = loop distance in km.

④ Longitudinal or Profile Levelling.

chainage → $\frac{0 + 000}{\text{km}}$ can write upto 0 + 999
Further, $\frac{1 + 010}{\text{m}}$ (always in 3 digits)

represented by 1500 km
2-1500m
Instrument
Design level
level.



Gradients: Min 1% Max 12%

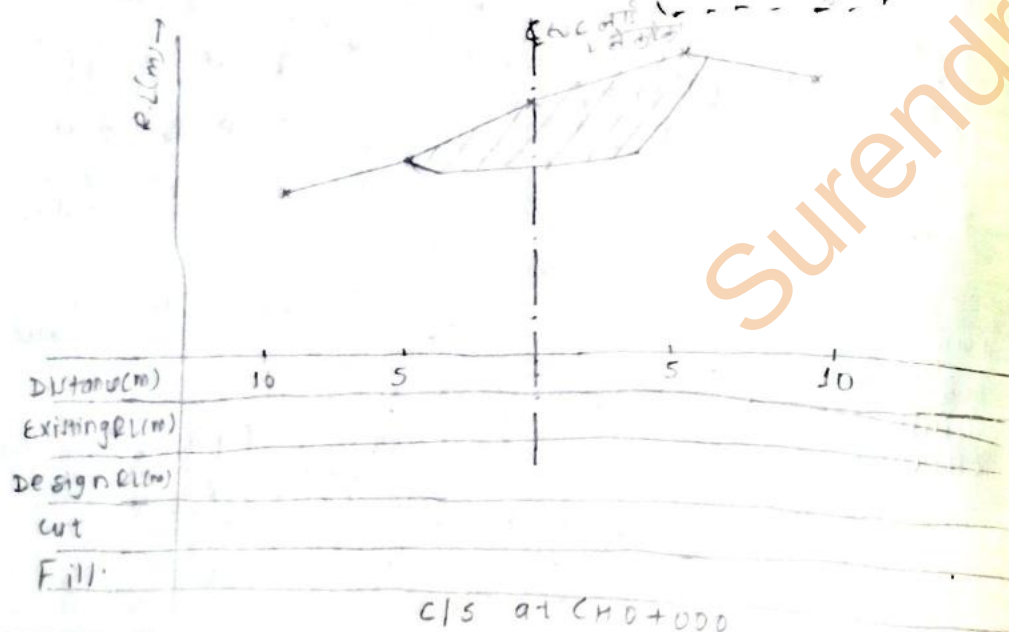
⑤ Cross-sectional Levelling

(X-Section)

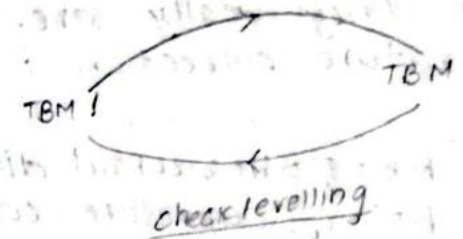
- Right angles to the longitudinal on either side.
- Taken at regular interval if ground is nearly equally inclined, If ground is drastically change taken R.L at all the points.
- For earthwork quantity estimation
- Scale: $H=1:200$
 $V=1:200$

eg: $H=1:100$
 $V=1:100$ } equal (In X-section)

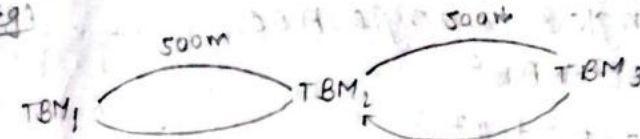
- In X-section, Always same H & V
- ↑ { estimation of earthwork quantity
easy Area calculation
(Boxes of equal size in CAD)



⑥ Check Levelling



Eg:

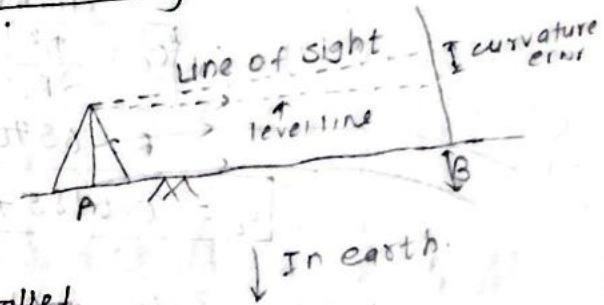


- every 500m to reduce minimize error
- Done for checking accuracy of day's work.

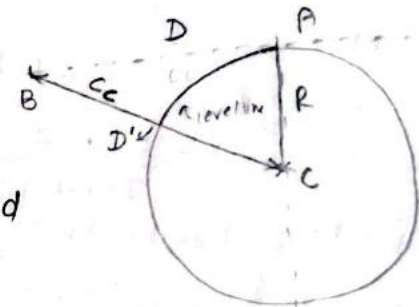
① Correction applied in levelling :-

① Curvature error

- For long sight, the curvature of Earth surface affects staff reading. (kms)
- ~~the~~ Line of sight parallel horizontal & level line parallel to mean spheroidal surface of earth.



- The vertical distance between line of sight & the level line at a particular place is called curvature correction (C_c).



C_c → curvature correction.

- Due to curvature objects ~~the~~ appear lower than they really are.
- Curvature correction is always negative (subtractive)

Let, $AB = D =$ horizontal distance in km.

$BD' = C_c =$ curvature correction.

From right angled triangle ABC ,

$$BC^2 = AC^2 + AB^2$$

$$\text{or, } (R + C_c)^2 = R^2 + D^2$$

$$\text{on, } R^2 + 2RC_c + C_c^2 = R^2 + D^2$$

neglecting C_c^2 in comparison to diameter.

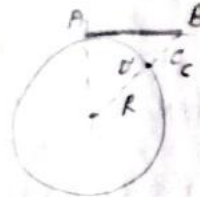
$$\therefore C_c = \frac{D^2}{2R}$$

$$R = 6370 \text{ km.}$$

$$C_c = 0.07857 D^2 \text{ m}$$

$D = \text{km}$

→ True staff reading = Observed staff reading - Curvature correction.



② Refraction Correction:-

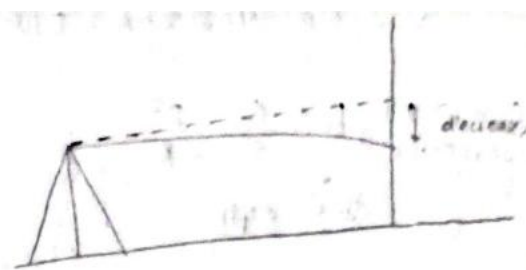
- Reading decreases
- Correction applied to the staff: positive
- varies with atmospheric condition (air density)
- Refraction always additive

$$C_r = \frac{1}{7} \cdot \frac{D^2}{2R}$$

$$= \frac{1}{7} \times 0.0785 D^2$$

$$C_r = 0.0112 D^2 \text{ m] positive}$$

True staff reading = Observed staff reading + Refraction correction.



③ Combined error correction

- combine effect of refraction & curvature.
- combined error = curvature correction + Refraction correction

$$= -0.0785 D^2 + 0.0112 D^2$$

$$= -0.0673 D^2$$

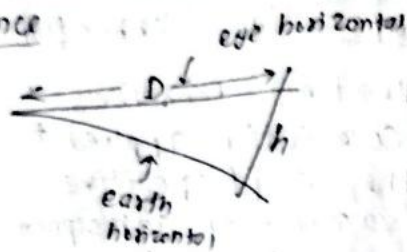
→ Combined ~~error~~ correction always subtractive.

$$\rightarrow \text{Combined correction} = \frac{D^2}{2R} - \frac{1}{7} \frac{D^2}{2R}$$

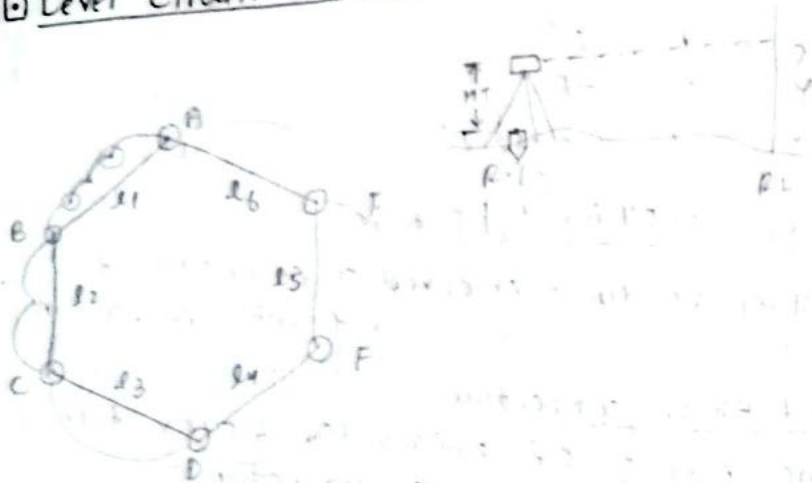
$$= \frac{6}{7} \frac{D^2}{2R} \text{ (negative)}$$

Visible Horizon Distance

Combined correction $h = \frac{6}{7} \frac{D^2}{2R}$
 ↓
 dist. depth.



Level circuit adjustment:-



→ RL of A = known

→ Calculated RL of A

Error (e) = Calculated RL - Known RL

→ Correction:

Permissible = $\pm 24\sqrt{K}$ mm

K = loop distance in km
 = perimeter

→ If distance known only mid wise reading taken.

0.6 to 3m
 with 10m
 levelling

Correction :-

① Correction for first RL = $\frac{l_1}{\sum l} \times e$

Correction for second RL = $\frac{l_1 + l_2}{\sum l} \times e$

Correction for 3rd RL = $\frac{l_1 + l_2 + l_3}{\sum l} \times e$

Correction for last RL (FS) = $\frac{\sum l}{\sum l} \times e$

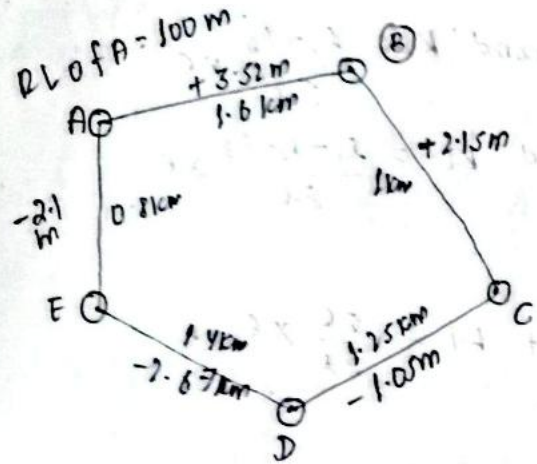
• For Number of Setup (No. of BS = No. of FS)



→ Error (e) = Calculated R.L. of Last point
 - Given RL of last point
 Permissible = $5\sqrt{N}$ mm } For number of setup
 N = Number of setup.

→ Correction for 1st setup = $\frac{1}{N} \times e$ N → No. of CP + 1
 corr = for 2nd setup = $\frac{2}{N} \times e$ CP → change pt.
 corr = for nth setup = $\frac{n}{N} \times e$

Example Figure show a closed circuit of level. Compute the adjusted values of level of diff. pts.



Solution

$$\begin{aligned} \text{Misclosure} &= \sum \text{Rise} - \sum \text{Fall} \\ &= 3.52 + 2.15 - 1.05 - 2.67 - 2.1 \\ &= -150\text{mm at total length} \end{aligned}$$

$$\begin{aligned} \text{Total length} &= 1.6 + 1.1 + 1.25 \\ &\quad + 1.4 + 0.8 \\ &= 6.05\text{km} \end{aligned}$$

$$\begin{aligned} \text{Permissible error} &= 24\sqrt{K} \\ &= 24\sqrt{6.05} \\ &= 59.03\text{mm} \end{aligned}$$

Thus,

$$\begin{aligned} \text{Elevation adjustment at B} &= 100 + 3.52 + \frac{0.15 \times 1.6}{6.05} \\ &= 103.559\text{m} \end{aligned}$$

$$\begin{aligned} \text{Elevation adjustment at C} &= 100 + 3.52 + 2.15 + \frac{(1.6 + 1.1) \times 0.15}{6.05} \\ &= 105.734\text{m} \end{aligned}$$

$$\begin{aligned} \text{Elevation adjustment at D} &= 100 + 3.52 + 2.15 - 1.05 + \frac{(1.6 + 1.1 + 1.25) \times 0.15}{6.05} \\ &= 102.080\text{m} \end{aligned}$$

$$\begin{aligned} \text{Elev. adj. at E} &= 100 + 3.52 + 2.15 - 1.05 - 2.67 \\ &\quad + \frac{(1.6 + 1.1 + 1.25 + 1.4) \times 0.15}{6.05} \\ &= 100\text{m} \end{aligned}$$

$$\begin{aligned} \text{Elev. adj. at A} &= 100 + 3.52 + 2.15 - 1.05 - 2.67 - 2.1 + \frac{\sum \text{Rise} - \sum \text{Fall}}{\sum \text{Length}} \\ &= 100\text{m} \end{aligned}$$

→ corrⁿ CP M M Apply on
→ ' ' → inverted reading; take it as it is { i.e. with min³ fly level.
→ Reciprocal levelling: $K = \frac{\text{bridge axis length} \times 2}{\text{level}}$
→ If T, M, B value given taken mean value
→ In two-peg test: If T, M, B value given take mid value

Transit and Theodolite

Transit → American

↳ closed box

Theodolite: basic surveying instrument for measuring horizontal and vertical angle.

↳ More precise than compass

Uses

- ↳ horizontal and vertical angle measurement
- ↳ locating points on line.
- ↳ prolonging survey line.
- ↳ Finding difference of level (Direct or Indirect Trigonometry)
- ↳ Setting out grades ($\frac{\text{vertical}}{\text{horizontal}}$) → If $\text{In } \% \Rightarrow \text{gradient}$
(cutting & filling)
- ↳ Ranging curves (curved road)
- ↳ Tacheometric survey

Telescope
doesn't move
up or down

Non-transit → telescope fixed

Transit → telescope can be revolved

} Based on horizontal axis

Vernier
Micrometer

Electronic digital

Optical

} Based on angle

Direction Theodolite] जहाँ दोनों को एक ही अक्ष पर 0 बनाते

Why transit / FL / FR? → s.t. both graduations reading can be taken

↳ Det. whether vertical axis is truly vertical or not

↳ (if not vertical $\Rightarrow$ eccentric so, reading might differ)

↳ independent check

Size of theodolite $\rightarrow$ dia of hor. circle / graduated circle

$\hookrightarrow$ greater the dia $\rightarrow$ greater accuracy.

$\hookrightarrow$ small dia $\rightarrow$ portable

Everest: 30cm / 1ft dia.

Mostly: 10cm dia.

Turning axis $\Rightarrow$ horizontal axis

Levelling Head $\left\{ \begin{array}{l} \text{Tribrach} \rightarrow \text{above foot screw} \\ \text{Trivet} \rightarrow \text{below foot screw} \end{array} \right.$

Big dia
 $\downarrow$
Four screw (glass)

Small dia
 $\downarrow$
3 screw

Transiting / plunging / reversing $\rightarrow$ turning telescope 180° in vertical plane

$\left\{ \begin{array}{l} \text{Bubble up / Bubble down} \\ \text{Face left / Face right} \end{array} \right.$
Vertical circle
left when looked at object

Line of collimation $\rightarrow$ line of sight
Line of sight $\rightarrow$ glass to glass centre
Axis of Telescope $\rightarrow$ Objective and eyepiece to glass centre in produce straight imaginary line
Fixed for given instrument.
 $\hookrightarrow$ $\perp$ to plumb line.

Line of sight & horizontal axis $\rightarrow$ combines

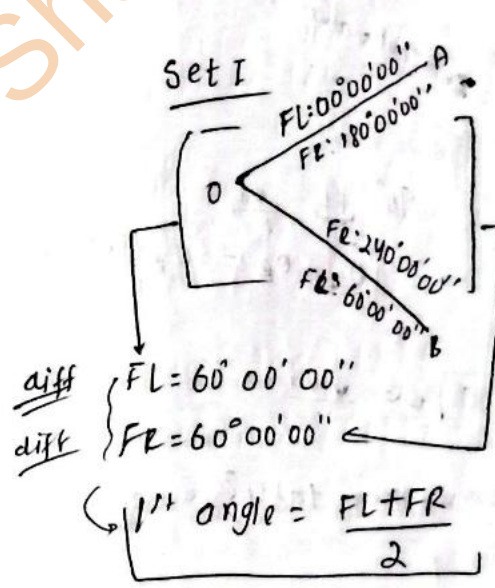
Horizⁿ axis $\rightarrow$ // to plate bubble

Face left $\rightarrow$ Telescope normal / Bubble down.
Face right $\rightarrow$ Telescope inverted / Bubble down.
Face left (FL) $\rightarrow$ L $\rightarrow$ D (Direct) $\rightarrow$ Zenithal angle near 90°
Face right (FR) $\rightarrow$ R $\rightarrow$ R (Reverse) $\rightarrow$ Zenithal angle near 270°
 $\uparrow$
for symmetrical angle.

Ordinary Method

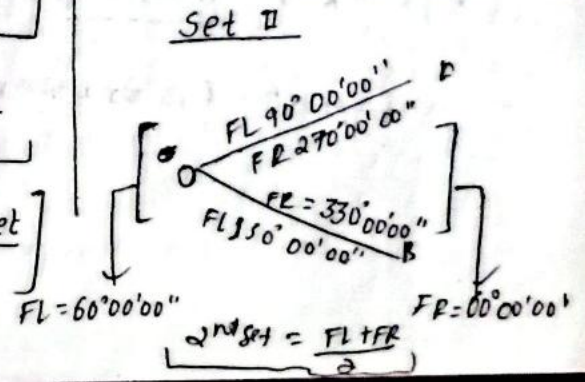
$\propto$ FL & FR difference in horizontal angle:

$\hookrightarrow 180^\circ \pm 2 \times L.C$
L.C. = Least count
 $\{ \text{Vertical: } \pm 20'' \rightarrow \text{bisection / centring, etc. also apply} \}$



Set angle = Mean of FL & FR
1st set $\rightarrow n=1 \Rightarrow \frac{180^\circ}{1} = 0^\circ$
2nd set $\rightarrow n=2 \Rightarrow \frac{180^\circ}{2} = 90^\circ$
3rd set $\rightarrow n=3 \Rightarrow \frac{180^\circ}{3} = 60^\circ$

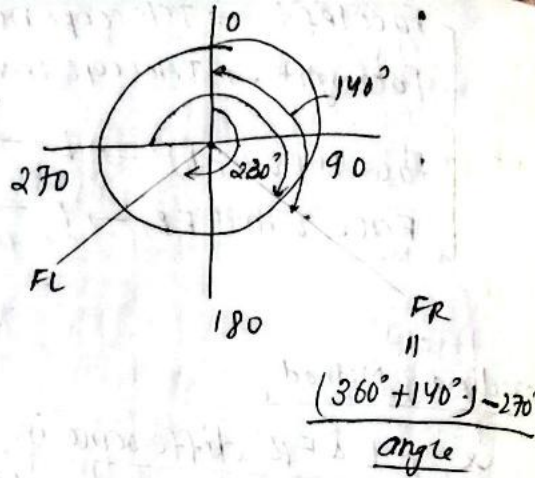
Mean Hor. angle = $\frac{1^{st} \text{ set} + 2^{nd} \text{ set}}{2}$



* Reading पढ़े Reading

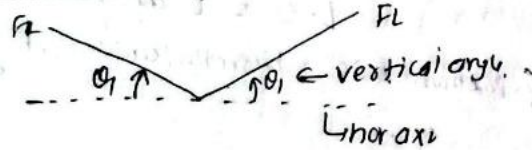
बेहतर छांटने वाले
it means circle cross
जहाँ 360° add
जाने

↳ usually seen in 2nd set.

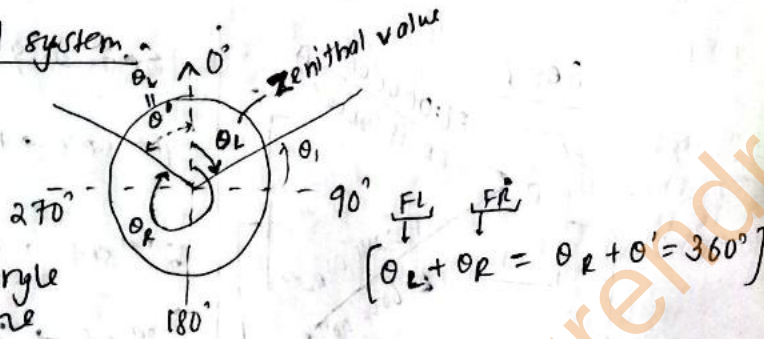


For vertical angle

→ w.r.t horizontal axis



→ In Zenithal system.
Zenith → sky → 0°



In Zenithal Angle system, the

relation between

FL & FR is, $\text{sum} = 360^\circ$

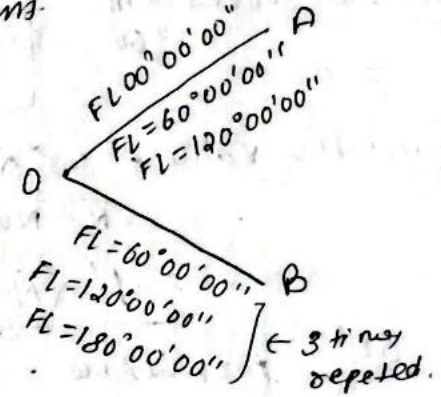
* Repetition & Retardation method table

Repetition Method

In Total station: REP

↳ TO minimize bisection error & graduation error
↳ repetitive angle measurements

$$\text{Mean Angle} = \frac{180^\circ 00' 00''}{3}$$



Theodolite: lower clamp rotate

तो फिर reading lined soon

→ 1st REP key.

Retardation method } when sev. angles are to be measured (2nd set).

Angle पढ़े Always long sight in zero गति

↳ So that good bisection

length पढ़े → short sight

Horizon close: एक pt. पर start जहाँ & end जहाँ

↳ while taking angle 270° starting point and ending point reading

taking process is called horizon close

done to find out error

↳ for independent check.

Observed sum
 $= \sum AOB + \sum BOC + \sum COA$

error = observed sum
 $- 360^\circ$
 $= e$

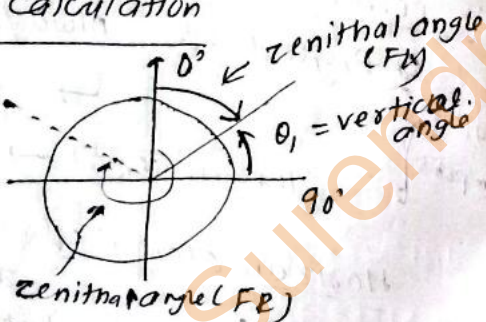
Permissible value = $30'' \sqrt{N}$ → For total station
 $N = \text{Number of angles}$
 $= 1' \sqrt{N}$ → theodolite

Corr permissible = $L.C. \sqrt{N}$ } L.C → least count
 $= C \sqrt{N}$ } C → constant (depend on type of work)

✗ Error: Equally distributed but not in fraction

Zenithal to Vertical Calculation

FL
 Vertical angle = $90^\circ - \text{Zenithal angle}$
 (θ_1)
 +ve → elevation angle
 -ve → Depression angle



For FR

$\theta_1 = \text{Zenithal} - 270^\circ$
 +ve → elevation
 -ve → Depression

Mean Direction Method → In vernier theodolite

LR LR
 Mean
 $-0'15''$
 ↑ ⊖
 L $0^\circ 00' 10''$
 R $179^\circ 59' 20''$
 L $60^\circ 20' 40''$
 R $240^\circ 20' 10''$
 10" वही अंश (00'00'00")
 (A) $\frac{+10'' - 40''}{2} = -15''$ (from $180^\circ 00' 00''$ to -ve)

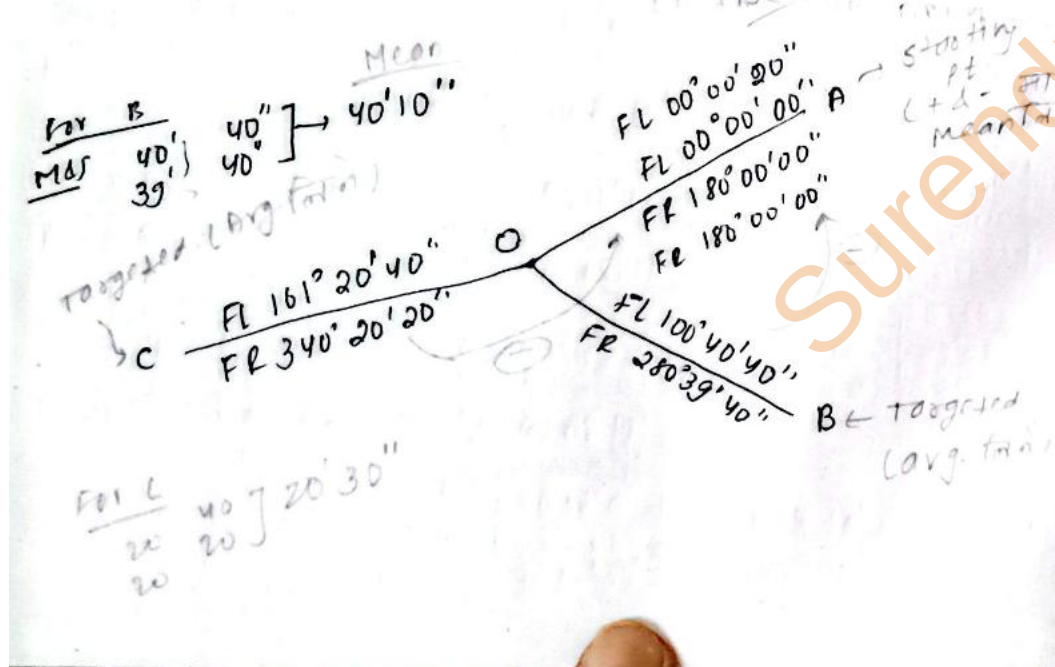
Mean = $-15''$

Mean angle: $60^\circ 20' 40''$

Mean angle → L-L, R-R लेखेंदा सविलो
 (Ordinary method)
 Mean Dis → L-R, L-R



Inst St. & HI	Sighted to	Face	OBSERVATION			MEAN		General			Angle	Corr ⁿ	Corr ^d Angle	Sum ang.
			O	I	"	'	"	O	'	"				
O & H.I 1.5	A	L	00	00	00	00	00	00	00	00				
		R	180	00	00									
	B	L	100	40	40	40	10	100	40	10	100°40'10"	-03"	100°40'07"	
		R	280	39	40									
	C	L	161	20	40	20	30	161	20	30	60°40'20"	-03"	60°40'17"	
		R	341	20	20									
	A	L	00	00	20	00	10	00	00	10	198°39'40"	-04"	198°39'36"	
		R	180	00	00									



$\Sigma \text{Sum (Angle)} = 100^\circ 40' 10'' + 60^\circ 40' 20'' + 198^\circ 39' 40''$

$= 360^\circ 00' 10''$

error = Observed sum - Theoretical

$= 00^\circ 00' 10''$ (error $\rightarrow +ve$, corrⁿ $\rightarrow -ve$)

Permissible error = $20'' \sqrt{3} = 34.64'' < 10''$

(Permissible limit)

Hence, error is within the permissible limit.

Therefore, distribute error equally.

corrⁿ in equal each angle = $\frac{00^\circ 00' 00''}{3}$

$= 3''$

3'' for 3 angles, and remain 1'' for largest angle.

Adjust 3'' in $\angle AOB$ and $\angle BOC$ & 4'' in $\angle COA$

Q. The following observation were recorded for traverse ABCDEA compute the mean horizontal angle and adjust them if necessary

Inst. Str	Target Str	H.C.R OBSERVED	(F.F) Reversed.
		Direct	
A	D	90°00'10"	269°59'40"
	B	200°25'40"	20°25'30"
B	A	89°59'30"	270°00'10"
	C	180°16'10"	00°16'00"
C	B	90°00'10"	270°00'10"
	D	179°08'40"	359°08'20"
D	C	89°59'50"	270°00'00"
	A	160°12'40"	340°12'30"

@ From Mean angle method.

Inst. Str.	Target Str.	Face.	H.C.R Observation	Face Angle	Mean Angle	Corr ^e	Corr ^e angle	Rem.
A	D	D (L)	90°00'10"	110°25'30"	110°25'40"	-44"	110°25'24" 56"	
	B	D (L)	200°25'40"					
	D	R	269°59'40"	110°25'50"				
	B	R	20°25'30"					
B	A	D	89°59'30"	90°16'40"	90°16'15"	-44"	90°15' 31"	
	C	D	180°16'10"					
	A	R	270°00'10"	90°15'50"				
	C	R	00°16'00"					
C	B	D	90°00'10"	89°08'30"	89°08'20"	-44"	89°07'36"	
	D	D	179°08'40"					
	B	R	270°00'10"	89°08'10"				
	D	R	359°08'20"					
D	C	D	89°59'50"	70°12'50"	70°12'40"	-43'	70°11' 57"	
	A	D	160°12'40"					
	C	R	270°00'00"	70°12'30"				
	A	R	340°12'30"					

$$\begin{aligned} \text{Error} &= \text{Observed sum of interior angle} - \text{Theoretical sum} \\ &= 360^\circ 2'55'' - 360^\circ \\ &= 2'55'' (+ve) \end{aligned}$$

So, correction should be -ve

$$\text{Correction} = \frac{2'55''}{4} = \frac{(60 \times 2 + 55)''}{4} = \frac{175''}{4}$$

$$\begin{aligned} &= 43.75 \\ &= 43'' + \frac{75 \times 4}{4} \\ &\text{for all} + 3'' \\ &\text{can allow for } \frac{3}{4} \end{aligned}$$

$$\text{Permissible} = 30'' \sqrt{4} = 60''$$

(< actual)

[least angle
is 43''
- 44'']

Observed sum = $110^{\circ}25'40'' + 50^{\circ}16'15''$
 $+ 89^{\circ}08'20'' + 70^{\circ}12'40''$

$$= 360^{\circ} 02' 55''$$

coll $\rightarrow$ -ve

total correction = $02'55''$

total correction = $0^{\circ} 55''$
correction to each angle = $-\frac{2^{\circ} 55''}{4} = -43.75''$

$$\overset{7}{(-75)} \rightarrow x4 = -3''$$

Mean Angle निर्धारण के लिए चर्क घूर्णन द्वारा निर्धारित
↓
degrees में

B After Mean calc EEP Initial, $90^{\circ}00'00''$
 $270^{\circ}00'00''$

1011P2 Coll. 172 E1012
 For same angle 1011P2

Chapter - 9

Computation of Area and Volume

↳ Planemetric.

$$1 \text{ dhur} = 182.25 \text{ sq. ft.}$$

$$1 \text{ anna} = 4 \text{ paisa} = 342.25 \text{ sq. ft.}$$

$$1 \text{ paisa} = 4 \text{ dom} = 85.56 \text{ sq. ft.}$$

$$1 \text{ sooni} = 16 \text{ anna}$$

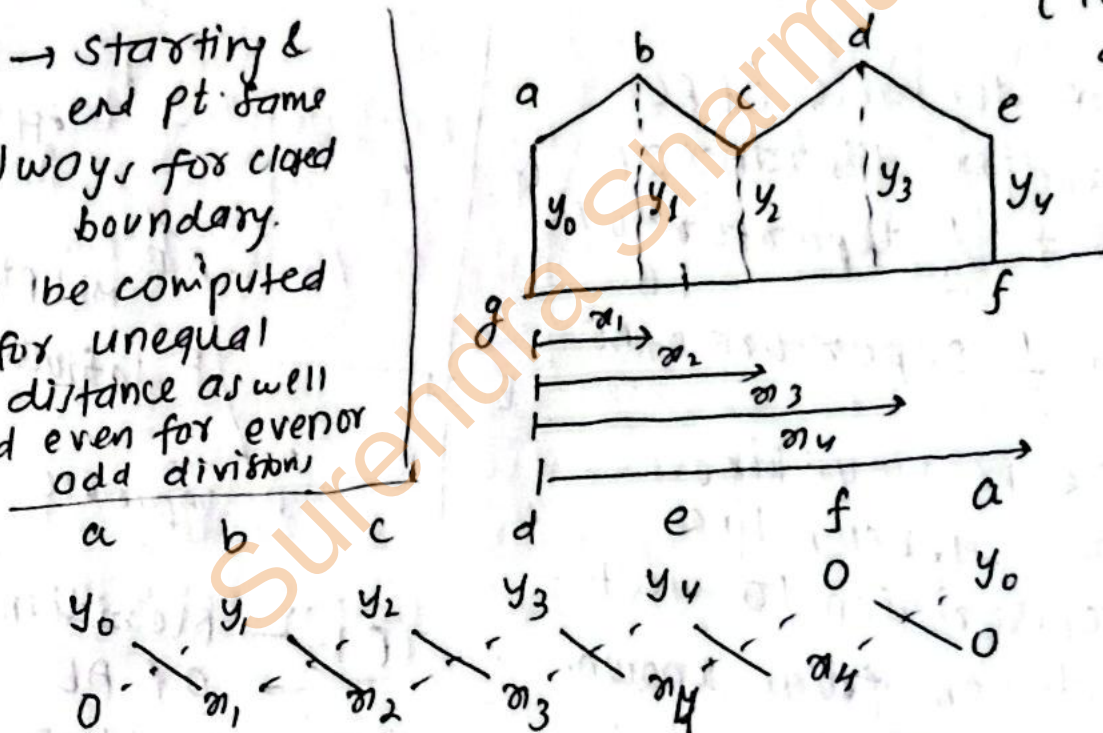
$$1 \text{ Bigha} = 20 \text{ katna} = 20 (20 \text{ dhur})$$

Co-ordinate Method of Finding Area.

area → starting & end pt. same

↳ always for closed boundary.

can be computed for unequal distance as well and even for even or odd divisions



|| to N
↑
(latitude & departure)
↓
N $\frac{E}{N}$
S $\frac{W}{S}$

Sum of product of solid line,

$$\Sigma P = y_0 x_1 + y_1 x_2 + y_2 x_3 + y_3 x_4 + 0 \cdot 0$$

↳ irregular interval

Sum of product along solid line

$$\Sigma Q = (0 \cdot y_1 + x_1 y_2 + \dots + 0 \cdot y_0)$$

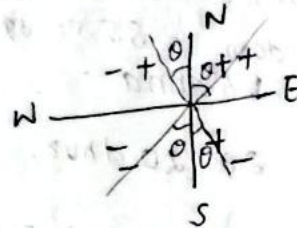
$$\left[\text{Required area} = \frac{1}{2} (\Sigma P - \Sigma Q) \right]$$

Altitude $\rightarrow$ // to Meridian
 Departure $\rightarrow$ $\perp$ to Meridian

$$\theta = \tan^{-1} \left(\frac{\text{Departure}}{\text{Latitude}} \right)$$

Gives Reduced Bearing.

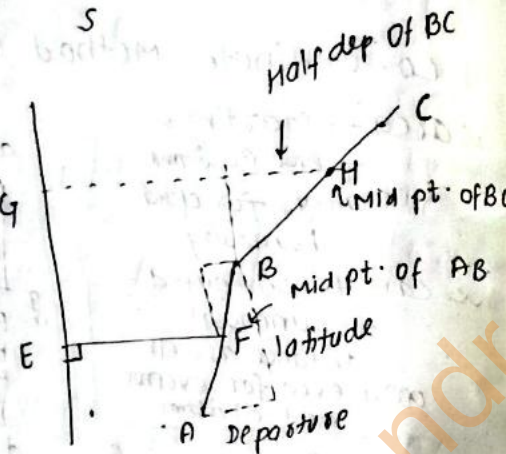
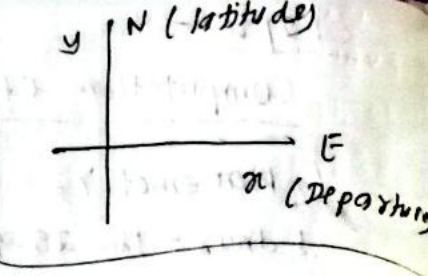
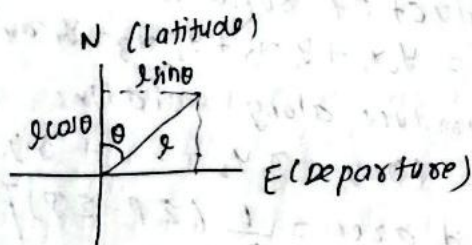
Convert to WCB.



Meridian distance of BC
 = Meridian distance of AB
 + $\frac{1}{2}$ departure of AB
 + $\frac{1}{2}$ departure of BC

(AB) $\leftarrow$ previous line
 (BC) $\rightarrow$ current line.

Take Meridian to west & calculation from known.



(EF) $\rightarrow$ Meridian distance of AB

(GH) $\rightarrow$ Meridian distance of BC.

Eg: Find the area for figure using DMD method.

Solution, computation of DMDs

Departure of AB = +517.44 = DMD of AB

Departure of AB = +517.44
X = 00
Y = 591.78

Departure of BC = +5.964
+ 1040.852 = DMD of BC

Departure of BC = +5.964

Departure of CD = +192.881

1239.697 = DMD of CD

Departure of CD = +192.881

Departure of DE = -590.571

+ 842.007 = DMD of DE

Departure of DE = -590.571

Departure of EA = -125.718

125.718

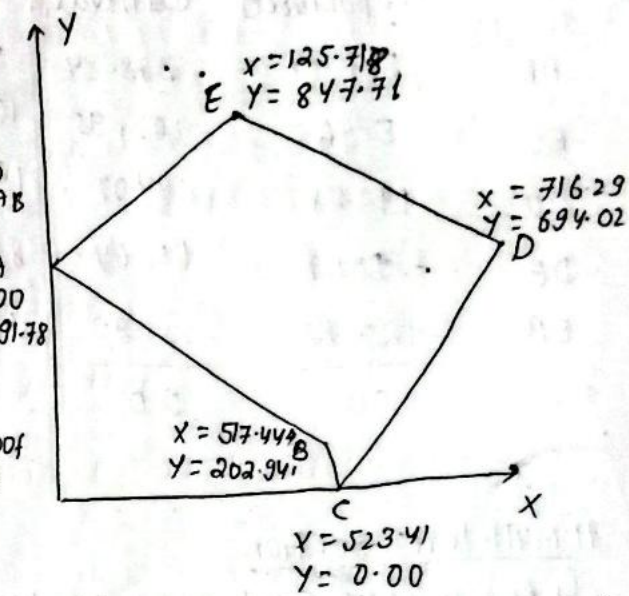


Fig. 1 traverse for computation of Area.

DMD = DMD of previous line
 + Departure of previous line + Departure of line itself.

Departure = अगला को - पछाडि को]

First line को departure $\downarrow$ DMD

Line	Balanced Departure	Balanced Latitude	Double Areas	Double Areas
			Plus	Minus
AB	517.44	-388.84	517.44	201,201
BC	5.96	-202.95	1040.84	211,238
CD	192.88	694.02	1239.68	860,363
DE	-590.57	153.69	841.99	129,405
EA	-125.72	-255.93	125.72	32,170
	00	00	$\Sigma = 989,768$	$\Sigma = -444,609$

Algebraic sum = 545159

Multilevel Section

↳ Best way to find area of multilevel section is by co-ordinate method. Origin assumed at middle of base width. Coordinates of points A, B, C, ... can be obtained from field notes.

↳ Coordinates: right → +ve
above → +ve.

• With proper sign of coordinates and including the base points A and B; usual form of recorded notes is written as

$$\begin{array}{c} y \rightarrow \frac{0}{-b_2} + \frac{h_1}{-w_1} + \frac{h_2}{-w_2} + \frac{h}{0} + \frac{h_3}{+w_3} + \frac{h_4}{+w_4} + \frac{0}{+b_2} \\ x \rightarrow \frac{0}{-b_2} + \frac{h_1}{-w_1} + \frac{h_2}{-w_2} + \frac{h}{0} + \frac{h_3}{+w_3} + \frac{h_4}{+w_4} + \frac{0}{+b_2} \end{array}$$

(A) (C) (D) (E) (F) (G) (B)

Methods to calculate area from above notes: (A) (B) (C) (D) (E) (F) (G) (B)

① Place opposite algebraic sign on opposite side of each lower term as shown above.

② Multiply each upper term by algebraic sum of two adjacent lower terms using signs facing the upper term.

③ Half the algebraic sum of these products will be the required area.

$$A = \frac{1}{2} \left[h_1 \left(\frac{b}{2} - w_2 \right) + h_2 (w_1 + 0) + h_3 (0 + w_4) + h_4 (-w_3 + \frac{b}{2}) \right]$$

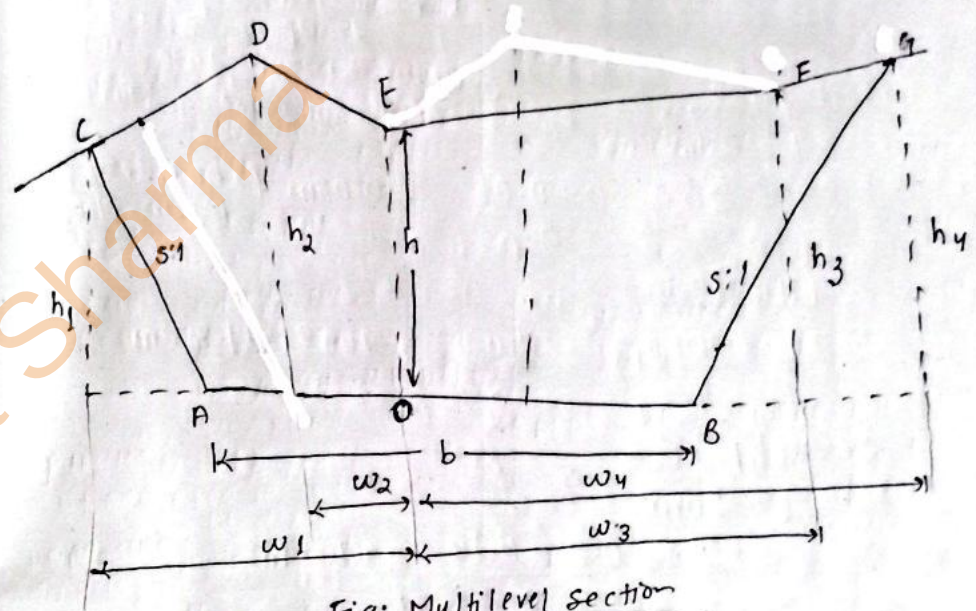


Fig: Multilevel Section

Distance measurement

Q. A line was measured by a 20m chain which was accurate before setting the day's work. After chaining 900m, the chain was found to be 6cm too long. After chaining a total distance of 1575m, the chain was found to be 14cm too long. Find true distance of the line.

→ Solⁿ:

First part:

$$L = 20\text{m}$$

$$L' = 20 + \frac{0 + 0.06}{2} = 20.03\text{m} \quad (\text{considering mean elongation})$$

$$ML = 900\text{m}$$

$$TL = \frac{L'}{L} \times ML = \frac{20.03}{20} \times 900 = 901.35\text{m}$$

Second part:

$$L = 20\text{m}$$

$$L' = 20 + \frac{0.06 + 0.14}{2} = 20.10\text{m}$$

$$ML = 1575 - 900 = 675\text{m}$$

$$TL = \frac{L'}{L} \times ML = \frac{20.1}{20} \times 675 = 678.375\text{m}$$

$$\begin{aligned} \text{True distance of line} &= 901.35 + 678.375 \\ &= 1579.725\text{m} \end{aligned}$$

Q. On a map drawn to a scale of 50m to 1cm, a surveyor measured distance between two stations as 3500m. But it was found that by mistake, he had used a scale of 100m to 1cm. Find true distance b/w the stations.

→ Solⁿ:

First method:

Surveyor used 100m to 1cm

$$\text{Distance b/w stations on map} = \frac{3500}{100} = 35\text{cm}$$

As actual scale of map is 50m to 1cm

$$\text{True distance on ground} = 35 \times 50 = 1750\text{m}$$

Second Method:

$$\text{True distance} = \frac{\text{RF of wrong scale}}{\text{RF of correct scale}} \times \text{Measured length}$$

$$\begin{aligned} &= \frac{\left(\frac{1}{100 \times 100}\right)}{\left(\frac{1}{50 \times 100}\right)} \times 3500 \\ &= 1750\text{m} \end{aligned}$$

Q. 6) A steel tape was exactly 30m long at 20°C when supported throughout its length under a pull of 10kg. A line was measured with this tape under a pull of 15kg and at mean temp of 32°C, and found to be 780m long. The c/s area of tape is 0.03cm² and its total wt = 0.693kg.
 $\alpha = 11 \times 10^{-6}/^\circ\text{C}$, $E = 2.1 \times 10^6 \text{ kg/cm}^2$

Compute true length of line if tape was supported during measurement (a) at every 30m.

(b) at every 15m.

→ Sol: - $L = 30\text{m}$ $A = 0.03\text{cm}^2$
 $T_0 = 20^\circ\text{C}$ $\alpha = 11 \times 10^{-6}/^\circ\text{C}$
 $P_0 = 10\text{kg}$ $E = 2.1 \times 10^6 \text{ kg/cm}^2$
 $P_m = 15\text{kg}$ $W = 0.693\text{kg}$
 $T_m = 32^\circ\text{C}$ $ML = 780\text{m}$

(a) When supported at every 30m.

Total correction per tape length was found out first

Here, $n = 1$. (said nothing $\Rightarrow$ no of span = 1)

$$C_t = \alpha (T_m - T_0) L = 0.00396\text{m}$$

$$C_p = \frac{P_m - P_0}{EA} L = 0.002381\text{m}$$

$$C_s = \frac{1}{n^2} \times \frac{1}{24} \left(\frac{W}{P_m} \right)^2 L = 0.002668\text{m}$$

Correct length, $L' = L + C_t + C_p - C_s$

$$= 30 + 0.00396 + 0.002381 - 0.002668 = 30.003673\text{m}$$

$$TL \text{ of line} = \frac{L'}{L} \times 780 = 780.095\text{m}$$

(b) When supported at Every 15m
 No. of span, $n = 2$.

Correction per tape.

$$C_t = 0.00396\text{m (as before)}$$

$$C_p = 0.002381 \text{ (as before)}$$

$$C_s = \frac{1}{n^2} \times \frac{1}{24} \left(\frac{W}{P_m} \right)^2 \times L$$

$$= 0.000667\text{m (-ve)}$$

$$\text{Correct tape length, } L' = 30 + 0.00396 + 0.002381 - 0.000667$$

$$= 30.005674\text{m}$$

$$TL \text{ of line} = \frac{L'}{L} \times 780$$

$$= 780.148\text{m}$$

Q. A 30m tape was standardised at a temperature of 20°C and under a pull of 5kg. The tape was used in catenary at a temperature of 25°C and under a pull of P kg. The cross area of tape is 0.02cm^2 , its wt - per unit length is 22g/m . $E = 2 \times 10^6 \text{ kg/cm}^2$, $\alpha = 11 \times 10^{-6}/^{\circ}\text{C}$. Find correct horizontal distance, if pull is equal to (a) 5kg (b) 11kg.

→ Sol:-

$$\begin{aligned} L &= 30\text{m} & A &= 0.02\text{cm}^2 \\ P_0 &= 5\text{kg} & E &= 2 \times 10^6 \text{ kg/cm}^2 \\ T_0 &= 20^{\circ}\text{C} & \alpha &= 11 \times 10^{-6}/^{\circ}\text{C} \\ T_m &= 25^{\circ}\text{C} & W &= 22\text{g/m} \\ P_m &= P & &= 22 \times 30 \\ n &= 1 & W &= 0.66\text{kg} \end{aligned}$$

(a) When applied pull, $P = 5\text{kg}$.

$$C_t = \alpha (T_m - T_0) L = 0.00165\text{m}$$

$$C_p = \frac{P_m - P_0}{EA} L = 0\text{m}$$

$$C_{ps} = \frac{1}{24n^2} \left(\frac{W}{P_m} \right)^2 \times L = 0.02178\text{m} (-)$$

$$\begin{aligned} \text{Corr. hor. dist} &= 30 + 0.00165 - 0.02178 \\ &= \underline{29.980\text{m}}. \end{aligned}$$

$$\text{When, } P = 11\text{kg}, \quad C_t = 0.0045\text{m}, \quad C_s = 0.0045\text{m}$$

$$\begin{aligned} \text{Corr. hor. dist} &= 30 + 0.00165 + 0.0045 - 0.0045 \\ &= 30.002\text{m}. \end{aligned}$$

Q. A 30m steel tape was standardised at a temp. of 20°C and under pull of 10kg. The tape was used in a catenary to fix a distance of 28m b/w two points at 40°C and under a pull of 5kg. Given cross area $A = 0.02\text{cm}^2$. Total wt = 470g. $E = 2.1 \times 10^6 \text{ kg/cm}^2$, $\alpha = 11 \times 10^{-6}/^{\circ}\text{C}$.

- (a) Find the correct distance b/w points and
(b) Find the value of pull for which measured distance would be equal to correct distance.

→ Sol:-

$$\text{Distance b/w two points} = 28\text{m}.$$

$$L = 28\text{m (span length)}$$

$$T_0 = 20^{\circ}\text{C} \quad A = 0.02\text{cm}^2$$

$$T_m = 40^{\circ}\text{C} \quad E = 2.1 \times 10^6 \text{ kg/cm}^2$$

$$P_0 = 10\text{kg} \quad \alpha = 11 \times 10^{-6}/^{\circ}\text{C}$$

$$P_m = 5\text{kg} \quad \text{Total wt} = 470\text{g}$$

$$\begin{aligned} \text{Wt for 28m} &= \frac{470}{30} \times 28 = 438.67\text{g} \\ &= 0.43867\text{kg} \end{aligned}$$

$$C_t = \alpha (T_m - T_0) L = 0.00616\text{m}$$

$$C_s = \frac{1}{24n^2} \left(\frac{W}{P_m} \right)^2 \times L = 0.00898\text{m} (-ve)$$

$$\begin{aligned} C_p &= \frac{P_m - P_0}{EA} \times L = -0.00333\text{m} \\ \text{(a) Correct length} &= 27.994\text{m} \end{aligned}$$

⑥ Now we have to find the normal tension at which the measured distance would be equal to correct distance. This condition is satisfied when

pull correction = sag correction

Let, normal pull = P_n

Then

$$\frac{(P_n - P_0)}{AF} L = \frac{L W^2}{24 P_n^2}$$

$$\text{or, } \frac{P_n - 10}{0.01 \times 2.1 \times 10^6} = \frac{(0.439)^2}{24 P_n^2}$$

$$\text{or, } 24 P_n^3 - 240 P_n^2 - 8095 = 0$$

$$\text{or, } P_n^3 - 10 P_n^2 - 337.3 = 0 \quad \text{--- (1)}$$

Now, value of P_n has to be found out by trial.

Method of Totals,

Putting assumed value of P_n in eqⁿ (1)

When, $P_n = 10$ { pull corr +ve when length inc.
when, $P_n > P_0$ start from 10 & go on

$$10^3 - 10 \times 10^2 - 337.3 = -337.3 \text{ (not acceptable)}$$

$$\text{When, } P_n = 11, -216.3 \neq 0$$

$$\text{When, } P_n = 12, -49.3 \neq 0$$

$$\text{When, } P_n = 12.25, +0.34 \text{ (which may be accepted, as eqⁿ is nearly satisfied)}$$

So, req. pull at which measured distance is correct is 12.25 kg.

Measurement of Volume

Cross section Method

Vertical volⁿ is divided into series of solids by the planes of c/s.

↳ For irregular ground, c/s taken close together.

Types of cross-section and Areas

b = subgrade (Constant formation) width

↳ finished roadway and depends upon project requirement.

$S:1$ = Side slope (s horizontal to 1 vertical)
(usually cut slopes $1:1$, Fill slopes $1.5:1$)

$n:1$ → horizontal slope of original ground
(n horizontal to 1 vertical)

h → depth of earthwork on centerline

d_1, d_2 → side widths

↳ distance from center line to the intersection of side slope with original surface.

h_1, h_2 → side heights

↳ vertical distance from subgrade to intersection of side slope and original surface.

w → formation width of excavation in side-hill section.

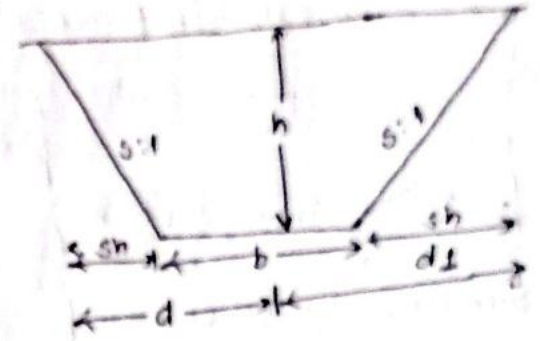
A → sectional area

Level section

$$d = d_1 = \frac{b}{2} + sh$$

$$A = \frac{1}{2} [b + (b + 2sh)]h$$

$$A = (b + sh)h$$



Two level section

Area ABEDC

$$= \text{Area ACF} +$$

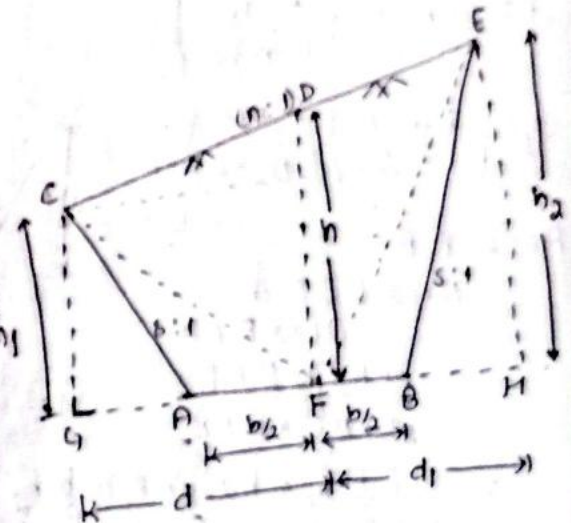
$$+ \text{Area CDF}$$

$$+ \text{Area DEF}$$

$$+ \text{Area FEB}$$

$$= \frac{1}{2} \left(\frac{b}{2} h_1 + hd + hd + \frac{b}{2} h_2 \right) h_1$$

$$A = \frac{b}{4} (h_1 + h_2) + \frac{h}{2} (d + d_1)$$



Using, $GA = GF - FA$

$$sh_1 = \frac{d - b/2}{n}$$

$$d = (h - h_1)n$$

$$h_1 = \frac{n}{n+s} \left(h - \frac{b}{2n} \right)$$

$$d = \frac{b}{2} + sh_1 = \frac{b}{2} + \frac{sn}{n+s} \left(h - \frac{b}{2n} \right)$$

$$\text{Similarly, } \left[h_2 = \frac{n}{n-s} \left(h + \frac{b}{2n} \right) \right] \left[d_1 = \frac{b}{2} + \frac{sn}{n-s} \left(h + \frac{b}{2n} \right) \right]$$

$$A = \frac{d d_1}{s} - \frac{b^2}{4s}$$

Substituting in 1 we get required area,

$$A = \frac{s(b/2)^2 + n^2(bh + sh^2)}{n^2 - s^2}$$

Three level section

Similar to two level section

$$d = \left(h + \frac{b}{2s}\right) \left(\frac{ns}{n+s}\right)$$

$$d_1 = \left(h + \frac{b}{2s}\right) \left(\frac{ns}{n-s}\right)$$

Area ABCDEF

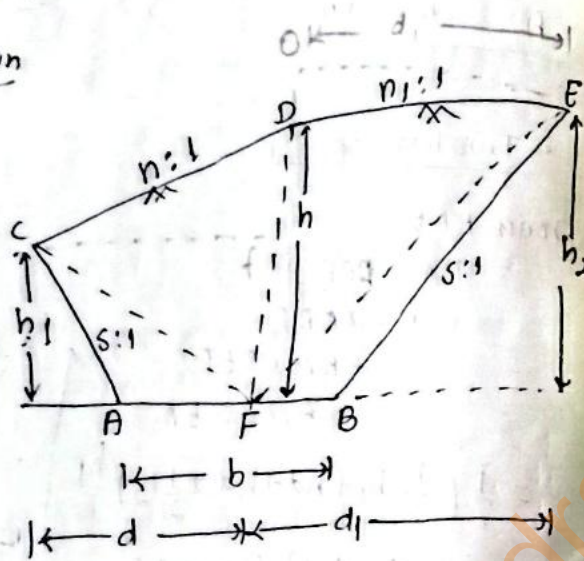
$$= \text{Area ACF} + \text{Area FCD} + \text{Area FDE} + \text{Area FEB}$$

$$= \frac{1}{2} \left(\frac{b}{2} h_1 + h d + h d_1 + \frac{b}{2} h_2 \right)$$

$$A = \frac{b}{4} (h_1 + h_2) + \frac{h}{2} (d + d_1)$$

$$h_2 = OD + DF = h + \frac{d}{n}$$

$$h_1 = DF - DF' = h - \frac{d}{n}$$



Side hill two level section

↳ Ground has transverse slope but the slope of ground cuts formation level partly in cutting and partly in filling.

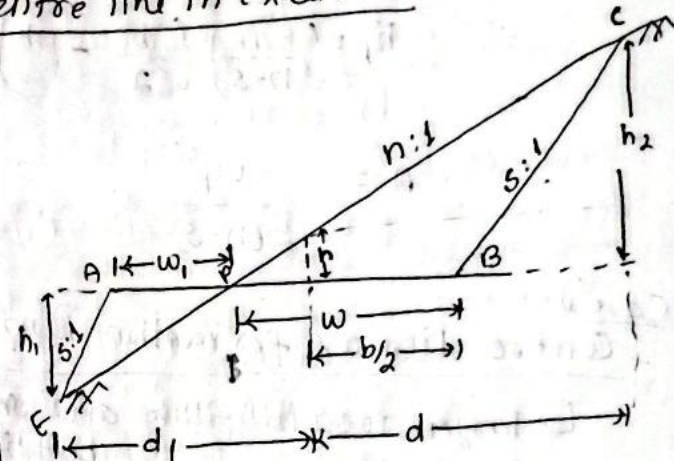
Case I: When centre line in excavation

$$w = \frac{b}{2} + nh$$

$$d = (h_2 - h)n$$

$$h_2 = \left(h + \frac{b}{2n}\right) \left(\frac{ns}{n-s}\right)$$

$$d = \frac{b}{2} + \frac{ns}{n-s} \left(\frac{b}{2n} + h \right)$$



Area in excavation, $A = \frac{wh_2}{2} = \frac{w^2}{2(n-s)}$

$$A = \frac{1}{2} \left(\frac{b}{2} + nh \right) \times \frac{\left(\frac{b}{2} + nh \right)^2}{2(n-s)}$$

$$= \frac{1}{2} \left(\frac{b}{2} + nh \right) \left(\frac{b}{2} + nh \right) \times \frac{1}{n-s}$$

$$= \frac{\left(\frac{b}{2} + nh \right)^2}{2(n-s)} = \frac{w^2}{2(n-s)}$$

Area of filling

$$A_2 = \frac{1}{2} w_1 \times h_1$$

$$w_1 = \frac{b}{2} - nh_I$$

$$h_f = \left(\frac{n}{n-s} \right) \left(\frac{b}{a} - nh \right)$$

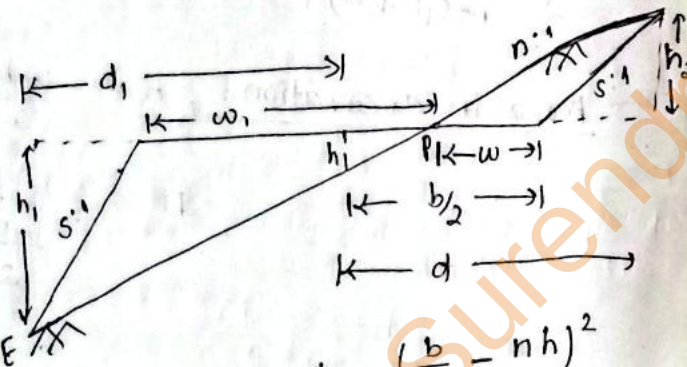
$$A = \frac{\omega_1^2}{2(n-s)}$$

Case II:
Centre line of formation is in excavation

क ११: Centre line of formation in excavation

↳ larger area in filling and smaller area in cutting

$$[w_1 = b - w]$$



Area cutting: $A_1 = \frac{1}{2} \times \frac{\left(\frac{b}{2} - nh\right)^2}{h-s}$

Area filling : $D_2 = \frac{1}{2} \frac{\left(\frac{b}{2} + nh\right)^2}{n-5}$ ✓

Volume Formula

Legend-area Formula

L₁ Norm

gen, ① yields less vol. than that by ②

Diff. b/w volⁿ obtained by (I) & (II) $\Rightarrow$ Prismoidal correction

- paraboloidal formula unjustified $\Rightarrow$ rock excavation concrete waste due to low precision in field data.

• **End-ore** → preliminary estimate
 primumoida → Final estimate

End Area Formula

↳ Trapezoidal formula.

$\therefore \text{vol} =$ assumed to be equal to avg. of area at two ends multiplied by distance b/w the

Vol^m = b/w first two section; $V_1 = \frac{A_1 + A_2}{2} \times L$

next two, $V_2 = \frac{A_2 + A_3}{2} y_L$

$$\text{Total vol}^n = v_1 + v_2 + \dots + v_n$$

$$V = \left[\frac{A_1 + A_n}{2} + A_2 + A_3 + \dots + A_n \right] \times L$$

$A_1, A_2, \dots, A_n \rightarrow$ end consecutive ones
 $L \rightarrow$ distance b/w them

Prismoidal Formula

↳ Simpson's rule for volume

L = length of prismoid

(distance b/w two end area)

(common) uniform distance
Total volume, b/w 2 cls

$$\# \left\{ \begin{aligned} V &= \frac{L}{3} [(A_1 + A_n) + 4(A_2 + A_4 + \dots + A_{n-1}) \\ &\quad + 2(A_3 + A_5 + \dots + A_{n-2})] \end{aligned} \right.$$

L ⇒ common distance

we need to have odd number of cross sections to use this formula.

$$\# V = \frac{d}{3} [A_1 + 4A_m + A_2]$$

A_m → Area of mid section

d → longitudinal distance b/w two adjacent section

Prism:

$$\text{Volume} = A \times L$$

A → End face area, L → length of pyramid

$$V = \frac{A \times h}{3}$$

A → Area of base

h → ht. of pyramid

Prismoidal correction (C_p)

↳ diff. b/w volⁿ computed by end area formula and prismoidal formula.

↳ If area calculated using end area
↳ C_p subtracted to get exact volume

$$C_p = \text{Vol}^n \text{ of Trap rule} - \text{Vol}^n \text{ by prism rule.}$$

$$C_p = \frac{A_1 + A_2}{2} \times D - \frac{D}{6} (A_1 + 4A_m + A_2)$$

Level section: $C_p = \frac{Ls}{6} (h-h')^2$

Two level section: $C_p = \frac{L}{6s} (d-d') (d_1 - d_1')$

Three-level section: $C_p = \frac{d}{12} (h-h') [(d+d_1) - (d'+d_1')]$

Side-hill two level section

Cut: $C_p = \frac{L}{12s} [d-d'] \left[\left(\frac{b}{2} + nh \right) - \left(\frac{b}{2} + n'h' \right) \right]$

Fill: $C_p = \frac{L}{12s} (d_1 - d_1') \left[\left(\frac{b}{2} - nh \right) - \left(\frac{b}{2} - n'h' \right) \right]$

d, d_1, h, n → refer to cls at one end

d', d_1', h', n' → refer to cls at other end

Curvature correction (c)

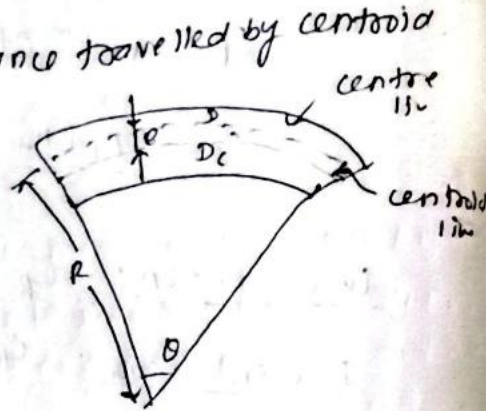
- ↳ Before computing volume it is normally assumed that sections are parallel to one another
- ↳ But, in curved path sections are radial and not parallel to one another

Volume = area $\times$ distance travelled by centroid
distance travelled by centroid

$$\begin{cases} D_c = (R - e) \times \theta \\ D_c = (R - e) \times \frac{D}{R} \end{cases}$$

When centroid away from centreline

$$D_c = (R + e) \times \frac{D}{R}$$



Volume by Pappus Theorem

$$V = \text{Area} \times \frac{D}{R} (R + e)$$

If Area not const;

$$\text{Avg. area} = \frac{A_1 + A_2}{2}$$

$$\bar{e} = \frac{e_1 + e_2}{2}$$

$$V = \frac{A_1 + A_2}{2} \times \frac{D}{R} (R \pm \bar{e})$$

Volume = volⁿ by trapezoidal $\pm$ correction rule

EDM

EDM is a method of determining the length b/w two points using em waves.
EDM commonly carried out with digit instrument called total station

Types (On basis of carrier wave)

- ↳ Microwave $\Rightarrow$ Long range; tellurometer
- ↳ visible light $\Rightarrow$ Medium range; Geodimeter
- ↳ Infrared wave $\Rightarrow$ Short range; Distomat

On basis of range of operation

- ↳ Short range $< 3\text{km}$; tellurometer
- ↳ Medium range $< 5\text{km}$
- ↳ Long range $< 100\text{km}$

Operation of EDM Instruments (Fundamental principle)

$\rightarrow$ EM wave propagate thr atmosphere based on equation

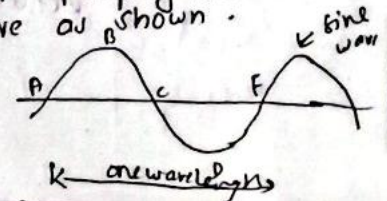
$$v = f\lambda = \frac{1}{T}\lambda$$

$v \rightarrow$ velocity of em energy in m/s

$f \rightarrow$ modulated frequency (in Hz)

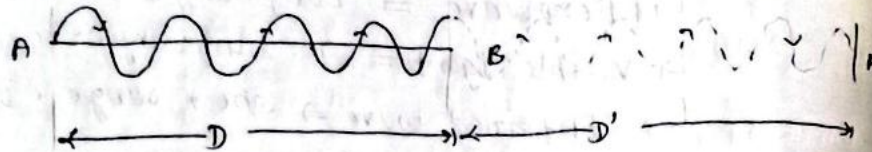
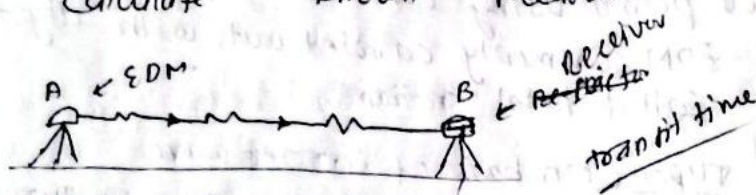
$\lambda \rightarrow$ wavelength of EDM (m)

$\rightarrow$ wave propagated can be represented like a sine wave as shown.



$$\text{Distance} = \text{Velocity} \times \text{Time}$$

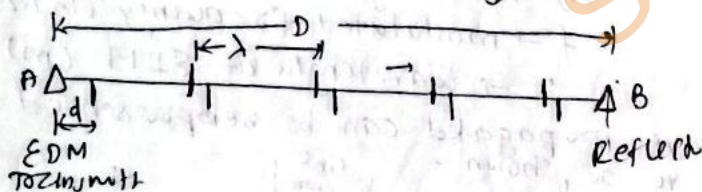
↓ ↓ ↓
Calculate Known Measured



Transit time measurement demonstration

Say AB is survey line to be measured having distance D. The EDM placed at end A, reflector at B.

$d = \frac{\Delta\phi \times \lambda}{360}$
 reflect → phase diff of received signal.
 $d = \frac{\Delta\phi}{360} \times \frac{\lambda}{2}$
 $D = \frac{1}{2} \left(n\lambda + \left(\frac{\Delta\phi}{360} \right) \lambda \right)$
 Double path



EDM Principles of actual Operation

Generation → Modulation → Transmission

↓
Propagation

Demodulation ← Reception ← Reflection

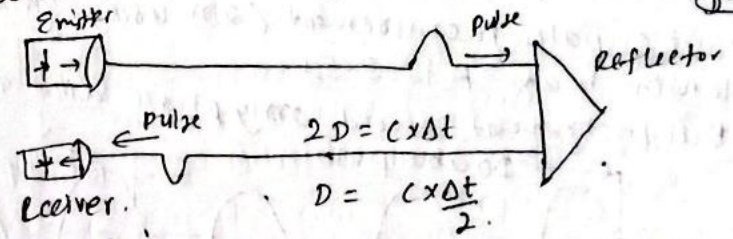
↓
Distance Measured

- × Transmitter sends out sine waves
- × Propagates thr. medium.
- × Prism - Reflect
- × Receiver - Receive
- × Received

Principle of EDM (Measuring Principle)

(1) Pulse Method

Pulse method means distance is equal to product of velocity (V) and time (t)



If time of departure of pulse from gate A is t_A and time of its reception at gate B is t_B .

$$\Delta t = t_B - t_A$$

c → velocity of light thr. medium thr. which it has travelled.

D = distance b/w instrument & target
 $D \rightarrow$ dependent on velocity of light in vacuum medium and accuracy of transit time

Taking approx value of $3 \times 10^8 \text{ m/s}$ for c
 & 10^{-10} s distance $\approx 15 \text{ mm}$

Distance measured function of power of pulse

② Phase-Difference Method

$\hookrightarrow$ Majority EDM whether infrared, light or microwave use this form.

Basic eqⁿ used:-

$$2D = M\lambda + \delta\lambda$$

M = Integer part of wavelength
 (eg. $M=3$ for wave A)

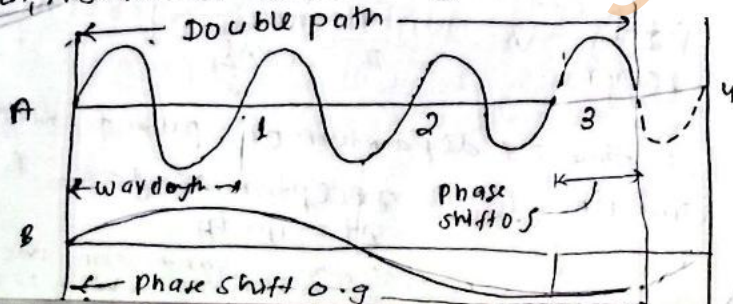
$\delta\lambda$ = fraction part of wavelength = $\frac{\phi}{2\pi} \lambda$

λ = wavelength

ϕ = phase difference

Double path measurement ($2D$) using smaller wavelength beam A is 3.5λ .

But instrument record only phase diff i.e. $0.5(\delta\lambda)$



To determine integer part, larger wavelength B (say 4 times A) is used which produce phase shift/diff of 0.9. In term of basic measuring unit $\Rightarrow$ i.e. $0.9 \times 4 \approx 3.6$ and hence M taken as 3

$$\text{And } 2D = 3\lambda + 0.5\lambda = 3.5\lambda$$

distance calculated knowing value of λ

Errors

Personal $\rightarrow$ Inaccuracy in initial setup of EDM and reflector

$\hookrightarrow$ wrong measurement

Instrumental $\rightarrow$ Calibration error

$\hookrightarrow$ errors shown by reflector

Natural $\rightarrow$ Atmospheric variation in temp, press and humidity. Microwave EDM most susceptible

$\hookrightarrow$ Multiple reflection of signal

EDM Principle of optical principle

(TCVT)

Look on generation and propagation of sine wave through fiber

Given

- ① Transmitted and received sine waves
- ② Chopper-splitter passes part of the wave energy reflecting some sine signal intensity for a reference signal
- ③ Variable filter attenuates internal reference signal wave to same level as externally received waves
- ④ Interference filter acts as polarizing filter eliminating undesirable wave
- ⑤ Finally, photometer measures difference of reflected wave with reference wave

Generation → Modulation → Transmission



Propagation



Reflection



Distance measured: Demodulation ← Reception

Spec of three cat of tailate

S.N.	Char.	First ord	2nd ord		3rd ord	
			I	II	I	II
1	spacing of pore	10km	1km	0.5	0.5	0.25
2	spacing of other	3km	1km	0.5	-	-
3	Min angle	25"	25"	20"	20"	15"
4	standard error length	1in 1×10^6	1in 7.5×10^5	4.5×10^5	2.5×10^5	1.5×10^5
5	Closure in por after satisfying geometric co	1×10^5	5×10^4	20×10^3	10×10^4	5×10^4

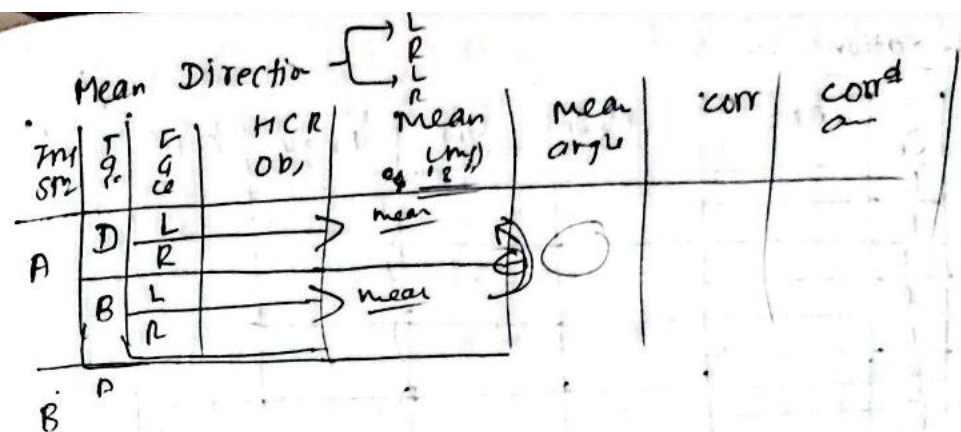
Surendra Sharma

General Specification of diff. types of Triangulation

S.No	Characteristics	First order	Second order	Third order
1	Length of base line	5km to 15km	2km to 5km	0.1 km to 2 km
2	Length of sides	16km to 150km	10km to 25km	1.5km to 10km
3	Avg error in triangulation	< 1"	3 to 3"	12"
4	Station closure	< 3"	8"	15"
5	Actual error of base line	1 in 50,000	1 in 25,000	1 in 10,000
6	Probable error of base line	1 in 10,00,000	1 in 5,00,000	1 in 2,50,000
7	Discrepancy between means	5Nk mm	15Nk mm	25Nk mm
8	Probable error of computed dist	1 in 50,000 to 1 in 250,000	1 in 20,000 to 1 in 50,000	1 in 5,000 to 1 in 20,000
9	Probable error in azimuth	0.5"	5"	10"

Basic Triang. closure of actual & probable error of base line
 discrepancy computed distance azimuth

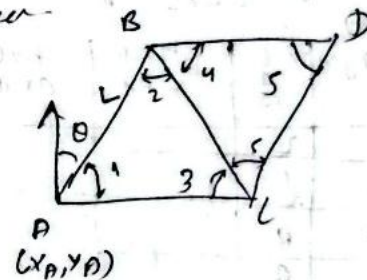
Triangulation:



Triangulation

- Method of fixing accurate control
- to go points
- one side & two angles of Δ known
- joined or over
- All angles mean

Length of all sides
 size law



#

Reiteration

Inst. Stn	Side	Face	Obs.	Mean	Gen.	Angle	Corr.	Corr.	R
O	A	L							
		R							
	B	L							
		R							
	C	L							
		R							
	A	L							
		R							

$$\text{err} = \text{Observed} - \text{Th.}$$

$$\text{Per.} = LC \sqrt{N}$$

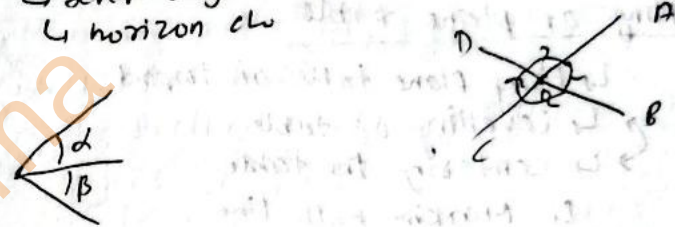
Mean Angle $\begin{matrix} \rightarrow L \cdot R \\ \rightarrow L \cdot R \end{matrix}$

Inst. Stn	Target	Face	HCR Obs.	Face angle	Mean An	Corr.	Corr.
A	D	L					
	B	L					
	D	R					
	B	R					
B	A						
	C						
	A						
	C						

Repetition $\rightarrow$ errors in good. minimized (diff. prob. orig.)
 Personal error of blisc. min
 error due to eccentricity of centres (both verns)
 Error due to LOC no. 18 to train an
 Horizontal circle reading
 Face left
 Face right
 obs. Total angle Mean | obs. Total angle Mean

Inst. Stn
HIReiteration

sever. angle measured at a past station
 horizon clw



Direct method Bookky for
 Set I
 Set II
 Inst. Stn S Fg CO Obj. Any. Mean | obs. Any. Mean

Inst. Stn

O

Mean Direction



Inst. Stn & HI	Target Stn	Face	Observation	mean cm)	Mean Hor. any	Corr.	Corr. R
O	A	L			X AOB		
		R					
	B	L					
		R					

Principle of PTS (Parallelism)

- Lines drawn during plotting is always parallel to the corresponding lines actually present on ground.
- Position of plane table must be identical at each station.

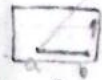
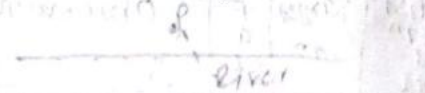
Setting of plane table

- Fixing Plane table on tripod
- Levelling of table
- Centering the table
- Marking north line
- Orientation → position occ. by board in var. station // to each
 - By magnetic Needle (lines on board // to lines on ground)
 - By Back sighting



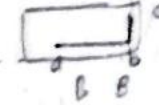
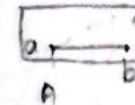
Method of plane table surveying

- Radiation method
 - Intersection method
 - Traversing
 - Resection
- } locate object features on
 } plot the network of stations
 } plot location of plane table and orient table



Traversing

- Plot traversing in case station have not been prev. plotted.
- Plotted using method of radiation by taking back sighting on preceding station & fore sight on following station.
- draw : tachometric method
- great care.



Adv

- Plan is drawn by surveyor himself when area to be surveyed is small.
- no possibility of omitting details.
- compare plotted work with actual feature on ground.
- Simple & cheap method.
- suitable for small scale map.
- NO great skill req.
- useful in rough

Dis.

- not intended for very accurate work.
- not suitable for monotonous work.
- not for large scale work.
- unfav. climate.
- plotting at diff. scales req. full ex.
- Quality dep. on skill of operator.
- many stations heavy, inconv. to move.

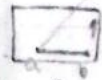
Int. Point

- obtained at the pt. of intersection of radial lines drawn from two diff. stations.
- For large distance

alidade

fiducial edge

adjust



→ well-conditioned triangle → leg ratio 1:2, 1:3
→ st distribute error equally.

→ Principle of chain survey

→ chain triangulation

→ Def: (tie line, Base line, check (proof line))

→ offset (types, description)

→ Field books.